MyDoctor24 · MYDR24

MYDR24 Unified Healthcare Ecosystem — Master Specification

Acute Care · Chronic Care · Virtual Hospitalization (Virtual IPD) · RPM & Elderly Care

Version 3.0 — Draft for approval
17 October 2026
Market: India-first
Contents: Master specification + Annexes A–D

Clinical content (thresholds, triage rules, programmes, response times) is draft and must be approved by the MYDR24 Clinical Governance Committee and regulatory counsel before use with patients. Screens are generated only after explicit approval of this document.

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MYDR24 Unified Healthcare Ecosystem — Master Specification v3.0

Acute Care · Chronic Care · Virtual Hospitalization (Virtual IPD) · Remote Patient Monitoring & Elderly Care

Item	Detail
Document type	Consolidated Stage-1 master specification. **For review and explicit approval.**
Version / date	v3.0-draft · 17 October 2026
Supersedes	Master v2.0 + Domain 2 v2.1 (now included as annexes)
Primary market	India (designed to extend to other regions)
Design source of truth	Existing MYDR24 Figma files (*Caregiver Flow Final*, *MVP-1 Instant Service Flow*), **not verified: no access (Section 17)**
Clinical content status	**Draft.** Every threshold, interval, triage rule and programme must be approved by the MYDR24 Clinical Governance Committee (CGC).
Screen generation	**On hold until this document is explicitly approved** (Section 21)

How to read this document — statement labels

Label	Meaning
[E] Evidence	Backed by a cited source (Evidence Register, Section 23)
[P] Proposal	Our recommended design. Needs approval.
[A] Assumption	Taken as true for design purposes. Needs confirmation.
[D] Decision needed	Open item, listed in Section 22

Document map

Part	Content
Main body (§1–§23)	Unified ecosystem: four pathways, shared record, transitions, Acute Care (new), Chronic Care, Virtual IPD, RPM/Elderly (new), unified dashboard, alert engine, governance, data/FHIR/APIs, security, design audit, complete screen inventory, Phase-1 Doctor app screen plan, acceptance tests, roadmap, decisions, evidence
Annex A	Evidence & market research: Virtual Hospitalization
Annex B	Platform & mobile ecosystem specification (architecture, devices, tech stack, NFRs, budget)
Annex C	Doctor app — Virtual IPD screen specification D-01…D-32
Annex D	Chronic Care Management & Longitudinal RPM domain specification (CC-*)

1. Executive Summary

MYDR24 will run **one longitudinal patient record** and **one authorised clinical team structure** across **four distinct clinical pathways**:

Pathway	Code	One-line purpose
A — Acute Care	ACU	Timely assessment and treatment of new or worsening problems, with an episode that is opened and closed
B — Chronic Care Management	CCM	Planned, long-term, programme-based management of chronic conditions
C — Virtual Hospitalization / Virtual IPD	VIPD	Inpatient-level hospital-at-home care for selected, formally admitted patients
D — Remote Patient Monitoring / Elderly Care	RPM	Device-based or manual observation and elderly support without necessarily treating, enrolling or admitting

Design principles:

1. **Pathway != diagnosis.** An acute illness doesn't automatically mean Virtual IPD, and a chronic condition doesn't make someone an inpatient. A clinician's assessment, safety criteria and the actual care capability available decide the pathway. [P]
2. **Separate lifecycles, shared record.** Each pathway has its own episode, eligibility, monitoring plan, care plan and closure. Every episode writes to the same record. [P]
3. **Explicit, clinician-governed transitions.** Every pathway change records the initiating event, rationale, authorising professional, timestamp, receiving pathway, outstanding tasks and handover status. AI can suggest a transition but can never carry one out. [P]
4. **Emergencies bypass everything.** Emergency symptoms or requests go to the emergency pathway at once. They never wait for routine timers, worklists or AI scoring. [P]
5. **Honest service claims.** Monitoring isn't presented as continuous clinical supervision unless staffing and response coverage actually support it (Section 8.3). [P]
6. **Nothing invented.** Thresholds, response times and triage logic come from validated guidance plus CGC-approved configuration. This document only defines *where* they plug in. [P]
7. **One design system.** The existing MYDR24 Figma theme is kept. New needs are met with documented, compatible component extensions (Section 17). [P]

2. Scope & Pathway Definitions

2.1 Pathway Comparison

Attribute	A — Acute (ACU)	B — Chronic (CCM)	C — Virtual IPD (VIPD)	D — RPM / Elderly (RPM)
Entry	Patient symptom report, referral, CCM/RPM trigger	Referral, post-discharge identification, enrolment	Formal admission after eligibility assessment (D-04)	Device enrolment order or elderly-care subscription with a clinician-approved plan
Eligibility owner	Triage protocol + clinician	Programme criteria + responsible clinician	Admitting doctor against VIPD criteria (clinical, staffing, equipment, home)	Ordering clinician (clinical rules); care coordinator (service)
Care plan	Acute treatment plan (short)	Longitudinal individual care plan (versioned)	Inpatient-level care plan, order sets	Monitoring plan + alert rules + notification preferences
Clinical cadence	As clinically required; follow-up and safety-net	Scheduled reviews per programme	Structured morning, evening and clinically required night reviews; nursing visits	Scheduled review of data per service level
Monitoring	Ad hoc observations; optional short-term device	Manual and/or IoT per programme	IoT and/or manual per tier (T1–T3)	Scheduled/continuous per device capability
Typical length	Days	Months–years	Days (~ 3–14)	Weeks–ongoing
Closure	Resolved / referred / escalated / transitioned	Graduated / withdrawn / closed	Discharge / transfer	Ended / transitioned
Primary FHIR	`EpisodeOfCare(type=ACU)` + `Encounter`	`EpisodeOfCare(type=CCM)` + `CarePlan`	`Encounter(class=HH)` + `EpisodeOfCare(type=VIPD)`	`EpisodeOfCare(type=RPM)` + `DeviceUseStatement` / `Device`
Annex detail	**New in this document (§5)**	Annex D	Annexes B and C	**New in this document (§8)**

2.2 Concurrency Rules (which episodes may be active together)

Combination	Allowed?	Rule
CCM + ACU	✅	Typical when a chronic patient has a new acute problem. CCM stays Active and the ACU episode is linked. CCM alerts carry on. [P]
CCM + VIPD	⚠️	CCM is **suspended, not closed**. VIPD orders take precedence. A Post-Acute Review is required before CCM resumes (Annex D §2.4). [P]
ACU → VIPD	✅ (transition)	The ACU episode closes with outcome "Admitted to VIPD" and is linked to the new VIPD encounter. [P]
RPM + CCM	✅	The RPM monitoring plan is merged into the CCM monitoring plan. Duplicate rules are reconciled, with **one** rule owner per parameter. [P]
RPM + VIPD	⚠️	RPM rules are paused for parameters the VIPD monitors and resume at VIPD discharge. [P]
Two VIPD encounters	❌	Only one active admission at a time.
Two ACU episodes	✅ (if different problems)	Each has its own problem, clinician and closure. [P]

2.3 Out of Scope (v3.0)

- Paediatric and obstetric pathways (need separate triage and programmes). [D]
- Mental-health crisis pathways beyond emergency referral. [D]
- Diagnosis or treatment decided autonomously by AI (prohibited, §10.6).

3. Shared Longitudinal Patient Record

All four pathways read and write the same record. The full field list and FHIR mapping are in **Annex D §4**. In brief:

Domain	Contents	Pathway-specific notes
Identity & demographics	Verified identity (ABHA/Aadhaar-verified flag, method, date)	Required before treatment orders in all pathways
History, allergies, problems	Conditions with status history; allergy/ADR	Acute problems are coded with onset and resolution; chronic problems are linked to programmes
Medications	History, current orders, reconciliation snapshots (`List`)	Reconciliation required at every pathway entry and exit [P]
Observations & trends	With provenance (device/manual/lab), freshness state	Freshness rules per pathway (Annex D §7.2)
Results	Labs, imaging, POC	Acknowledgement tracked per result
Notes & consultations	SOAP, reviews, specialist reports	Linked to episode
Specialist recommendations	Stored apart from orders	Need authorisation to become orders
Care plans & goals	Versioned; never overwritten	One per episode type
Nursing records	Visits, eMAR, assessments	Qualification captured
IoT devices	Assignments, lifecycle, readings	Device validation registry

Domain	Contents	Pathway-specific notes
Alerts & escalations	Event log with event type (§10)	Shared infrastructure
Referrals & transfers	`ServiceRequest`, handover bundles	—
Admissions & discharges	VIPD encounters, hospital ADT feeds	—
Follow-up plans	`Appointment`, `Task`	—
Consent, permissions, audit	Consent ledger per scope and pathway; `AuditEvent`; `Provenance`	DPDP-aligned (§16)

Rule [P]: every record carries the **current pathway(s) and responsible clinician**. These show on the patient header of every screen (component `PathwayChip`, §17.4).

4. Care Transitions

4.1 Mandatory Transition Record (every transition)

Field	Description
Initiating event	Alert ID, patient request, clinician decision, discharge notice, etc.
Clinical rationale	Structured reason + free text
Authorising professional	Name, role, registration number
Timestamp	Initiated / authorised / completed
From → To pathway	Episode IDs
Outstanding tasks	Pending results, meds, visits, devices
Handover status	Not required / Pending / Sent / Accepted (receiver, time)
Patient/family informed	Yes/No + channel
Consent	Consent check for the receiving pathway

FHIR representation [P]: a `Task` (code=`pathway-transition`) with `focus` = new episode, `basedOn` = trigger, plus `Provenance` for authorisation and a `Composition` for any handover.

4.2 Transition Matrix

#	Transition	Who may initiate	Who must authorise	Mandatory checks	System effects
TR-01	CCM → Acute assessment	Rule engine (suggestion), any care-team member, patient	Clinician opening the ACU episode	Red-flag screen first (§5.3)	ACU episode linked; CCM stays Active
TR-02	Acute → Routine follow-up	Treating clinician	Treating clinician	Safety-net advice documented; follow-up booked	ACU closed "Resolved – follow-up"
TR-03	Acute → Virtual IPD assessment	Treating clinician	**Admitting VIPD doctor** after D-04	VIPD eligibility (clinical, staffing, equipment, home), consent	ACU closed "Admitted VIPD"; VIPD encounter opened
TR-04	Chronic → Virtual IPD assessment	Rule (suggestion), care team	Admitting VIPD doctor after D-04	As TR-03	CCM suspended (Annex D T4)
TR-05	Virtual IPD → Chronic follow-up	Discharging doctor	Discharging doctor + CCM responsible clinician acceptance	Discharge summary, med reconciliation, CCM plan review	CCM resumes as a new plan version
TR-06	Virtual IPD → Routine monitoring (RPM)	Discharging doctor	Discharging doctor; RPM ordering clinician named	Monitoring plan + review owner + service level	RPM episode opened (e.g. 14-day post-discharge)
TR-07	Acute → Emergency hospital transfer	Any clinician; patient/caregiver SOS	**No authorisation delay**: emergency first, documentation after	Emergency pathway (§12)	ACU closed "Transferred"; handover bundle
TR-08	Chronic → Emergency hospital transfer	As TR-07	As TR-07	As TR-07	CCM suspended for hospital
TR-09	Post-discharge → Chronic Care or RPM	Discharging team / CCM intake	Responsible clinician (CCM) or ordering clinician (RPM)	Eligibility; consent	New episode
TR-10	RPM → Acute / Chronic / VIPD	Assigned professional	Clinician per receiving pathway	As receiving pathway	RPM continues or pauses per §2.2
TR-11	Any → Closed/Withdrawn	Clinician; patient (withdrawal)	Clinician (closure); patient (withdrawal)	Handover letter if care transfers	Devices/tasks stopped

Prohibitions [P]:

- AI can't admit, discharge, close or transfer anyone.
- No transition can complete with an unacknowledged critical alert unless the transition *is* the emergency response.
- Every transition shows up as an event on the patient timeline.

5. Pathway A — Acute Care (new)

5.1 Evidence Basis for Triage Design

Finding	Source	Implication
Structured ED triage systems have moderate, context-dependent accuracy, and clinically important under-triage is still possible.	[E] EV-31, EV-32	Triage output **supports** clinicians, never replaces them. Track under-triage as a safety KPI.
The Manchester Triage System showed low safety in one review because of under-triage.	[E] EV-33	Validate any protocol locally before use.
The **AIIMS Triage Protocol (ATP)** was developed in India by Delphi consensus and prospectively validated for ED mortality/ICU admission.	[E] EV-34, EV-35	A candidate basis for in-person and nurse triage. **It wasn't validated for remote/tele-triage.** [A]
Online symptom checkers vary widely: triage accuracy roughly 49–90% and low diagnostic accuracy.	[E] EV-36	**Patient-facing symptom checkers can't decide home-treatment suitability.** They only screen for red flags and route to a human.
NMC Telemedicine Practice Guidelines 2020: in emergencies, teleconsultation should be limited to first aid, life-saving advice, referral and counselling when in-person care is available.	[E] EV-37	Acute tele-consults must route emergencies to in-person or emergency services.

[D] Decision: choose the approved remote-triage protocol. Options: (a) adapt ATP for tele-triage with local validation; (b) license a validated telephone-triage system; (c) a CGC-authored protocol with prospective audit. *Recommendation:* (a) plus mandatory clinician review, with an audit of under-triage events.

5.2 Acute Care Workflow (13 steps)

Step	Description	Actor	System support	Screen IDs
1	Patient reports symptoms, or a referral/CCM/RPM trigger arrives	Patient, family, clinician, rule engine	Symptom intake; referral inbox	AC-P-01, AC-D-01
2	Identity and relevant consent verified	Coordinator / clinician	ABHA/OTP check; tele-consult consent	AC-D-02
3	**Immediate emergency red-flag screen**	System (CGC-approved list) + triage nurse	Red-flag positive → **Emergency pathway now** (§12)	AC-P-02, AC-D-02
4	Urgency category selected using the approved triage protocol	Triage nurse/clinician	Protocol engine (versioned); category + target time from service policy	AC-D-03
5	Qualified clinician review as required	Doctor	Queue by urgency; SLA timers	AC-D-04
6	History, meds, allergies, observations collected	Doctor/nurse/patient	Pre-filled from longitudinal record; med reconciliation	AC-D-04, AC-D-05
7	Examination, tests, diagnostics ordered	Doctor	Video exam guide; field-nurse visit; CPOE/labs	AC-D-06
8	Assessment and treatment plan recorded	Doctor	Structured acute note (problem, differential, plan, risk)	AC-D-07
9	Prescriptions authorised by a qualified professional	Doctor (RMP)	e-Rx with registration number, digital signature, NMC drug-list checks	D-17 (shared)
10	Nursing, home-care or specialist services arranged	Doctor/coordinator	Service request; dispatch	AC-D-08
11	Follow-up and safety-net instructions documented	Doctor	Safety-net templates (CGC-approved), follow-up booking	AC-D-09
12	Reassessment when indicated	Doctor/nurse	Scheduled reassessment task; patient check-in	AC-D-10
13	Close: resolved / referred / escalated (VIPD, hospital)	Doctor	Closure form; transition record (§4)	AC-D-11

5.3 Red-Flag Screening (structure only — content is CGC-owned)

Element	Specification
Source	CGC-approved red-flag list, versioned, with evidence reference per item [D]
Examples (illustrative, **not a protocol**)	Suspected stroke signs; chest pain suggesting ACS; severe breathing difficulty; major bleeding; severe altered consciousness/unresponsiveness; seizure; anaphylaxis signs; severe injury
Behaviour	Any positive item → emergency screen with **Call 108/112**, simultaneous command-centre E-event, caregiver guidance. **Nothing waits for further questions.**
Uncertainty	"Not sure" counts as positive for routing to an immediate human call-back [P]
Audit	Every screen result is stored with protocol version

5.4 Exception Pathways (mandatory)

Exception	Detection	Required handling [P] (timings from service policy)
Deterioration during consultation	Clinician observes it, or vitals cross a critical rule mid-call	One-tap *Emergency* from the consult screen (D-25 equivalent); keep the patient on the line; dispatch; notify ED; document after
Can't contact patient (call/video fails, no answer)	Missed call-back, no response to an urgent task	Retry ladder (alternate number → caregiver → emergency contact) → field-team welfare visit if risk warrants → police/emergency services per CGC policy. Each attempt timestamped.

Exception	Detection	Required handling [P] (timings from service policy)
Home-care arrangement fails (nurse unavailable, device not delivered, caregiver absent)	Dispatch failure, task overdue	Re-plan: alternative provider, or upgrade to an in-person facility visit, or hospital. The doctor must re-confirm the plan is still safe.
Emergency resources unavailable (ambulance delay, ED diversion)	Dispatch ETA beyond policy, or decline	Escalate to on-call doctor and ops lead; alternative transport/facility list; continuous tele-support; incident report
Patient declines advice (e.g. refuses transfer)	Clinician records it	Capacity assessment note, informed-refusal documentation, safety-net, scheduled re-contact, senior review
Diagnostic delay (lab not collected/reported)	Overdue result task	Escalate to coordinator; clinician decides whether the plan is still safe

5.5 Acute Episode Status Model

From	Event	To
(start)	Symptom report / referral	Intake
Intake	Red flag positive	**EmergencyRouted** (terminal for ACU; emergency episode continues)
Intake	Red flags negative	Triaged (urgency category)
Triaged	Clinician picks up	InAssessment
InAssessment	Plan signed	Treatment
Treatment	Reassessment due	Reassessment
Reassessment	Improving	Treatment / ReadyToClose
Reassessment	Worsening	Escalated (VIPD assessment / hospital / specialist)
ReadyToClose	Closure form signed	Closed (Resolved / Referred / Transitioned)
Any	Can't contact patient	ContactFailure (exception ladder)

6. Pathway B — Chronic Care Management (summary)

Full specification: Annex D. The brief's 17-step workflow maps to Annex D like this:

Step	Description	Annex D section	Screens
1	Referral, enrolment, post-discharge identification	§2.3–2.4, TR-09	CC-D-03
2	Diagnosis & eligibility verification	§3	CC-D-03
3	Baseline clinical & functional assessment	§3, §4	CC-D-06
4	Medication reconciliation	§9	CC-D-07
5	Individual care-plan creation & approval	§5	CC-D-08
6	Monitoring frequency & targets	§5, §7	CC-D-09
7	Doctor/nurse/specialist assignment	§5.1 #16, §11	CC-D-08
8	IoT setup or manual instructions	§7, §17.2	CC-N-04, CC-P-06
9	Scheduled observations & symptom reporting	§7	CC-P-04/06/08
10	Adherence & side effects	§9	CC-P-07
11	Lab & diagnostic follow-up	§6 worklists	CC-D-02
12	Planned doctor & MDT reviews	§11	CC-D-15, CC-D-14
13	Care-plan progress evaluation	§5.2	CC-D-15
14	Worsening symptoms / abnormal trends	§8	CC-D-11
15	Reassessment & escalation	§2.4 T3, TR-01/04	CC-D-12
16	Care-plan modification by an authorised clinician	§5.3	CC-D-08
17	Periodic outcome assessment	§14	CC-D-16

Launch programmes recommended: Hypertension, Type 2 Diabetes, Heart Failure, and Post-discharge continuity. These need CGC sign-off (Annex D §21). [D]

7. Pathway C — Virtual Hospitalization / Virtual IPD (summary & updates)

Full specification: Annexes B and C. Updates in v3.0:

Area	Specification
Admission	Formal admission only after D-04 (clinical inclusion, hard exclusions, **staffing capacity check**, **equipment availability**, home environment, ambulance radius) and consent. An acute or chronic diagnosis alone never activates VIPD. [P]
Episode	Virtual ward + bed assignment; responsible doctor; care team; code status (D-07)

Area	Specification
Structured reviews	**Morning review** (full round, plan), **evening review** (progress, overnight plan), **night review when clinically required** (by rule or nurse request). The US CMS AHCAH waiver requires daily physician evaluation and at least two in-person RN/paramedic visits a day [E] EV-38. The MYDR24 cadence is configurable per tier and set by the CGC. [P]
Monitoring	IoT or manual per tier; freshness rules (Annex C global rules)
Specialist opinions	Through the referral workflow (§13)
Deterioration	NEWS2 + rules; emergency bypass; escalation → transfer (D-24…D-27)
Discharge	3-step discharge (D-32) → transition TR-05 / TR-06 / closed
Capacity guard	If staffing or equipment capacity is reached, the admission CTA is disabled with a "Capacity reached — refer" message. **No over-admission.** [P]

8. Pathway D — Remote Patient Monitoring & Elderly Care (new)

8.1 Workflow

Step	Description	Owner
1	Order/subscription: clinical question, parameters, duration, **ordering clinician**, **assigned reviewing professional**	Ordering clinician
2	Consent: device data, notifications to family (scoped), service-level disclosure (what is and isn't monitored, response hours)	Coordinator
3	Device enrolment: validated device registry, pairing, test reading	Field tech / nurse
4	Monitoring plan: measurement schedule per device capability; manual fallback	Ordering clinician
5	Alert rules: CGC-approved templates; individual thresholds set by a clinician	Ordering clinician
6	Data review per service level; freshness monitoring	Assigned professional
7	Notifications to patient and authorised family per consent	System
8	Escalation to the assigned professional, then clinician	Per rule
9	Clinical reassessment when indicated	Clinician
10	Transition to ACU / CCM / VIPD (TR-10) or end of episode	Clinician

8.2 Elderly-Care Features [P]

- Scheduled wellbeing check-ins (voice/video).
- Fall-detection sensors or reports.
- Inactivity alerts (where the device supports them).
- Medication reminders (with dose-event states: Annex D §9.2).
- Caregiver burden check, cognitive-change reports, social contact log.
- One-tap emergency button (Home Hub/pendant).

8.3 Service Levels and Honest Claims (mandatory)

Service level	Monitoring	Human review	Response coverage	Allowed marketing claim	Prohibited claim
RPM-L1 Self-monitoring	Device + app	Patient/family; clinician at scheduled reviews	No real-time response	"Readings shared with your doctor at your reviews"	"Monitored 24×7"
RPM-L2 Business-hours review	Device + rules	Assigned professional during stated hours	Stated hours only; emergencies → 108/112	"Reviewed by our care team [hours]"	"Continuous supervision"
RPM-L3 Extended / 24×7 staffed	Device + rules	Command centre staffed per roster	24×7, only when the roster and SLA are verified monthly	"24×7 monitored by clinical staff"	Any claim when coverage gaps exist

Rules [P]:

- The service level is shown on the patient's app and in consent.
- If coverage drops (e.g. staffing gap), the system automatically shows a **coverage notice** to affected patients and routes alerts to the backup policy.

9. Unified Doctor Dashboard

9.1 Structure (no undifferentiated queue)

- **Top level:** pathway tabs `All · Acute · Chronic · Virtual IPD · RPM/Elderly` . Each tab has its own counts and worklists.
- **Always-visible strip:** Critical alerts & unresolved escalations, across pathways, pinned above everything.

9.2 Worklists

Worklist	Pathways	Inclusion
Critical alerts & unresolved escalations	All	E/U class events not closed
Acute patients (by urgency)	ACU	Triaged, InAssessment, Reassessment
Chronic patients (by programme)	CCM	Active, ReassessmentRequired

Worklist	Pathways	Inclusion
Active Virtual IPD patients	VIPD	Admitted; NEWS2-sorted
RPM & elderly patients	RPM	Active; data-gap and alert badges
Awaiting clinical review	All	Review task due/overdue
Pending diagnostic results	All	Ordered, not resulted / resulted, not acknowledged
Upcoming consultations	All	Appointments next 24 h / 7 d
Specialist referrals	All	Awaiting response / recommendations to authorise
Medication review tasks	All	Reconciliation due, renewal requests
Nursing visits	All	Scheduled/overdue/failed
Hospital transfer coordination	ACU, VIPD, CCM	Transfer in progress
Discharge planning	VIPD (and ACU closure)	Discharge-ready
Overdue follow-ups	All	Past due date

Filters: pathway, assigned clinician, urgency, care-plan status, monitoring mode (IoT/Manual/Hybrid), next review, unresolved tasks, programme, location.

Every patient row and header shows: `PathwayChip` (one or more), responsible clinician, urgency/severity, freshness, code status (if VIPD/elderly).

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10. Alert & Event Engine (shared infrastructure, separate rule sets)

10.1 Event Taxonomy (distinct event types — never conflated)

Event type	Meaning	Example	Default handling [P]
`EMERGENCY_REQUEST`	Patient/caregiver SOS or emergency symptom	Red-flag positive	**Immediate emergency routing; no ack timer wait**
`CLINICAL_DETERIORATION`	Clinical rule fired on valid data	NEWS2 rise; trend rule	Severity per rule
`CRITICAL_OBSERVATION`	Single critical value	SpO_2 below critical rule	Per rule
`CRITICAL_RESULT`	Lab critical value	K^+ critical	Per lab policy
`ADVERSE_REACTION`	Suspected severe reaction	Anaphylaxis symptoms → emergency	Per rule
`REASSESSMENT_DUE`	Clinically required reassessment not done	Acute 24-h review	Clinical task
`DATA_MISSING`	Expected clinically important reading not received	Missed weight (HF)	Adherence/data class
`DEVICE_FAILURE`	Device fault, battery, sensor error	Cuff error	Technical
`CONNECTIVITY_LOSS`	Hub/device offline	No heartbeat	Technical → clinical only if the plan requires live data
`NOTIFICATION_DELIVERY_FAILURE`	Alert/notification couldn't be delivered to the recipient	Push failed, phone unreachable	**Re-route immediately to the next channel/recipient**; never counted as "acknowledged"
`ACK_TIMEOUT`	Recipient didn't acknowledge in time	—	Escalation sequence
`TASK_OVERDUE`	Care task not done	Visit missed	Coordinator

10.2 Rule Sets by Pathway

Rule set	Typical triggers (CGC to define)	Evaluation
ACU	Emergency symptoms; rapid deterioration; critical observations/results; severe adverse reactions; urgent reassessment	Real-time + task timers
CCM	Sustained deviation from baseline; outside targets; worsening symptoms; missing important observations; missed follow-ups/investigations; medication concerns; disease-specific trends	Batch windows + data-sufficiency (Annex D §8)
VIPD	NEWS2 bands, hard thresholds, trend rules, device alarms	Real-time stream (Annex B §9)
RPM	Template rules per device and service level	Real-time or batch per device; routed per service level (§8.3)

10.3 Rule Definition Schema (all fields mandatory)

Trigger & rationale · clinical protocol & version · event type · severity · pathway · responsible recipient (role) + fallback · required response · acknowledgement deadline (service-policy reference) · escalation sequence · resolution criteria · suppression/de-dup rules (**never by low risk score**) · audit events · CGC approval ID/date.

10.4 Routing

- Route by **pathway + monitoring plan + service level + current roster**.
- If there's no rostered recipient (coverage gap), the alert goes to the **backup policy**. A `NOTIFICATION_DELIVERY_FAILURE` /coverage event is raised and the gap is logged for quality review.

10.5 Emergency Bypass

`EMERGENCY_REQUEST` and E-class rules skip queues and acknowledgement timers. The command-centre critical channel, the emergency guidance screen and the caregiver call all fire **in parallel**.

10.6 AI Boundaries (all pathways)

AI can summarise, flag, prioritise within limits, and suggest. AI **can't**:

- diagnose, prescribe, change or stop treatment;
- decide home-treatment suitability;
- suppress or downgrade alerts;
- admit, discharge or transition a patient.

Full register: Annex D §13.

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11. Clinical Governance & Patient Safety

Element	Specification
Clinical Governance Committee (CGC)	Medical director (chair), nursing lead, clinical safety officer, pharmacist, programme leads, quality lead, patient representative. Approves protocols, rules, thresholds, service policies and content. [D] members
Clinical safety case	Hazard log, risk controls, residual-risk acceptance, using ISO 14971 risk management and IEC 62304 / IEC 82304-1 software lifecycle and health-software safety [P]. A safety case per release.
Change control	Clinical-content changes (rules, thresholds, triage lists, education) are versioned, peer-reviewed, CGC-approved and released with an effective date. Running episodes keep a record of the version applied.
Incident management	Near-miss and adverse-event reporting from every app; root-cause analysis; reports to partner hospitals and regulators as required
Audit programme	Monthly: under-triage audit (ACU), alert ack performance, transitions without complete handover, coverage gaps (RPM-L3), medication-reconciliation completion
Training & credentialing	Role-based training; qualification stored per user; task-level qualification enforcement (Annex D §10.2)

11.1 Top Hazards (initial hazard log extract)

ID	Hazard	Cause examples	Key controls
HZ-01	Delayed emergency response	Emergency treated as routine; notification failure	Red-flag bypass; parallel routing; `NOTIFICATION_DELIVERY_FAILURE` re-route; drills
HZ-02	Under-triage in Acute Care	Unvalidated protocol; patient misreporting	Validated protocol; clinician review; "not sure" = escalate; under-triage audit
HZ-03	Inappropriate home admission	Pressure on beds; capacity exceeded	D-04 hard stops; capacity guard; senior override audit
HZ-04	Missing data treated as normal	UI interpolation; averages without n	Freshness states; gaps visible; "not assessable"
HZ-05	Wrong patient	Similar names	Identity verification; two identifiers on every order
HZ-06	Medication error across transitions	No reconciliation	Mandatory reconciliation at entry and exit
HZ-07	Over-claimed monitoring	Marketing beyond coverage	Service levels (§8.3); coverage notice
HZ-08	Lost handover	Transition without acceptance	Handover status required; overdue acceptance alert
HZ-09	AI error influencing care	Hallucinated summary	Labelling; sign-off; prohibited actions enforced
HZ-10	Privacy breach to family	Wrong consent scope	Scope enforcement; revocation <= 5 min

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12. Emergency Escalation & Hospital Transfer (all pathways)

Step	Action	Notes
1	Trigger: red flag, SOS, E-rule, clinician declares emergency	Any pathway, any screen
2	Parallel: show **Call 108/112** to the patient/caregiver; command-centre critical event; on-call doctor paged	No sequential waiting
3	Code-status check (D-07) shapes guidance (Full/DNR/Limited)	
4	Dispatch: ambulance (108/112/private contract), field team if nearer	ETA tracked; resources-unavailable exception (§5.4)
5	Caregiver guidance (CPR / recovery position) only if code status allows and protocol approved	Annex C D-25
6	Receiving ED pre-alert + handover bundle (FHIR `Bundle`)	Annex C D-27
7	Transition record (TR-07/TR-08); episode suspended or closed	§4
8	Post-event review: incident if any delay	§11

13. Specialist Consultation & Third Opinion

Element	Specification
Request types	Specialist referral (routine/urgent), **e-consult** (asynchronous advice), **second/third opinion** (patient- or clinician-requested independent review)
Mandatory package	Clinical question, urgency, longitudinal summary, meds, allergies, observations, investigations, previous opinions, patient consent for sharing
Statuses	Draft → Sent → Accepted/Declined → Scheduled → Completed (report) → Recommendations pending authorisation → Incorporated / Not adopted (reason)
Third-opinion rules [P]	Reviewer is independent of the previous opinions; sees all opinions; disagreement is shown side by side; the responsible clinician documents the decision and discusses it with the patient
Authority	Recommendations != orders until authorised by the responsible clinician (or the specialist, if formally delegated)
Screens	CC-D-14, CC-S-01/02, plus new **SP-D-01 Third-Opinion Request** and **SP-S-01 Third-Opinion Review** (§18)

14. Data Model & APIs (unified)

14.1 Core Resources (FHIR R4 / ABDM profiles)

Concept	Resource	Key elements
Patient & identity	`Patient`, `RelatedPerson`, `Provenance`	ABHA, verification
Pathway episode	`EpisodeOfCare`	`type` in {ACU, CCM, VIPD, RPM}; `status`; `careManager`; `team`; links
Visit / consult / admission	`Encounter`	class AMB/VR/HH/EMER; `episodeOfCare`
Transition	`Task` (code= `pathway-transition`) + `Provenance` + `Composition`	§4.1 fields
Triage	`QuestionnaireResponse` (protocol version) + `RiskAssessment` (urgency)	Red-flag results
Problems	`Condition`	category problem-list/encounter-diagnosis
Plans & goals	`CarePlan`, `Goal`, `PlanDefinition` (programmes, order sets)	Versioned
Orders	`MedicationRequest`, `ServiceRequest`	Signed
Administration / adherence	`MedicationAdministration`; dose-event extension (Annex D §9.2)	
Observations	`Observation` (incl. `valueSampledData`) + `Device`	Provenance, quality, freshness
Results	`DiagnosticReport`, `ImagingStudy`	Ack task
Alerts/events	`Flag`, `DetectedIssue`, `Communication`, `CommunicationRequest`	Event type (§10.1) extension
Tasks	`Task`	Qualification requirement extension
Consent	`Consent`	Scope per pathway & family
Audit	`AuditEvent`	All reads/writes

14.2 API Surface (additions to Annexes B and D)

Endpoint	Purpose
`POST /episodes` · `PATCH /episodes/{id}/status`	Create/transition any pathway episode (validated against status models)
`POST /transitions` · `PATCH /transitions/{id}`	Transition record lifecycle (initiated → authorised → handover accepted)
`POST /acute/intake` · `POST /acute/{id}/redflag-screen` · `POST /acute/{id}/triage`	Acute intake, red flags, triage (protocol version required)
`POST /acute/{id}/closure`	Close with outcome
`POST /rpm/orders` · `PUT /rpm/{id}/service-level`	RPM order and service level
`GET /dashboard/worklists?pathway=&filters=`	Unified worklists
`GET/POST /events` (`eventType`, `severity`, `pathway`, `ruleId`, `ruleVersion`)	Shared event engine
`POST /events/{id}/ack` · `POST /events/{id}/resolve`	Ack/resolution with criteria
`GET /coverage/roster`	Coverage for routing and RPM-L3 claims
`POST /opinions` (type=referral/e-consult/second/third)	Specialist workflows

Event streaming topics (extends Annex B §11.3): `episodes.events`, `transitions.events`, `triage.events`, `events.clinical`, `events.technical`, `events.delivery`.

15. HL7 FHIR & ABDM Interoperability

- **FHIR R4 (4.0.1)** pinned. ABDM profiles are used for health records shared via HIP/HIU (Prescription, DiagnosticReport, OPConsultation, DischargeSummary, WellnessRecord). [E] ABDM (Annex B §10.2)
- Retired STU3 resources are not used (`ReferralRequest`, `ProcedureRequest`). Mapping corrections are in Annex C Part C.
- Partner HIS: HL7 v2 ADT (admission/discharge notices trigger TR-05/08/09), ORU (results), FHIR APIs where available.
- Terminology: SNOMED CT (India has an NRC), LOINC for observations, UCUM units, and a national drug list for medications. [A] Confirm licensing.
- Claims: NHCX readiness (Annex B).

16. Security, Privacy & Consent

Area	Specification
DPDP Act 2023 & Rules 2025	Rules were notified 13 Nov 2025 with phased commencement: Consent-Manager provisions from Nov 2026, core obligations from May 2027 [E] EV-16. **Build for full compliance at launch.** [P]
Consent scopes	Per pathway (ACU tele-consult; CCM programme; VIPD admission; RPM device data and service level), per data category, per family member; withdrawal is honoured system-wide
Telemedicine	NMC Telemedicine Practice Guidelines 2020: RMP identification, consent, prescribing lists, emergency limits [E] EV-37
Access control	RBAC + ABAC (care-team/roster/pathway); break-glass with reason; session controls
Encryption & security ops	TLS 1.3/mTLS, AES-256 at rest, CERT-In incident directions, VAPT, ISO 27001 path (Annex B §13)
Medical-device software	CDSCO SaMD classification for alerting/triage/AI components that drive clinical action [D]
Data residency	India regions (Annex B §8.3)
Patient rights	Access, correction, erasure (with clinical-record retention limits), grievance, nominee

17. Existing MYDR24 Design-System Audit

17.1 Audit Scope & Limitation (please read)

- **Figma files couldn't be opened.** Both shared links need a Figma login (HTTP 403). This audit is based **only** on the three doctor-app storyboard canvases (raster images) shared earlier.
- Colours, type sizes and spacing below are **visual observations, not verified tokens**. **Exact design fidelity isn't claimed.** [A]
- **[D]** To complete the audit: grant view access, share a Figma library export, or allow a Figma API token/MCP connection.

17.2 Observed Foundations (unverified)

Foundation	Observation from canvases
Brand	"MyDoctor24 — Healthcare for Every Community" logo, green/blue cross mark
Primary colour	Saturated medium blue used for primary buttons, active tabs and headers
Status colours	Red (Critical, Red alert, Live Emergency banner), Amber/orange (Amber alerts, "Due"), Green (Stable, "Given"), Blue (Info, PRN)
Surfaces	White cards on a light grey/blue background, rounded corners, light shadow
Typography	Sans-serif; bold screen titles; medium labels; large numeric vitals
Navigation	Bottom tab bar (Home · Patients · Rounds/Chart · Alerts · More); back-arrow headers
Iconography	Outline icons; coloured document/med icons in lists

17.3 Observed Components (candidate reuse)

Component (proposed name)	Seen on	Reuse for
`PatientHeaderCard`	Chart, alerts, handover	All patient-context screens
`VitalTile`	Chart, baseline, alert detail	All vitals displays
`StatusPill`	Census (Stable/Critical/Pending), MAR (Given/Due/PRN)	Status everywhere
`AlertRow` + severity chip	Alert inbox	Event inbox (all pathways)
`SegmentedTabs`	Chart, consent, CPOE	Pathway tabs, sub-views
`Stepper`	Admission eligibility/consent	Acute intake, CCM enrolment
`ChecklistRow`	Eligibility, discharge checklist	Red-flag screen, closure checklists
`DocumentRow` (View)	Consent review	Consent scopes
`OrderLine` / `MARLine`	CPOE, e-Rx & MAR	Orders across pathways
`TrendChartCard`	Real-time vitals, progress analytics	Longitudinal trends
`VideoCallScreen`	Tele-round console	Acute consults, reviews
`ChatThread`	Care-team chat	All messaging
`EmergencyBanner`	Emergency console	Emergency screens
`BottomNav` , `PrimaryButton` , `KPIcard`	Dashboard	All

17.4 Required Extensions (compatible proposals — styled from existing tokens)

New component	Purpose	Based on	Design uncertainty
`PathwayChip` (ACU / CCM / VIPD / RPM)	Show pathway on every row and header	`StatusPill` shape, **neutral outline + icon** (so it doesn't clash with severity colours)	Icon set and label language [D]
`ResponsibleClinicianTag`	Show clinical responsibility	Avatar + name text style	—

New component	Purpose	Based on	Design uncertainty
`FreshnessIndicator` (Live/Current/Due/Missing/Stale)	Data-freshness states	Small dot + caption	Grey shade for stale
`DataGapBadge`	Data gap != risk	Outline badge, technical blue	—
`CodeStatusChip`	Full/DNR/Limited	`StatusPill`	Wording per legal review
`ServiceLevelBadge` (RPM-L1/L2/L3)	Honest coverage display	`StatusPill`	Marketing copy approval
`EventClassChip` (E/U/R/A/T + technical/delivery)	Event taxonomy	Existing severity chips + new technical/delivery variants	Colour for delivery failure [D]
`TransitionCard`	Show initiated/authorised/handover status	Card + stepper	—
`RedFlagScreen` template	Emergency-first intake	`ChecklistRow` + `EmergencyBanner`	Must be usability-tested
`PathwayTabs` on dashboard	No undifferentiated queue	`SegmentedTabs` (scrollable)	Tab overflow on small screens

Rule: no new colours, fonts or visual identity per pathway. Pathways are told apart by label and icon, not by theme. [P]

18. Complete Screen Inventory (all roles, all pathways)

18.1 ID Scheme

`<Pathway>-<Role>-<nn>`. Pathway: **UD** unified · **AC** acute · **CC** chronic · **D/VN/VP** virtual IPD (existing) · **RM** RPM/elderly · **SP** specialist/opinion · **EM** emergency. Role: D doctor · N nurse/professional caregiver · P patient/family · A allied · S specialist · M manager/admin.

18.2 Totals

Role	Unified/shared	Acute	Chronic	Virtual IPD	RPM/Elderly	Specialist/Opinion	Total
Doctor	4 (UD-D)	11 (AC-D)	16 (CC-D)	32 (D-01…32)	5 (RM-D)	1 (SP-D)	**69**
Nurse / professional caregiver	2 (UD-N)	6 (AC-N)	12 (CC-N)	10 (VN)	4 (RM-N)	—	**34**
Patient / family	2 (UD-P)	6 (AC-P)	18 (CC-P)	8 (VP)	6 (RM-P)	—	**40**
Allied professionals	—	—	8 (CC-A)	—	—	—	**8**
Specialist	—	—	2 (CC-S)	—	—	1 (SP-S)	**3**
Manager / admin	3 (UD-M)	1 (AC-M)	5 (CC-M)	—	1 (RM-M)	—	**10**
Total	**11**	**24**	**61**	**50**	**16**	**2**	**164**

18.3 Per-screen Specification Standard

Every screen is specified with: Screen ID · role · purpose · entry point · data displayed · actions & permissions · validation · alert states · loading/empty/offline/error states · API dependencies · navigation · clinical safety requirements · acceptance criteria · Figma components to reuse (§17.3/17.4).

- **Global defaults** (states, audit, accessibility, freshness, emergency access) apply to all screens (Annex C B2, Annex D §15.1). Tables list only what's specific to each screen.
- **Existing detailed specs:** D-01…D-32 → Annex C; CC-* → Annex D §15.

18.4 New Screens — Unified (UD)

ID	Role	Screen	Purpose	Entry	Key data / actions	Safety / validation	APIs	Acceptance criteria	Reuse
UD-D-01	Doctor	**Unified Clinical Dashboard**	Pathway-separated worklists (§9)	Login	Pathway tabs, critical strip, worklist cards, filters	Critical strip always on top; no mixed queue	`GET /dashboard/worklists`	E/U events visible <= 5 s; counts per tab correct	KPIcard, PathwayTabs, AlertRow
UD-D-02	Doctor	Unified Patient Header & Pathway Summary	Show all active episodes, responsibility, code status	Any patient	Episodes list, responsible clinicians, open transitions	Header present on every patient screen	`GET /patients/{id}/episodes`	Header shows >= 1 PathwayChip when any episode is active	PatientHeaderCard, PathwayChip
UD-D-03	Doctor	Transition Request & Authorisation	Create/authorise transitions (§4)	Header action, alert, closure	TR type, rationale, receiving pathway, outstanding tasks, handover	Required fields; AI suggestion labelled; authoriser role check	`POST/PATCH /transitions`	Can't complete without the receiving acceptance where required	TransitionCard, Stepper
UD-D-04	Doctor	Unified Event Inbox	All event types with filters (§10.1)	Alerts tab	Event class, type, pathway, rule, ack timer	Delivery failures shown separately; E bypass	`GET /events`	`NOTIFICATION_DELIVERY_FAILURE` never shows as acknowledged	AlertRow, EventClassChip
UD-N-01	Nurse	Unified My Day	Visits across pathways	Login	Visits with PathwayChip, tasks	Qualification locks	`GET /visits`	Pathway visible per visit	PathwayChip

ID	Role	Screen	Purpose	Entry	Key data / actions	Safety / validation	APIs	Acceptance criteria	Reuse
UD-N-02	Nurse	Transition Handover Inbox	Accept handovers	Notification	Handover package	Acceptance required	PATCH /transitions/{id}	Overdue acceptance → alert	TransitionCard
UD-P-01	Patient	Home (all my care)	Active pathways in plain language	App open	Cards per active pathway; emergency button	Service level shown for RPM	GET /me/episodes	Emergency reachable in 1 tap	KPIcard, EmergencyBanner
UD-P-02	Patient	My Care Team & Responsibility	Who is responsible for what	Home	Clinicians per pathway, hours	"Not for emergencies" banner	GET /me/careteam	Matches current roster	ChatThread
UD-M-01	Manager	Coverage & Roster Monitor	Verify coverage for routing and RPM-L3 claims	Admin	Roster vs required coverage	Coverage gap → coverage notice workflow	GET /coverage/roster	Gap creates an event within 1 min	—
UD-M-02	Manager	Transition & Handover Audit	Quality review	Admin	Transitions with missing handover/fields	—	GET /transitions?audit	Exportable	—
UD-M-03	Manager	Clinical Content & Rule Governance	Versioned protocols, rules, red-flag lists	Admin	Draft → review → CGC approve → publish	Only approved versions execute	GET/POST /rules, /protocols	Draft can't run in production	—

18.5 New Screens — Acute Care (AC)

ID	Role	Screen	Purpose	Entry	Key data / actions	Safety / validation	APIs	Acceptance criteria	Reuse
AC-D-01	Doctor	Acute Intake Queue	New symptom reports & referrals by urgency	UD-D-01 Acute tab	Urgency, wait time vs target, red-flag result, source	Red-flag-positive cases never sit here (already routed)	GET /acute?status=	Sorted by urgency then wait	AlertRow, StatusPill
AC-D-02	Doctor	Identity, Consent & Red-Flag Review	Steps 2–3	Open case	Identity check, tele-consult consent, red-flag answers	Positive → Emergency (EM) immediately	POST /acute/{id}/redflag-screen	Emergency route <= 2 taps	ChecklistRow, RedFlagScreen
AC-D-03	Doctor/Nurse	Triage (approved protocol)	Step 4	AC-D-02	Protocol questions, urgency category, rationale	Protocol version stored; clinician can up-triage freely; down-triage needs a reason	POST /acute/{id}/triage	Category + version saved	Stepper
AC-D-04	Doctor	Acute Consultation Workspace	Steps 5–6, 8	Queue	History, meds, allergies, observations, video, notes	Deterioration → Emergency button always visible	GET /patients/{id}/longitudinal	Prefilled from record	VideoCallScreen, VitalTile
AC-D-05	Doctor	Acute Med Reconciliation	Step 6	Workspace	Current meds vs reported	Required before e-Rx	POST /medrec	Blocks e-Rx if not done	MARLine
AC-D-06	Doctor	Acute Orders (tests, visits)	Step 7	Workspace	Labs, imaging, field visit, POC	Home-ineligible orders blocked	POST /servicerequests	Urgent order → field task	OrderLine
AC-D-07	Doctor	Acute Assessment & Plan	Step 8	Workspace	Problem, differential, plan, risk	Sign required	POST /compositions	Plan signed before closure	SegmentedTabs
AC-D-08	Doctor	Arrange Services	Step 10	Plan	Nursing/home-care/specialist requests	Provider availability check; failure → exception	POST /servicerequests	Failure raises a task	ChecklistRow
AC-D-09	Doctor	Safety-Net & Follow-up	Step 11	Plan	Approved safety-net template, follow-up booking	Template from approved library	POST /appointments	Patient receives instructions in their language	DocumentRow
AC-D-10	Doctor	Reassessment & Exceptions	Step 12, §5.4	Task	Reassessment form; contact-failure ladder; declined advice	Each attempt timestamped	PATCH /acute/{id}	Ladder can't be skipped without a reason	Stepper
AC-D-11	Doctor	Acute Closure / Escalation	Step 13	Workspace	Outcome: resolved / referred / VIPD assessment (→ D-04) / hospital (→ EM)	Transition record required	POST /acute/{id}/closure	Closure blocked with an unacknowledged critical event	TransitionCard
AC-N-01	Nurse	Acute Visit	Assessment visit	My Day	Observations, exam aid, POC	Qualification	POST /observations	—	VitalTile
AC-N-02	Nurse	Triage Call (nurse-led)	Protocol triage by phone	Queue	Protocol, urgency	As AC-D-03	POST /acute/{id}/triage	—	Stepper
AC-N-03	Nurse	Welfare Check (can't contact)	Contact-failure visit	Exception task	Outcome, escalation	Emergency button	PATCH /acute/{id}	Outcome required	EmergencyBanner

ID	Role	Screen	Purpose	Entry	Key data / actions	Safety / validation	APIs	Acceptance criteria	Reuse
AC-N-04	Nurse	Acute Medication Administration	Give prescribed meds	Task	eMAR, scan	5 rights	POST /medadmin	—	MARLine
AC-N-05	Nurse	Acute Escalation (SBAR)	Escalate	Any	SBAR	—	POST /escalations	Ack tracked	AlertRow
AC-N-06	Nurse	Emergency Documentation	Record emergency events	EM	Timeline, handover	—	POST /compositions	—	—
AC-P-01	Patient	Report Symptoms	Start acute care	Home	Symptom entry, duration, photos	Red-flag screen first	POST /acute/intake	Red flags shown before other questions	Stepper
AC-P-02	Patient	Emergency Help (red flag)	Immediate action	Red flag positive / SOS	Call 108/112, care-team notified, guidance	Works offline (dialer)	POST /events (EMERGENCY)	Command-centre event <= 10 s	EmergencyBanner
AC-P-03	Patient	Waiting & Call-back Status	Expectations	After intake	Urgency category explanation, expected contact window, "get worse? do this"	No diagnosis shown	GET /acute/{id}	—	StatusPill
AC-P-04	Patient	Consultation (video/audio)	Consult	Notification	Video, chat	Fallback to audio/PSTN	—	—	VideoCallScreen
AC-P-05	Patient	My Acute Plan & Instructions	Follow the plan	After consult	Meds, safety-net, follow-up	Plain language	GET /acute/{id}/plan	—	DocumentRow
AC-P-06	Patient	Acute Check-in	Reassessment PROs	Scheduled	"Better / same / worse" + symptoms	"Worse" → clinical task; red flags → AC-P-02	POST /questionnaireresponses	—	ChecklistRow
AC-M-01	Manager	Acute Service Monitor	Wait-time vs target, under-triage audit	Admin	KPIs	—	GET /metrics?pathway=ACU	—	TrendChartCard

18.6 New Screens — Virtual IPD Nurse & Patient (VN/VP)

(The doctor side is D-01…D-32 in Annex C. The platform features are in Annex B §6.)

ID	Role	Screen	Purpose / key safety
VN-01	Nurse	VIPD Round Schedule (morning/evening/night)	Planned visits and reviews; missed review → event
VN-02	Nurse	Home Setup & Device Pairing	Kit checklist, hub test, connectivity
VN-03	Nurse	VIPD Observations & NEWS2	Auto NEWS2; escalation band shown
VN-04	Nurse	eMAR (barcode)	5 rights; high-alert double check
VN-05	Nurse	IV / Infusion Record	Record-only; home-ineligible infusions blocked
VN-06	Nurse	Specimen Collection	Chain of custody, POC import
VN-07	Nurse	Deterioration Escalation (SBAR)	Direct to D-24/D-25
VN-08	Nurse	Shift Handover (I-PASS)	Mirrors D-28
VN-09	Nurse	Emergency & Transfer Documentation	Timeline, handover bundle
VN-10	Nurse	Discharge Teaching & Device Pickup	Teach-back, asset return
VP-01	Patient	My Hospital Room at Home	Plan, next visit, live vitals with freshness
VP-02	Patient	Emergency / Call Nurse (SOS)	1-tap; PSTN fallback
VP-03	Patient	Medicines Today	Reminders; dose events
VP-04	Patient	Video Rounds	Morning/evening rounds
VP-05	Patient	Device Help	Guided fixes; technician request
VP-06	Patient	My Reports	Results, prescriptions
VP-07	Family	Family Updates (consented)	Scope-limited
VP-08	Patient	Discharge & What Next	Summary, follow-up, red flags

18.7 New Screens — RPM & Elderly Care (RM)

ID	Role	Screen	Purpose / key safety
RM-D-01	Doctor	RPM Order & Monitoring Plan	Clinical question, parameters, duration, reviewer, service level
RM-D-02	Doctor	RPM Panel & Data Review	Patients by service level; freshness; data gaps != risk
RM-D-03	Doctor	RPM Patient Trends	Longitudinal charts; annotations
RM-D-04	Doctor	RPM Alert Rules (patient)	Template rules + individual thresholds; CGC bounds
RM-D-05	Doctor	RPM Reassessment & Transition	TR-10 to ACU/CCM/VIPD

ID	Role	Screen	Purpose / key safety
RM-N-01	Nurse	Device Enrolment & Validation	Registry check; test reading
RM-N-02	Nurse	Data-Gap & Technical Outreach	Device/connectivity events (technical class)
RM-N-03	Nurse	Elderly Wellbeing Check-in	Structured check; escalate
RM-N-04	Nurse	Fall / Inactivity Response	Call, welfare visit, emergency
RM-P-01	Patient	My Monitoring	What's monitored, by whom, when (service level)
RM-P-02	Patient	Take a Reading	Guided manual/device reading
RM-P-03	Patient	My Devices	Battery, sync, help
RM-P-04	Patient	Reminders & Check-ins	Med reminders (dose states), wellbeing
RM-P-05	Family	Family Notifications & Settings	Scoped alerts; quiet hours never mute emergencies
RM-P-06	Patient	Help / Emergency Button	Pendant/Hub/app SOS
RM-M-01	Manager	RPM Service-Level Compliance	Claims vs coverage; coverage notices

18.8 New Screens — Specialist / Third Opinion (SP)

ID	Role	Screen	Purpose / key safety
SP-D-01	Doctor	Opinion Request (referral / e-consult / second / third)	Mandatory package; patient sharing consent
SP-S-01	Specialist	Independent Opinion Review	Sees all prior opinions; structured response; disagreement shown side by side

19. Phase 1 — Doctor Application Screen Plan (ready to generate once approved)

Phase 1 scope: 69 doctor screens. Recommended build order: 6 sprints of design + clinician testing.

Sprint	Module	Screens
D1	Unified dashboard & patient context	UD-D-01, UD-D-02, UD-D-04, D-01, D-02, D-03
D2	Acute Care workflow	AC-D-01 … AC-D-11
D3	Virtual IPD admission & ward	D-04 … D-12, D-14, D-15
D4	Assessment, documentation, orders	D-13, D-16 … D-22, AC-D-07
D5	Chronic Care & RPM	CC-D-01 … CC-D-16, RM-D-01 … RM-D-05
D6	Alerts, escalation, transfer, discharge, opinions	D-23 … D-32, UD-D-03, SP-D-01

Phase 1 design deliverables (after approval):

- Figma frames using the existing MYDR24 components and the §17.4 extensions.
- Interactive prototypes for the five critical journeys:
  1. Acute intake → red flag → emergency
  2. Acute → VIPD admission
  3. Chronic deterioration → acute episode
  4. VIPD red alert → transfer
  5. VIPD discharge → CCM/RPM
- States for every screen.
- Clinician usability test (>= 5 doctors).

> **Status:** visual screens, mockups and code are **not generated yet**, as the document-first rule requires. Reply **"Approved"** (or list changes) and Phase 1 generation starts with Sprint D1.

20. Clinical Requirements & Acceptance Tests (cross-pathway)

Pathway-specific tests are in Annex C (D-xx acceptance criteria) and Annex D §18 (CC-SR-01…16).

ID	Requirement	Acceptance test	Priority
X-01	Emergency bypass in every pathway	Trigger red flag from AC-P-01, CC-P-08, VP-02, RM-P-06 → emergency screen + command-centre E-event <= 10 s; no routine task created first	P0
X-02	Separate worklists	Dashboard shows four pathway tabs; no patient list mixes pathways without a PathwayChip	P0
X-03	Pathway visible on every record	Every patient-context screen shows PathwayChip(s) + responsible clinician	P0
X-04	Transition record completeness	Attempt a transition missing rationale/authoriser → blocked	P0
X-05	No AI-executed transitions	AI suggestion produces a draft only; status unchanged until a human authorises	P0

ID	Requirement	Acceptance test	Priority
X-06	VIPD activation only via D-04	Acute or chronic diagnosis alone can't create a VIPD encounter (API returns 422)	P0
X-07	Capacity guard	At configured capacity, admission is disabled with a referral option	P0
X-08	Event-type separation	Simulate device failure, push failure and clinical deterioration → three different event types, different routing	P0
X-09	Delivery failure re-route	Push to on-call fails → next channel/recipient within policy; event never "acknowledged" by failure	P0
X-10	Triage protocol versioning	Every triage stores protocol ID/version; down-triage needs a reason	P0
X-11	Contact-failure ladder	No patient response → ladder steps logged; can't close the episode without an outcome	P0
X-12	Med reconciliation at transitions	ACU→VIPD, VIPD→CCM transitions need a reconciliation snapshot	P0
X-13	RPM honest claims	Patient app shows the service level; RPM-L3 badge hidden automatically when a coverage gap is detected	P0
X-14	Concurrency rules	Creating a second active VIPD encounter is rejected; CCM suspended on VIPD admission	P1
X-15	Consent per pathway	Family member with CCM-only scope can't see VIPD data	P0
X-16	Design consistency	Visual regression: new screens use only approved tokens/components (after Figma access)	P1
X-17	End-to-end journeys	The five critical journeys (§19) pass in a simulation with clinicians	P0

21. Implementation Roadmap

Phase	Scope	Gate
0 — Approval (now)	This document; Figma access; CGC formation	**Explicit approval**
0b — Clinical content	Triage protocol decision; red-flag list; launch programmes (HTN, T2DM, HF, post-discharge); VIPD criteria; service policies; RPM service levels	CGC sign-off packs
1 — Doctor app	69 screens (§19)	Doctor usability test; safety review
2 — Nursing/Caregiver app	34 screens	Nursing usability test
3 — Patient/Family app	40 screens	Elderly + regional-language usability test
4 — Allied professionals	8 screens (+ specialist 3)	Allied usability test
5 — Integration & validation	Cross-pathway transitions, shared record, permissions, alert ack, med/diagnostic continuity, transfer/discharge, IoT and non-IoT, safety case, E2E acceptance (§20)	Clinical safety case sign-off → pilot
Pilot	1 city, 1 partner hospital, ACU + VIPD (T1) + CCM (HTN/T2DM/HF) + RPM-L2	Safety KPIs met

Platform build timeline, team and budget: Annex B §19–21. These need to be re-estimated for the added Acute and RPM scope. [D]

22. Assumptions & Open Decisions

ID	Type	Item	Recommendation / note
OD-01	[D]	Remote-triage protocol (§5.1)	Adapt ATP for tele-triage + local validation + clinician review
OD-02	[D]	CGC membership & approval process	Name members; monthly cadence + urgent route
OD-03	[D]	Figma access for the design audit	View access or library export
OD-04	[D]	Launch programmes	HTN, T2DM, HF, post-discharge
OD-05	[D]	VIPD review cadence per tier (morning/evening/night)	CGC to set; at least daily physician review
OD-06	[D]	RPM service levels offered at launch	L1 + L2; L3 only with a verified 24×7 roster
OD-07	[D]	Ack deadlines/escalation timers (service policies)	Ops + CGC per event class/pathway
OD-08	[D]	SaMD classification for triage/alerting/AI	Early regulatory consultation
OD-09	[D]	Paediatric/obstetric scope	Out of v3.0; plan v4
OD-10	[D]	Terminology licensing (SNOMED CT India)	Confirm with NRC
OD-11	[A]	Partner hospital provides ED acceptance and ADT feeds	Confirm in contracts
OD-12	[A]	Ambulance coverage via 108/112 + private contracts in pilot city	Confirm SLA data availability

ID	Type	Item	Recommendation / note
OD-13	[A]	"Caregiver Flow" Figma = professional caregiver/nurse app	Confirm
OD-14	[D]	Mobile framework (brief mentions React Native; Annex B recommends Flutter)	Decide before Phase 1 build; design is framework-neutral
OD-15	[D]	Budget re-estimate for Acute + RPM scope	After approval

23. Evidence Register (master)

EV-01…EV-17 are in Annex D §20. EV-31 onwards are listed here. Annex A has the Virtual Hospitalization market evidence.

ID	Source	Used for
EV-31	Diagnostic accuracy of 5-level ED triage systems, systematic review & meta-analysis — PubMed 42398236	Triage accuracy and under-triage risk
EV-32	Zachariasse et al., *Performance of triage systems in emergency care*, BMJ Open 2019 — BMJ Open	Variable triage performance
EV-33	Manchester Triage System validity/reliability review — PubMed 24613653	Under-triage safety concern
EV-34	AIIMS Triage Protocol (ATP) — PMC7472824	Indian triage protocol
EV-35	Prospective validation of ATP — PubMed 36353399	ATP validation (ED setting)
EV-36	Diagnostic & triage accuracy of digital symptom checkers, systematic review — npj Digital Medicine 2022	Limits of symptom checkers
EV-37	Telemedicine Practice Guidelines of India 2020, implications — PMC8106416; emergency limits summary — Lexology	Tele-consult rules
EV-38	CMS Acute Hospital Care at Home (Federal Register notice, Apr 2026) — Federal Register; CMS QualityNet	VIPD review/visit benchmark
EV-01…17	Chronic care, guidelines, DPDP, CMS RPM — Annex D §20	Chronic Care
Annex A	Hospital-at-home RCTs, Medicare outcomes, NHS virtual wards	Virtual IPD

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Annex A — Evidence & Market Research: Virtual Hospitalization

Virtual Hospitalization: IPD & ICU-Level Care at Home

Research brief — October 2026

1. Definitions — three models that get mixed up

Model	What it is	Acuity	Typical length	Evidence
Hospital at Home (HaH) / Virtual IPD	Patient who *would be admitted* to a ward is admitted at home instead. Physician rounds every day, nurses/paramedics visit in person, vitals monitored remotely.	Ward-level (general medicine)	3–7 days, then 30 days of follow-up	Strong (RCTs and meta-analyses)
Virtual Ward (UK NHS term)	Wider label: covers both "step-up" (avoiding admission) and "step-down" (early discharge) care, often run by community teams with remote monitoring.	Low to ward-level	Days to weeks	Mixed
ICU at Home / Home Critical Care (common in India)	ICU equipment (ventilator, multipara monitor, suction, syringe pumps) plus 24×7 ICU-trained nurse at home, with an intensivist overseeing remotely or on visits.	Chronic critical illness: long-term ventilated, tracheostomy, post-ICU step-down, palliative	Weeks to months	Weak (case series, vendor data)

Main finding: no proven model exists today for *acute* ICU care at home, such as starting vasopressors, unstable shock, or a fresh intubation. Every major HaH program lists these as **exclusion criteria**. A program can check for ICU-type changes *(vitals, waveforms, AI early-warning scores)* and keep stable *chronic* ICU patients at home. Patients who become acutely unstable still need a defined plan to move them to a real ICU.

2. How strong is the evidence?

- **Brigham & Women's RCT (Levine et al., Annals of Internal Medicine, 2020):** cost per acute episode was 38% lower, there were far fewer lab tests and imaging studies, and 30-day readmission was 7% vs 23% — Harvard Health, Annals abstract.
- **Remote vs. in-home physician RCT (2022):** after the first home visit, daily *video* rounding (with the nurse in the room) did as well as daily physician home visits. This is what makes a "virtual physician + in-person nurse" model possible — PubMed.
- **Medicare propensity-matched study, about 16,000 patients (JAMA Netw Open, 2026):** in-hospital mortality was **0.4% vs 3.6%**, with fewer 30-day ED visits, fewer ICU escalations and complications, and no difference in readmissions — AHA summary, AJMC. *Caveat: the study is observational, so selection bias is likely.*
- **Older meta-analysis (61 RCTs):** lower mortality (OR 0.81), lower readmissions (OR 0.75), lower cost, and higher satisfaction — PubMed.
- **UK virtual wards:** the evaluations are more cautious. Some reviews find no significant change in 30-day survival or readmission — NIHR poster, Strategy Unit 2025 report.

Summary: the best results come from programs where a physician leads care and patients are selected carefully. Remote-monitoring-only "virtual wards" show smaller benefits.

3. Regulatory & market landscape

United States

- The CMS **Acute Hospital Care at Home waiver** has been extended to **30 Sept 2030** (Consolidated Appropriations Act 2026) — CMS QualityNet, AMA.
- Waiver rules:
  - Patients must be admitted from the ED or transferred from an inpatient bed.
  - At least 2 in-person RN or paramedic visits per day.
  - Daily physician rounding.
  - Payment is the same as a normal inpatient DRG (Federal Register, Apr 2026).
- **Financial headwinds:** several systems (for example Cone Health) closed their programs in 2026. They cited reimbursement, operational complexity and sustainability — Becker's. Programs have to be funded from operating budgets rather than capital, which makes them hard to start — Penn LDI.

United Kingdom

- About **20 virtual-ward beds per 100,000 people** in England (March 2025) — UK Parliament POST.

India

- No specific HaH regulation and no insurance code. Most home ICU care is paid out of pocket, and insurers generally don't cover home aftercare — The Hindu, 2025.
- Main players: **Apollo HomeCare**, **Portea**, **HealthCare atHOME (HCAH)**, plus Tele-ICU hub-and-spoke networks for rural hospitals (Guardian). Portea claims home ICU costs **30–50% of the cost of a hospital ICU** (Portea). This is a vendor claim.
- Rules that apply:
  - **Telemedicine Practice Guidelines 2020** (NMC)
  - **Medical Devices Rules 2017** (CDSCO registration for monitors and AI software, which counts as SaMD)

- **DPDP Act 2023** (health data protection)
- **ABDM** (ABHA IDs, FHIR-based record exchange)
- **NABH** home-care standards (voluntary accreditation)

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4. Patient selection (clinical guardrails)

Typical inclusions (virtual IPD): CHF exacerbation, COPD/asthma exacerbation, community-acquired pneumonia, cellulitis, UTI/pyelonephritis, dehydration, DVT/PE in low-risk patients, selected post-surgical recovery, and stable COVID or other viral illness.

Typical exclusions: needing vasopressors, blood products, telemetry for arrhythmia, new positive-pressure ventilation, continuous heparin or insulin drips, active GI bleed, unstable vitals, unsafe home, no caregiver, or living outside the response radius (West Health playbook, ASPE/PTAC, Mount Sinai manual).

Home-ICU (chronic) candidates: long-term ventilator or BiPAP dependence, tracheostomy care, neuro-rehab (stroke, TBI, ALS), post-ICU step-down, and end-of-life care with high monitoring needs.

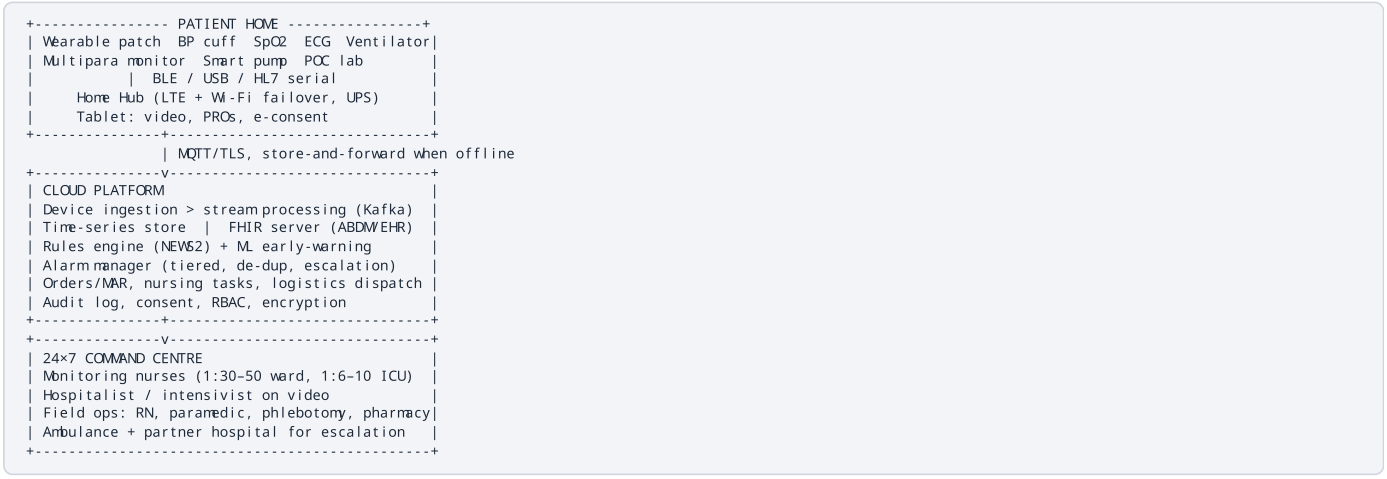
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5. "Full hospital-level" monitoring — what's achievable at home

Hospital capability	Home equivalent	Gap
Continuous vitals (HR, RR, SpO_2, Temp)	Medical-grade wearable patch or armband (e.g. Biofourmis; similar devices from VitalConnect and BioIntelliSense)	Motion artifacts; skin adhesion
Blood pressure	Automated cuff, every 15–60 min	Not continuous or invasive
Cardiac telemetry	1–3 lead ECG patch with arrhythmia detection	Not a 12-lead; review delay
ICU multipara monitor	Bedside monitor (Philips, Mindray, GE) with cloud gateway	Needs reliable power and network
Ventilator / BiPAP	Home ventilators with remote telemetry	Pressures and alarms must be streamed to the command center
Labs	Point-of-care testing (i-STAT: electrolytes, gases, Hb, troponin) plus phlebotomy	24×7 lab access not practical
Imaging	Portable X-ray, POCUS (handheld ultrasound), mobile vendors	CT/MRI need transport
IV therapy	Elastomeric or smart pumps; home infusion pharmacy	No high-risk drips
Early-warning system	NEWS2/MEWS plus AI deterioration models (personalized baselines; Biofourmis platform)	Alarm fatigue; validation required
Code / rapid response	Escalation SLA: virtual MD within 5 min, paramedic within 30 min, ambulance transfer	No crash team at the bedside

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6. Reference architecture — Virtual Hospital platform



Must-have software modules:

1. Referral & eligibility screening (clinical score plus home-safety checklist)
2. Virtual admission: orders, care plan, medication reconciliation
3. Device fleet management (provisioning, battery and connectivity health)
4. Real-time monitoring dashboard with waveforms
5. Tiered alarm/escalation engine with acknowledgement SLAs
6. Video rounding plus in-home visit scheduling and route optimization
7. eMAR and home-pharmacy/infusion coordination
8. Clinical documentation and DRG/claim generation
9. Discharge, then 30-day remote follow-up
10. Quality analytics: escalation rate, mortality, readmission, LOS, patient experience

Non-functional requirements:

- Alarm latency under 10 seconds end-to-end
- 99.95% uptime
- Offline buffering on the hub
- Dual-SIM failover
- Encryption at rest and in transit
- DPDP/HIPAA compliance
- Full audit trail
- Device integration via HL7/IEEE 11073/FHIR
- If the AI is used for clinical decisions, it is regulated as SaMD (CDSCO Class B/C; FDA 510(k))

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7. Operating model & economics

- **Staffing:** a remote MD covers about 10–15 HaH patients. Field RNs make 4–6 visits per day. Response time sets the service radius, usually 30–45 min.
- **Cost drivers:** staff (about 60–70%), logistics, devices, and the 24×7 command center fixed cost. **Scale matters.** The 2026 Medicare study looked at facilities above and below about 149 admissions, and small programs struggle financially.
- **Revenue (US):** inpatient DRG through the waiver. **India:** self-pay per day, corporate/insurer pilots, and hospital partnerships for early discharge, which frees up beds.
- **Key KPIs:**
  - Escalation back to hospital (benchmark under 10%)
  - Unplanned ICU transfer
  - Mortality
  - 30-day readmission
  - Mean time to acknowledge an alarm
  - Cost per episode vs. inpatient

---

8. Risks & open problems

1. **Patient safety on escalation.** Ambulance and transfer times are the hard limit on acuity.
2. **Caregiver burden.** Families take on the overnight observation role.
3. **Alarm fatigue.** False alerts from wearables wear down command-center staff.
4. **Equity.** Requires stable housing, power and connectivity.
5. **Reimbursement uncertainty.** Applies especially in India, where there is no insurance code.
6. **Liability.** Remote clinicians practicing across states or countries; device failure.
7. **Evidence gap.** No RCTs of *acute* ICU care at home.

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9. Recommended path (if building a program or product)

1. **Start with virtual IPD (ward-level)** for 4–6 conditions (CHF, COPD, pneumonia, cellulitis, UTI) through a hospital partner's ED/early discharge.
2. **Add chronic home-ICU** (ventilator/trach patients) on the same command center and device stack.
3. **Treat "ICU-level" as a monitoring tier, not a treatment tier.** Use continuous waveforms and AI deterioration alerts, with a guaranteed 30-minute transfer pathway.
4. Build on **FHIR/ABDM** from day one. Get CDSCO/FDA clearance for any diagnostic AI. Pursue NABH home-care accreditation.
5. Measure outcomes rigorously so you have evidence to show insurers.

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Annex B — Platform & Mobile Ecosystem Specification

Virtual Hospital at Home — Ecosystem & Mobile Application Development Specification

Item	Detail
Document	Product, Technical & Clinical-Operations Specification
Version	1.0 (Draft for review)
Date	17 October 2026
Primary market	India (designed to extend to US/UK/GCC)
Companion doc	virtual-hospitalization-research.md (evidence & market research)
Status	Ready for stakeholder review → architecture sign-off → sprint planning

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1. Executive Summary
2. Vision, Scope & Care Tiers
3. Ecosystem Map & Stakeholders
4. Clinical Operating Model & Patient Journey
5. Product Suite (Mobile, Web, Edge)
6. Detailed Mobile App Specifications
7. Medical Device Ecosystem & Integration
8. System Architecture
9. Monitoring, Early-Warning & Escalation Engine
10. Data Model & Interoperability (FHIR / ABDM)
11. API & Event Design
12. Technology Stack
13. Security, Privacy & Regulatory Compliance
14. Non-Functional Requirements
15. Low-Connectivity & Power-Failure Strategy
16. UX & Accessibility Guidelines
17. AI/ML Roadmap
18. Testing, Validation & Clinical Pilot
19. Delivery Roadmap
20. Team Structure
21. Indicative Budget
22. Business Model & KPIs
23. Risk Register
24. Appendices

1. Executive Summary

The platform lets a hospital, or a home-care provider working with one, **admit patients to a virtual bed at home** and run their care to the same clinical standard as an inpatient ward. Patients who need intensive monitoring get an ICU-grade monitoring tier. The ecosystem has:

- **6 apps:**
  - Patient app
  - Caregiver app
  - Field Clinician app (nurse/paramedic/phlebotomist)
  - Doctor app (hospitalist/intensivist)
  - Command Centre web console
  - Partner & Admin portals
- A **Home Hub** edge device that collects data from medical devices and stays connected even when the internet drops.
- A **cloud clinical platform** that handles vitals streaming, early-warning scoring, orders and medication administration (eMAR), logistics, billing, and FHIR/ABDM interoperability.
- A **24×7 virtual command centre** staffed by monitoring nurses and doctors.

Clinical guardrail (non-negotiable): the platform *monitors* at ICU level. It does **not** replace an ICU for acutely unstable patients. Every patient has a guaranteed escalation-and-transfer pathway (Section 9.4).

Target outcomes (Year 1 pilot):

- Cost per episode >= 30% lower than inpatient.
- Unplanned escalation < 10%.

- Mean time for staff to acknowledge a critical alarm < 60 seconds.
- Patient satisfaction (NPS) > 60.
- No preventable deaths attributable to delayed escalation.

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2. Vision, Scope & Care Tiers

2.1 Vision

"Any bed can be a hospital bed." Deliver safe, hospital-grade care at home by combining continuous monitoring, virtual physicians, in-person clinical teams and fast escalation.

2.2 Care Tiers

Tier	Name	Typical patients	Monitoring	Human touchpoints	Typical stay
T0	Remote Monitoring / Post-discharge	Post-discharge for CHF, COPD, diabetes, post-op	Spot checks 2–4×/day	Tele-consult as needed	14–90 days
T1	**Virtual IPD (Ward-level)**	Pneumonia, cellulitis, UTI/pyelo, CHF/COPD exacerbation, dehydration, low-risk DVT/PE	Continuous wearable plus BP every 1–4 h	Daily video MD round; >= 2 RN/paramedic visits/day	3–7 days + 30-day follow-up
T2	**High-Dependency / Step-down**	Early ICU discharge, post-major surgery, stable NIV users	Continuous multipara monitor with waveforms, ECG, capnography (optional)	MD round twice a day; 12–24 h nurse at home	3–14 days
T3	**Chronic Home ICU**	Long-term ventilator or tracheostomy, neuro-rehab (stroke/TBI/ALS), palliative high-dependency	Ventilator telemetry, multipara monitor, continuous SpO_2/$EtCO_2$	24×7 ICU-trained nurse; intensivist round every day or two	Weeks–months
⛔	**Out of scope (must stay or go to hospital ICU)**	Vasopressors, new invasive ventilation, active bleeding needing blood products, unstable arrhythmia, continuous high-risk drips	—	—	—

2.3 In Scope (v1.0)

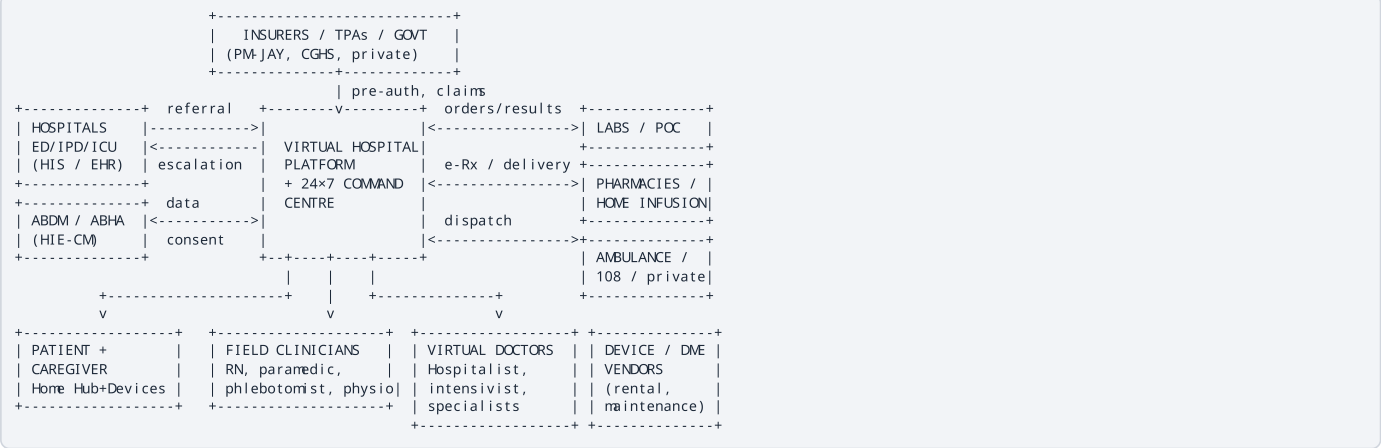
- Referral, eligibility screening, virtual admission, daily care delivery, monitoring, escalation, discharge, follow-up.
- Equipment logistics, medicine and lab logistics, billing/claims, analytics.
- Integration with hospital HIS/EHR, labs, pharmacies, ambulance services and ABDM.

2.4 Out of Scope (v1.0)

- An in-house e-pharmacy licence. The platform uses partner pharmacies.
- Diagnostic AI that makes decisions on its own. AI only gives *decision support*, with a clinician in the loop.
- Paediatric and obstetric tiers (planned for v2.0).

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3. Ecosystem Map & Stakeholders



Stakeholder	Needs	Touchpoint
Patient	Safety, comfort, clarity, a quick way to get help	Patient App, Home Hub tablet, SOS button
Caregiver / family	Updates, task guidance, emergency help	Caregiver App
Field clinician	Route, tasks, documentation, offline use	Field Clinician App
Virtual doctor	Patient list, trends, orders, video	Doctor App + Command Centre
Command-centre nurse	Real-time alarms, triage, escalation	Command Centre Console
Hospital partner	Bed capacity, referral, outcomes, revenue	Partner Portal, HIS integration
Lab / pharmacy / ambulance	Orders, dispatch, status	Partner Portal / APIs

Stakeholder	Needs	Touchpoint
Insurer / TPA	Pre-auth, documentation, outcomes	Claims API / Portal
Ops / admin	Inventory, staffing, SLAs, finance	Admin Portal
Regulator / auditor	Audit trail, consent, data protection	Audit exports

4. Clinical Operating Model & Patient Journey

4.1 End-to-End Journey

```
Referral -> Eligibility -> Consent -> Home Setup -> Virtual Admission -> Daily Care Loop -> Discharge -> 30-day Follow-up
 (ED/IPD/    (clinical +     (e-consent   (< 4 h: devices,  (MD admission     (rounds, visits,     (summary,    (T0 RPM,
  OPD/GP)     home safety)    + ABHA link)  hub, meds, kit)   note, orders)     monitoring, labs)    e-Rx, pickup) tele-consult)
                                                                    |
                                                                    v
                                                          Escalation -> Hospital transfer
```

4.2 Step Detail & SLAs

Step	Owner	SLA	System support
Referral received	Hospital ED/IPD doctor	—	Referral form (Partner Portal/HIS), auto-fetch of records
Eligibility decision	Admitting physician	<= 60 min	Eligibility checklist (Appendix A), risk score, geo-fence check
Consent & onboarding	Coordinator	<= 30 min	e-Consent (video + OTP/Aadhaar e-sign), ABHA linking
Home setup	Field team	<= 4 h from decision	Kit checklist, device pairing wizard, connectivity test
Admission note & orders	Physician	Before patient leaves hospital, or within 1 h of arrival	Order sets per condition
First in-home RN visit	Field RN	<= 2 h after arrival	Field App task
Daily MD round	Physician	1×/day (T1), 2×/day (T2)	Video round, trends summary
In-person visits	RN/paramedic	>= 2/day (T1)	Route optimisation
Labs	Phlebotomist / POC	Results <= 4 h (routine), <= 60 min (urgent POC)	Lab orders & results (FHIR)
Meds delivery	Pharmacy partner	<= 2 h (urgent), <= 6 h (routine)	e-Rx, delivery tracking
Discharge	Physician	—	Discharge summary, device pickup <= 24 h
Follow-up	Care team	Day 2, 7, 14, 30	T0 monitoring + tele-consult

4.3 Staffing Ratios (starting assumptions)

Role	T1 Virtual IPD	T2 Step-down	T3 Home ICU
Command-centre monitoring nurse	1 : 30–40 patients	1 : 8–12	1 : 10–15 (with on-site nurse)
Virtual hospitalist / intensivist	1 : 12–15	1 : 8–10	1 : 20–30
Field RN (visits/shift)	5–7 visits	—	On-site 12/24 h nurse
Paramedic rapid-response unit	1 per 25–30 km radius	Same	Same

5. Product Suite

#	Product	Platform	Primary users	MVP
1	**Patient App**	Android, iOS, plus a kiosk build for the Home Hub tablet	Patient	✅
2	**Caregiver App**	Android, iOS (shared codebase with Patient App, role-based)	Family	✅
3	**Field Clinician App**	Android first, then iOS (offline-first)	RN, paramedic, phlebotomist, physio	✅
4	**Doctor App**	iOS, Android, tablet-optimised	Hospitalist, intensivist, specialists	✅
5	**Command Centre Console**	Web (multi-monitor; video-wall mode)	Monitoring nurses, MDs, ops	✅
6	**Partner Portal**	Web	Hospitals, labs, pharmacies, ambulance, insurers	Phase 2
7	**Admin & Ops Portal**	Web	Ops, inventory, finance, HR rostering	✅ (core)
8	**Home Hub Firmware/App**	Android (locked-down kiosk) or Linux gateway	— (headless + patient UI)	✅
9	**Logistics Driver App**	Android (module inside Field App)	Equipment/med delivery staff	Phase 2

6. Detailed Mobile App Specifications

> Priority key: **P0** = MVP must-have, **P1** = Phase 2, **P2** = later.

6.1 Patient App

Goals: feel safe, know what's happening, get help in one tap, and make device use easy.

Module	Features	Priority
Onboarding	Language selection (EN, HI + 8 regional), OTP login, ABHA create/link, e-consent with video explainer, permissions walkthrough	P0
Home / "My Hospital Room"	Today's care plan timeline, next visit ETA, next medicine, live vitals card, care team photos	P0
SOS / Call Nurse	Large red button. Opens instant voice/video to the command centre; shares location and latest vitals; falls back to a PSTN call	P0
Vitals	Live and trend charts, guided spot measurements (BP, glucose, weight, temp), symptom check-ins	P0
Medications	Schedule, reminders, "taken" confirmation, photo of pill pack, side-effect report	P0
Video Rounds	Scheduled and on-demand video (WebRTC), screen share for results, interpreter add-on	P0
Care Team	Who's on duty, chat (async, triaged), visit history	P0
Device Help	Pairing status, battery, "device not working" guided fix, request a technician	P0
Reports	Lab results, discharge summary, prescriptions (PDF + FHIR), share via ABHA	P0
Education	Condition-specific videos and red-flag symptoms	P1
Feedback	Daily comfort score, PROMs/PREMs, NPS at discharge	P1
Payments	Estimate, deposit, UPI/card, insurance status	P1
Voice Assistant	Voice symptom logging and reading the care plan aloud (regional languages)	P2

Key screens: Splash → Language → Login → Consent → Home → Vitals → Meds → Video → SOS → Reports → Profile/Settings.

6.2 Caregiver App (role inside the Patient App codebase)

Feature	Detail	Priority
Multi-patient linking	One caregiver, several patients (e.g. elderly parents)	P0
Real-time status	Vitals summary, alerts relevant to the caregiver (with the patient's consent)	P0
Task guidance	Step-by-step videos: repositioning, suction (T3), feeding, mobility	P0
Shift handover	Night-observation checklist, log observations	P1
SOS on behalf of patient	Same escalation path as the patient	P0
Remote family	View-only access for relatives living elsewhere or abroad	P1

6.3 Field Clinician App (offline-first)

Goals: fast, accurate documentation at the bedside, with or without network.

Module	Features	Priority
My Day	Route-optimised visit list, ETA sharing with the patient, navigation	P0
Visit Workflow	Check-in (geo + QR on Home Hub), assessment template per tier, vitals capture (BLE auto-import), wound photos, notes (voice-to-text)	P0
eMAR	Barcode scan of patient + medication (5 rights), IV start/stop, pump programming record	P0
Lab Collection	Order list, label print/scan, chain of custody, POC device result import (e.g. blood gas analyser)	P0
Device Setup	Kit checklist, BLE pairing wizard, hub connectivity test, asset tag scan	P0
Escalation	One-tap "Deteriorating patient" → command centre MD on video; structured SBAR handoff	P0
Tele-assist	Pull in a remote MD/specialist during the visit; POCUS image streaming	P1
Offline Sync	Local encrypted DB, conflict resolution, background sync, queued uploads	P0
Safety	Lone-worker check-in timer, panic button, shift check-in/out	P0
Inventory	Consumables used, restock request	P1

6.4 Doctor App (Hospitalist / Intensivist)

Module	Features	Priority
Census	Virtual ward list sorted by acuity (NEWS2 / AI risk), filters by tier, day-of-stay, pending tasks	P0
Patient Chart	Summary ("One-liner + overnight events"), vitals trends, waveforms (T2/T3), labs, meds, notes, images	P0
Orders (CPOE)	Condition order sets, meds, labs, imaging, visit frequency, monitoring thresholds per patient, drug-interaction checks	P0
e-Prescription	Telemedicine-Guideline-compliant e-Rx with digital signature, sent to partner pharmacy	P0
Video Round	Multi-party (patient + RN at home + family + interpreter), in-call vitals overlay	P0
Alerts Inbox	Escalated alarms, acknowledgement, actions, audit	P0
Documentation	Admission, progress, and discharge templates; AI-drafted summaries (clinician signs)	P0 / P1 (AI)
Escalate / Transfer	One-tap transfer request → ambulance dispatch + receiving hospital handoff (FHIR bundle)	P0
Handover	Shift handoff list (I-PASS format)	P1
Analytics	Personal outcomes, LOS, escalation rates	P2

6.5 Command Centre Console (Web)

Area	Features
Live Board	Tile grid of all patients, colour-coded by acuity and alarm state; sort by risk; video-wall mode
Patient Focus View	Real-time vitals and waveforms, 24 h trends, NEWS2 history, device health, last visit, care plan
Alarm Queue	Prioritised queue, acknowledge/assign/escalate, snooze with reason, alarm-fatigue analytics
Communication	Click-to-video/voice (patient, caregiver, field RN), secure chat, broadcast
Dispatch	Map of field staff and paramedics, assign urgent visit, ambulance request, ETA tracking
Device Fleet	Connectivity, battery, data gaps, firmware, remote reboot of the hub
Shift Ops	Staff on duty, load balancing, handoff
Audit	Full event timeline per patient (who saw what, when)

6.6 Home Hub (Edge Gateway)

Feature	Detail
Hardware	Rugged Android tablet (kiosk mode) **or** Linux gateway (Raspberry Pi CM4/industrial) + small screen; dual-SIM 4G/5G + Wi-Fi; BLE 5.x; USB/RS-232 for monitors and ventilators
Power	UPS / battery backup >= 8 h; power-loss alert to the command centre
Functions	Device data collection, normalisation, timestamping; **local alarm evaluation** (so alarms still fire offline); store-and-forward buffer (>= 72 h); video endpoint; SOS button (hardware)
Management	MDM (zero-touch enrolment), remote config, OTA updates, tamper detection
Security	Device certificate (mTLS), encrypted storage, secure boot

6.7 Cross-App Feature Matrix

Capability	Patient	Caregiver	Field	Doctor	Command
Live vitals	✅	✅ (consented)	✅	✅	✅
Waveforms	—	—	✅	✅	✅
SOS / escalate	✅	✅	✅	✅	✅
Video	✅	✅	✅	✅	✅
Orders	—	—	View/execute	Create	View
eMAR	View	View	Administer	Order	View
Offline mode	Partial	Partial	**Full**	Read cache	—
Device pairing	Guided	—	✅	—	Remote

7. Medical Device Ecosystem & Integration

7.1 Device Kit by Tier

Device	T0	T1	T2	T3	Interface
Wearable vitals patch/armband (HR, RR, SpO_2, temp, activity, posture)	○	●	●	○	BLE → Hub or vendor cloud API
BP monitor (automatic, scheduled)	●	●	●	●	BLE (Bluetooth BLP profile)
Pulse oximeter (continuous)	○	●	●	●	BLE (PLX profile)
1–3 lead ECG patch / Holter	—	○	●	○	BLE / vendor API
Glucometer	○	●	●	●	BLE (GLS profile)
Thermometer	●	●	●	●	BLE (HTS profile)
Weighing scale (CHF)	●	●	—	—	BLE (WSS profile)
Multiparameter monitor (ECG, NIBP, SpO_2, RR, Temp, $EtCO_2$)	—	—	●	●	HL7 v2 / serial / vendor gateway
Ventilator / BiPAP with telemetry	—	—	○	●	Vendor cloud API / serial
Oxygen concentrator (flow, alarms)	—	○	●	●	Smart plug / vendor API
Infusion pump (smart)	—	○	●	●	Vendor API (record only)
Suction, nebuliser, air mattress	—	—	○	●	Asset tracking only
POC blood analyser (e.g. blood gas/electrolytes)	—	● (carried by RN)	●	●	USB/Wi-Fi export → Field App
Handheld ultrasound (POCUS)	—	● (carried by MD/paramedic)	●	○	DICOM / vendor app
Fall-detection / bed-exit sensor	—	○	○	●	Zigbee/BLE
Smart camera (with consent, privacy mode)	—	—	○	○	RTSP/WebRTC

● Standard ○ Optional

7.2 Integration Strategy

1. **Standard BLE GATT profiles first** (BP, glucose, SpO_2, thermometer, weight) using a generic driver layer.
2. **Vendor cloud-to-cloud APIs** for wearables and ventilators whose vendors don't allow raw BLE access.
3. **HL7 v2 / IEEE 11073 SDC** for bedside multipara monitors, with an edge parser on the Hub.
4. **Normalisation layer** that converts all data to FHIR `Observation` with LOINC codes, UCUM units, device ID and quality flags.
5. A **device certification lab** in-house. Each new device model passes an integration test suite covering accuracy, latency, reconnection and battery.

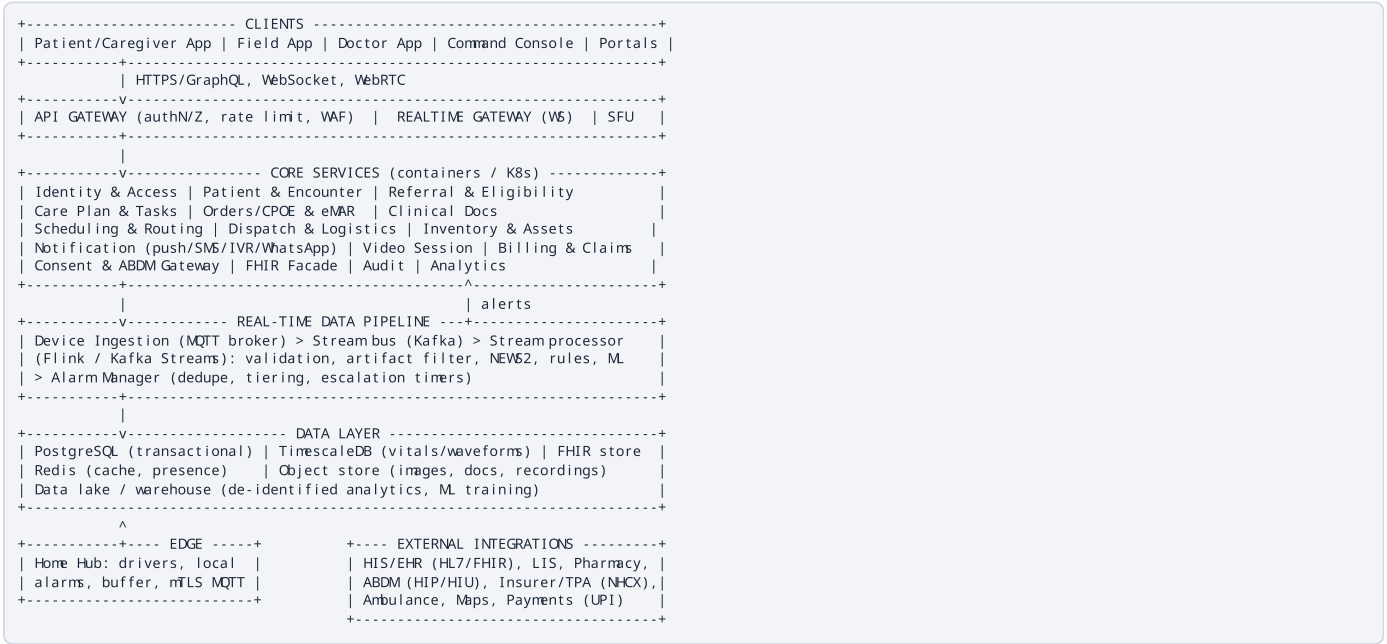
7.3 Device Selection Criteria

- Regulatory: CDSCO-registered (India), plus FDA/CE where applicable.
- Clinical accuracy (SpO_2 accuracy across skin tones; ISO 81060-2 for BP).
- Open data access (SDK/API), battery >= 3 days for wearables.
- Rental/refurbish economics, vendor service SLA in your cities.

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8. System Architecture

8.1 High-Level Architecture



8.2 Critical Data Flow: Vital Sign → Alarm → Action

1. A device sends a reading to the Home Hub (BLE), and the Hub timestamps it and attaches the device ID.
2. The Hub runs a **local threshold check**. A critical value fires a local alarm (siren plus screen) right away.
3. The Hub publishes the reading over MQTT (QoS 1, mTLS) to the cloud and buffers it if the connection is down.
4. Ingestion validates the reading and writes it to Kafka topic `vitals.raw`.
5. The stream processor filters artifacts, calculates NEWS2, applies patient-specific thresholds and runs the ML risk score, then writes to `vitals.scored`.
6. The alarm manager creates an alarm, dedupes it, assigns a tier, and starts an escalation timer.
7. The Command Console receives a WebSocket push. The nurse acknowledges within the SLA, or the alarm auto-escalates.
8. The nurse calls the patient by video, dispatches a paramedic, or escalates to the MD. Every action is audited.

Latency budget: device to Hub < 2 s, Hub to cloud < 3 s, processing < 2 s, push to console < 1 s. **Total < 10 s.**

8.3 Deployment

- Cloud: AWS **ap-south-1 (Mumbai)** primary and **ap-south-2 (Hyderabad)** for DR, which keeps data in India. Azure Central/South India works equally well.
- Kubernetes (EKS/AKS), Infrastructure-as-Code (Terraform), GitOps (ArgoCD).
- Multi-AZ for all stateful services. RPO <= 5 min, RTO <= 30 min. The real-time alarm path is active-active.

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9. Monitoring, Early-Warning & Escalation Engine

9.1 Scoring Layers

1. **Hard thresholds:** device and patient-specific, set by the MD (e.g. SpO_2 < 90% for 60 s).
2. **NEWS2:** calculated automatically from every complete set of vitals (RR, SpO_2, O_2 use, systolic BP, HR, consciousness, temp).
3. **Trend rules:** e.g. HR up >= 20% from baseline over 4 h, weight up >= 2 kg over 3 days (CHF), rising RR trend.

4. **ML deterioration index (Phase 2):** personalised baseline model. Used for decision support only and needs SaMD clearance before it drives any clinical action.

9.2 NEWS2 Response Mapping (adapted for home)

NEWS2	Risk	Platform response
0–4	Low	Routine monitoring; RN reviews at next visit
3 in any single parameter	Low–medium	Command nurse video check <= 30 min; MD informed
5–6	Medium	**Urgent**: MD video review <= 15 min; consider urgent RN/paramedic visit <= 60 min
>= 7	High	**Emergency**: MD <= 5 min; paramedic dispatch; prepare transfer; ambulance on standby

9.3 Alarm Tiers & SLAs

Tier	Example	Acknowledge SLA	Auto-escalation if not acknowledged
Critical (Red)	SpO_2 < 85% sustained, HR < 40 / > 150, apnoea, ventilator disconnect, SOS	<= 60 s	60 s → charge nurse + MD; 3 min → ops lead + auto-call patient
Urgent (Amber)	NEWS2 5–6, SBP < 90, new arrhythmia	<= 5 min	5 min → second nurse; 15 min → MD
Advisory (Yellow)	Trend deviation, missed medication, missed vitals	<= 30 min	Queue to next round
Technical (Blue)	Device offline, low battery, signal loss, hub on battery	<= 15 min	Tech support → field visit if > 60 min data gap (T2/T3: 15 min)

Alarm-fatigue controls:

- Artifact filtering (motion, poor perfusion index).
- Delay windows (sustained duration before an alarm fires).
- Dedupe.
- Per-patient threshold tuning by the MD.
- Weekly alarm analytics review.

9.4 Escalation & Transfer Pathway (guaranteed)

```
Deterioration -> Command nurse assess (video) -> MD review -> Decision:
   +- Treat at home (new orders, urgent RN/paramedic visit)
   +- Upgrade tier (T1>T2: add monitor, increase visits)
   +- TRANSFER: ambulance dispatch (target door-to-door < 30-45 min)
                + FHIR handoff bundle to receiving ED/ICU
                + family notified + bed pre-alert at partner hospital
```

Contracts with ambulance providers and partner hospitals must guarantee response times. Each patient's escalation plan, including the nearest ICU, is fixed at admission.

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10. Data Model & Interoperability

10.1 FHIR R4 Resource Mapping

Domain	FHIR resource(s)
Patient, caregiver	`Patient`, `RelatedPerson`
Staff	`Practitioner`, `PractitionerRole`
Virtual admission	`Encounter` (class = `HH` home health / `VR` virtual), `EpisodeOfCare`
Location / virtual bed	`Location` (mode = kind, type = virtual bed)
Referral	`ServiceRequest`
Care plan, tasks	`CarePlan`, `Task`, `Goal`
Vitals & device data	`Observation` (LOINC), `Device`, `DeviceMetric`
Alarms	`Flag`, `DetectedIssue`, `Communication`
Orders	`MedicationRequest`, `ServiceRequest` (lab/imaging/visits)
Med administration	`MedicationAdministration`
Labs	`DiagnosticReport`, `Specimen`
Notes	`DocumentReference`, `Composition` (discharge summary)
Consent	`Consent`
Billing	`Claim`, `Coverage`, `ChargeItem`
Audit	`AuditEvent`, `Provenance`

10.2 ABDM Integration (India)

- **ABHA** creation/linking during onboarding.

- Register as a **HIP** (share records) and **HIU** (fetch prior records with consent) through the ABDM Gateway and Consent Manager.
- Health records published as FHIR documents following the ABDM FHIR profiles (DischargeSummary, Prescription, DiagnosticReport, OPConsultation, WellnessRecord).
- Claims via **NHCX** (National Health Claims Exchange) once insurers onboard.

10.3 Key Internal Entities (non-FHIR operational)

`VirtualBed`, `Kit`, `Asset`, `HubDevice`, `Visit`, `Route`, `Shift`, `AlarmEvent`, `EscalationTimer`, `ThresholdProfile`, `Dispatch`, `Invoice`.

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11. API & Event Design

11.1 REST/GraphQL (illustrative)

Method	Endpoint	Purpose
POST	`/v1/referrals`	Create referral (from HIS/Partner Portal)
POST	`/v1/referrals/{id}/eligibility`	Submit eligibility checklist → decision
POST	`/v1/encounters`	Virtual admission
GET	`/v1/encounters/{id}/summary`	Chart summary (Doctor App)
PUT	`/v1/encounters/{id}/thresholds`	Patient-specific alarm thresholds
POST	`/v1/orders`	Create order (med/lab/visit)
POST	`/v1/med-admin`	Record administration (Field App)
GET	`/v1/patients/{id}/observations?code=&from=&to=`	Vitals trends
POST	`/v1/alarms/{id}/ack`	Acknowledge alarm
POST	`/v1/alarms/{id}/escalate`	Escalate
POST	`/v1/transfers`	Initiate hospital transfer
POST	`/v1/video/sessions`	Create video room token
POST	`/v1/devices/pair`	Register device to patient/hub
GET	`/fhir/r4/...`	Standards-based FHIR facade

11.2 MQTT Topics (Hub ↔ Cloud)

```
hub/{hubId}/obs          # normalised observations (QoS1)
hub/{hubId}/waveform     # compressed waveform chunks (T2/T3)
hub/{hubId}/alarm        # locally-detected alarms (QoS1, retained)
hub/{hubId}/status       # heartbeat, battery, power, connectivity (every 30 s)
hub/{hubId}/cmd          # cloud > hub: config, thresholds, reboot, OTA
```

11.3 Kafka Topics

`vitals.raw`, `vitals.scored`, `alarms.events`, `encounter.events`, `orders.events`, `tasks.events`, `dispatch.events`, `audit.events`, `notifications.outbound`

11.4 Sample Observation Event

```
{
  "eventId": "8b2f…",
  "patientId": "pt_123",
  "encounterId": "enc_456",
  "hubId": "hub_789",
  "deviceId": "dev_spo2_001",
  "code": { "system": "http://loinc.org", "code": "59408-5", "display": "SpO2" },
  "value": 89, "unit": "%",
  "quality": { "perfusionIndex": 1.2, "artifact": false },
  "effectiveTime": "2026-10-17T10:42:05.120Z",
  "receivedTime": "2026-10-17T10:42:07.004Z"
}
```

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12. Technology Stack (Recommended)

Layer	Choice	Rationale
Mobile apps	**Flutter** (Dart) with native modules (Kotlin/Swift) for BLE and background services	One codebase for 4 apps and the tablet kiosk; strong offline/UI performance. *Alternative: React Native + TypeScript if the team is JS-heavy.*
BLE	`flutter_reactive_ble` or a custom native plugin; Android foreground service	Reliable background device connections
Offline DB (mobile)	SQLite + SQLCipher (via Drift)	Encrypted local store, sync queue
Video	WebRTC via **LiveKit** (self-hostable) or a managed CPaaS (Agora / 100ms)	Low latency, recording, multi-party, India PoPs
Web (Console/Portals)	React + TypeScript, Next.js; uPlot/ECharts for real-time charts; WebGL waveforms	High-frequency charting performance

Layer	Choice	Rationale
Backend services	**Kotlin + Spring Boot** (clinical core) and **Go** (ingestion/real-time) *or* Node.js/NestJS	Type safety, mature healthcare libs (HAPI FHIR on JVM)
FHIR server	**HAPI FHIR** (JPA, Postgres)	Open source, ABDM-compatible profiles
Messaging	EMQX/HiveMQ (MQTT) + **Apache Kafka** (MSK/Confluent)	Device scale + durable event streaming
Stream processing	Apache Flink or Kafka Streams	Windowed rules, NEWS2, CEP
Databases	PostgreSQL, **TimescaleDB** (vitals), Redis	Relational + time-series + cache
Search	OpenSearch	Patient/notes search, logs
Object storage	S3 (SSE-KMS)	Images, documents, recordings
Analytics	Warehouse (Redshift/BigQuery/Snowflake) + dbt + Metabase/Superset	KPIs, outcomes
ML	Python, scikit-learn/XGBoost/PyTorch, MLflow, SageMaker	Model lifecycle
Auth	Keycloak / Cognito (OIDC), MFA, device binding	RBAC + ABAC
Notifications	FCM/APNs, SMS, WhatsApp Business API, IVR	Multi-channel for low-literacy users
Maps/routing	Google Maps Platform / MapmyIndia (Mappls)	Routing, geofence
Payments	UPI via Razorpay/PayU/Cashfree	Indian payment rails
DevOps	GitHub Actions, Terraform, ArgoCD, Docker, EKS	CI/CD, GitOps
Observability	OpenTelemetry, Prometheus, Grafana, Loki/ELK, Sentry, PagerDuty	End-to-end tracing of the alarm path
MDM (Hub & staff devices)	Android Enterprise MDM (e.g. Esper / Scalefusion / Intune)	Kiosk, OTA, remote wipe

13. Security, Privacy & Regulatory Compliance

13.1 Regulatory Map

Area	India	International (if expanding)
Data protection	**DPDP Act 2023** & DPDP Rules: consent notices, purpose limitation, data principal rights, breach notification, Data Protection Officer	HIPAA (US), GDPR/UK GDPR
Telemedicine	**Telemedicine Practice Guidelines 2020** (NMC): RMP identification, consent, e-Rx rules, drug lists	State licensure (US), CQC (UK)
Medical devices / software	**Medical Devices Rules 2017** (CDSCO): risk-classify alarm and AI modules as SaMD (likely Class B/C)	FDA 510(k)/De Novo, EU MDR, UKCA
Digital health	**ABDM** compliance and certification (HIP/HIU, sandbox → production)	ONC, NHS DSPT
Quality	**NABH** home-care standards; ISO 9001	Joint Commission, CMS AHCAH waiver conditions
Software lifecycle	IEC 62304 (software lifecycle), ISO 14971 (risk), IEC 62366-1 (usability), ISO 13485 (QMS) for SaMD components	Same
Info security	ISO 27001, CERT-In directions (6-hour incident reporting, log retention 180 days)	SOC 2 Type II, HITRUST
Clinical establishment	Clinical Establishments Act / state equivalents; pharmacy & drug licensing via partners	—

> Get legal/regulatory counsel to confirm classification and licensing before launch.

13.2 Security Controls

- **Identity:**
  - OIDC with MFA for all clinicians.
  - Device binding and biometric unlock on mobile.
  - Session timeout (clinician: 15 min idle).
- **Access:**
  - RBAC plus ABAC: a clinician only sees patients on their care team or roster.
  - Break-glass access with justification and audit.
- **Encryption:**
  - TLS 1.3 in transit and mTLS for Hub/devices.
  - AES-256 at rest (KMS).
  - Field-level encryption for sensitive identifiers (Aadhaar/ABHA).
- **Mobile hardening:**
  - SQLCipher, certificate pinning, root/jailbreak detection, screenshot blocking on PHI screens, remote wipe, OWASP MASVS L2.
- **Audit:**
  - Immutable audit log (`AuditEvent`) for every read and write of PHI, plus alarm actions.
  - Logs retained according to the regulatory policy.

- **Testing:**
  - SAST/DAST/SCA in CI.
  - Annual third-party penetration test plus a pre-launch VAPT (CERT-In empanelled auditor).
- **Privacy:**
  - Consent ledger, data minimisation, de-identification for analytics/ML, configurable retention, export and erasure workflows.

14. Non-Functional Requirements

Category	Requirement
Availability	99.95% core clinical services; **99.99% alarm pipeline** (active-active)
Latency	Device→console alarm <= 10 s (p99); API p95 <= 300 ms; video setup <= 5 s
Scalability	Phase 1: 500 concurrent virtual beds; design target 10,000 beds, 50k devices, ~5k obs/s, waveforms 250 Hz × 3 leads for T2/T3
Data retention	Clinical records per Indian medical-records norms (typically >= 3 years for outpatient-type records and longer for inpatient; confirm with counsel); raw waveforms 30 days hot, summarised thereafter
Offline	Field App fully functional offline >= 12 h; Hub buffers >= 72 h; local alarms always on
Power	Hub runs on battery >= 8 h; power-loss alert <= 60 s
Accessibility	WCAG 2.2 AA; font scaling 200%; voice prompts
Localisation	EN, HI + Tamil, Telugu, Kannada, Malayalam, Marathi, Bengali, Gujarati (phase-wise)
Device compatibility	Android 9+ (Field/Patient), iOS 16+; low-end Android (2 GB RAM) supported for Patient App
Observability	100% distributed tracing on the alarm path; synthetic "canary patient" running 24×7
DR	RPO <= 5 min, RTO <= 30 min; quarterly DR drills

15. Low-Connectivity & Power-Failure Strategy (India-critical)

Failure	Mitigation
Home Wi-Fi down	Hub automatically switches to 4G/5G (dual-SIM, two different carriers)
All data down	Hub local alarms (siren + screen + caregiver phone over BLE); **SMS fallback** for critical alarms; command centre flags a "data gap" after 2–5 min (by tier) and calls by PSTN
Power cut	Hub UPS; ventilator/concentrator backup batteries and inverter checks during home-safety assessment; power-loss alert
Low bandwidth	Adaptive bitrate video → audio-only → PSTN; vitals prioritised over waveforms; waveform compression
App killed by OS	Android foreground service, battery-optimisation whitelisting (MDM-enforced on Hub)
Patient phone unavailable	Hub tablet is the primary patient device; the patient's phone is optional

16. UX & Accessibility Guidelines

- **Design for elderly and low-literacy users:**
  - Big touch targets (>= 48 dp).
  - Icon + text + voice on every screen.
  - At most 3 primary actions per screen.
- **SOS is always one tap away** on every patient and caregiver screen, plus a physical button on the Hub.
- **Calm technology:** the Patient App shows reassurance ("Your nurse is watching your vitals"), not raw alarms.
- **Clinician UX:**
  - Acuity-first sorting.
  - "What changed since I last looked" diffs.
  - Keyboard shortcuts on the console.
  - Dark mode for night shifts.
  - Consistent colour semantics (Red/Amber/Yellow/Blue).
- **Design system:** one shared component library (Flutter + React tokens), clinical colour palette checked for colour-blind safety.
- **Usability engineering** per IEC 62366-1: formative tests with real nurses and elderly patients; a summative test before launch.

17. AI/ML Roadmap

Phase	Capability	Regulatory note
1	Rules + NEWS2, artifact filtering	Low risk; document under QMS
1	AI-drafted clinical notes and discharge summaries (LLM, clinician signs off)	Decision support; PHI must stay in-region; human sign-off
2	Personalised deterioration index (multivariate vitals)	SaMD; needs clinical validation & CDSCO registration

Phase	Capability	Regulatory note
2	Readmission risk at discharge	Decision support
2	Alarm-fatigue optimiser (suggests threshold tuning)	Clinician approves changes
3	Computer vision: fall detection, wound-healing tracking	Consent, privacy
3	Capacity forecasting & staff rostering	Operational
3	Multilingual voice assistant for symptom check-ins	Decision support

MLOps:

- Versioned datasets and models (MLflow).
- Bias testing across age, sex and skin tone (SpO_2).
- Drift monitoring.
- Model cards.
- A predetermined change control plan.

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18. Testing, Validation & Clinical Pilot

Layer	Approach
Unit/integration	>= 80% coverage on clinical core; contract tests for every external integration
Device integration	Device-in-the-loop lab rig; simulators for vitals and waveforms; reconnect/out-of-range/battery tests
Alarm pipeline	Chaos testing (kill broker/zone); end-to-end latency tests; synthetic patients 24×7 in production
Load	k6/Gatling: 10k beds, 5k obs/s, 1k concurrent video
Security	SAST/DAST/SCA, mobile MASVS testing, VAPT
Usability	IEC 62366 formative and summative tests
Clinical simulation	Tabletop and live drills: hypoxia, sepsis, fall, power cut, ventilator disconnect
Pilot	1 partner hospital, 2–3 conditions, 50–100 patients, 3 months; compare against matched inpatient controls on mortality, escalation, 30-day readmission, LOS, cost, and patient/caregiver satisfaction; ethics committee approval if publishing

Go-live readiness gates:

- Clinical governance committee sign-off.
- All Red-tier alarm SLAs passing in drills.
- Escalation contracts signed.
- VAPT closed.
- DPDP notices and consents live.
- ABDM certification complete.

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19. Delivery Roadmap

Phase	Duration	Scope	Exit criteria
0. Discovery & Design	Months 0–2	Clinical protocols, order sets, eligibility, device selection, UX research, architecture, regulatory classification, QMS setup	Signed PRD, clickable prototypes, architecture & risk file
1. MVP Build	Months 2–7	Patient/Caregiver, Field, Doctor apps; Command Console; Hub; ingestion + NEWS2 + alarms; orders/eMAR; video; HIS referral integration; Admin core	Internal alpha, device lab validated
2. Pilot (T1 Virtual IPD)	Months 7–10	1 city, 1 hospital, 2–3 conditions; 24×7 command centre live	Safety KPIs met; >= 50 discharged patients
3. Scale & T2/T3	Months 10–15	Multipara monitor/ventilator integration, waveforms, Partner Portal, billing/claims (NHCX), ABDM HIP/HIU production, regional languages	3 hospitals, 200 concurrent beds
4. Intelligence & Expansion	Months 15–24	ML deterioration index (validated), logistics optimisation, voice assistant, multi-city, international readiness (HIPAA/GDPR)	1,000+ concurrent beds; SaMD registration filed

MVP Sprint Plan (2-week sprints, Phase 1)

Sprint	Focus
1–2	Foundations: IAM, CI/CD, IaC, design system, FHIR server, patient/encounter services
3–4	Device ingestion (MQTT/Kafka), Hub app v0, BLE drivers (BP, SpO_2, temp, glucose), vitals store
5–6	NEWS2 + rules engine, alarm manager, Command Console live board & alarm queue
7–8	Doctor App: census, chart, orders, e-Rx; video service
9–10	Field App: visits, offline sync, eMAR barcode, lab collection; routing
11–12	Patient/Caregiver App: home, vitals, meds, SOS, video; onboarding/consent; ABHA link

Sprint	Focus
13–14	Referral/eligibility, HIS integration, discharge & follow-up, notifications (SMS/WhatsApp/IVR)
15–16	Hardening: security, load, chaos, usability tests, clinical drills, documentation; alpha release

20. Team Structure (Phase 0–2)

Function	Roles	FTE
Product	Head of Product, Product Manager (clinical), Product Manager (platform)	3
Clinical	Medical Director (intensivist/internist), Nursing Lead, Clinical Informaticist, Clinical Safety Officer	4
Design	UX Lead, UX/UI Designers, UX Researcher	3
Mobile	Flutter engineers (incl. lead), native Android/iOS (BLE) engineer	5
Web	React engineers	3
Backend	Backend engineers (clinical core, integrations), FHIR/ABDM specialist	6
Data/Real-time	Streaming engineers, data engineer	3
IoT/Edge	Embedded/Edge engineer, device integration engineer	2
ML	ML engineer / data scientist (Phase 2+)	1–2
QA	QA lead, automation engineers, device-lab tester	4
DevOps/SRE/Sec	SRE, DevSecOps	3
Regulatory/QA (QMS)	Regulatory & quality manager	1
Total		**~ 38–40**

Operations staff (command-centre nurses, doctors, field teams, logistics) scale with the number of beds and are budgeted separately.

21. Indicative Budget (Technology Build, India-based team)

> These are rough planning estimates for early budgeting only. Get vendor quotes before committing.

Item	Phase 0–2 (~ 10 months)	Notes
Product & engineering team (~38 FTE)	₹6.0 – 9.0 Cr (~ US$0.7–1.1M)	Blended Indian market rates
Cloud & infrastructure	₹40 – 80 L	Dev/stage/prod, DR, observability
Video/CPaaS, SMS/WhatsApp/IVR	₹15 – 40 L	Usage-based
Device lab & pilot device kits (~ 60 kits)	₹60 L – 1.5 Cr	Depends on T1 vs T2/T3 mix; rental models reduce capex
Security (VAPT, ISO 27001 readiness)	₹15 – 30 L	
Regulatory, legal, QMS, ABDM certification	₹20 – 50 L	
Command-centre setup (fit-out, video wall, telephony)	₹30 – 75 L	
Contingency (15%)		
Indicative total	**~ ₹9 – 14 Cr (~ US$1.1–1.7M)**	Excludes clinical operations payroll

Per-bed unit economics (to model):

- Device kit amortisation, consumables, field visits (number × cost), command-centre share, MD share, logistics, and connectivity.
- Compare against the inpatient bed-day cost and payer tariff.

22. Business Model & KPIs

22.1 Revenue Models

1. **Hospital partnership (B2B):** platform plus operations as a service. Per-virtual-bed-day fee or revenue share. The hospital gains bed capacity.
2. **Direct-to-consumer:** per-day packages by tier (out-of-pocket).
3. **Payer/insurer:** bundled episode rates; pre-authorised virtual admissions.
4. **Corporate/government:** PM-JAY / state schemes / CGHS pilots; employer health plans.
5. **SaaS licence:** platform-only for large hospital groups that run their own operations.

22.2 KPI Dashboard

Category	KPI	Target
Safety	Unplanned escalation to hospital	< 10%
Safety	Unplanned ICU transfer within 24 h of admission	< 2%
Safety	Mortality (unexpected)	<= matched inpatient

Category	KPI	Target
Safety	Critical alarm acknowledgement (median / p95)	< 30 s / < 60 s
Quality	30-day readmission	<= matched inpatient (aim -20%)
Quality	Hospital-acquired complications (falls, delirium, CLABSI)	Below inpatient baseline
Experience	Patient NPS / caregiver burden score	> 60 / monitored
Operations	Referral-to-home-setup time	<= 4 h
Operations	Device data completeness	>= 95%
Operations	Alarms per patient-day (actionable %)	Track; actionable >= 30%
Financial	Cost per episode vs inpatient	>= 30% lower
Financial	Beds freed for partner hospital (bed-days/month)	Track

23. Risk Register

#	Risk	Likelihood	Impact	Mitigation
1	Delayed escalation → patient harm	M	Critical	Strict eligibility, local alarms, SLA auto-escalation, ambulance contracts, drills
2	Connectivity/power failure	H	High	Dual-SIM, UPS, SMS/PSTN fallback, data-gap alarms
3	Alarm fatigue	H	High	Artifact filtering, tuning, analytics, staffing ratios
4	Regulatory reclassification of software as a higher-risk device	M	High	Early CDSCO consultation, QMS from day 0, modular SaMD boundary
5	Reimbursement/insurer adoption slow	H	High	Hospital B2B model first, outcome evidence, NHCX readiness
6	Device vendor lock-in / API changes	M	Medium	Abstraction layer, standards-based profiles, multi-vendor
7	Data breach	L–M	Critical	Zero-trust, encryption, VAPT, DPDP compliance, incident response plan
8	Workforce shortage (critical-care nurses)	H	High	Training academy, tele-mentoring, efficient ratios via tech
9	Caregiver burden & home unsuitability	M	Medium	Home-safety assessment, caregiver training, respite visits
10	Low digital literacy	H	Medium	Hub-first design, voice, regional languages, assisted onboarding
11	Clinician liability across jurisdictions	M	High	Licensed RMPs, malpractice cover, clear protocols, documentation
12	Program unsustainable at small scale	M	High	Hub-and-spoke command centre shared across hospitals; scale targets

24. Appendices

Appendix A — Eligibility Checklist (T1 Virtual IPD)

Clinical (all must be ✅):

- ☐ Requires inpatient-level care for an approved condition
- ☐ Haemodynamically stable: SBP >= 90, HR 50–120, RR <= 24, SpO_2 >= 92% (or >= 88% for known COPD) on <= 4 L O_2
- ☐ NEWS2 <= 4 (or <= 6 with MD approval)
- ☐ No vasopressors, blood products, new NIV/invasive ventilation, continuous high-risk infusions
- ☐ No acute delirium without a 24 h caregiver
- ☐ Diagnosis is clear and the treatment plan is defined

Home & social (all must be ✅):

- ☐ Inside the service radius (ambulance <= 30–45 min to partner hospital)
- ☐ Reliable electricity (or UPS feasible) and mobile network coverage verified
- ☐ Caregiver available (mandatory for T2/T3; strongly preferred for T1)
- ☐ Safe home environment (no fall hazards that can't be fixed, sanitation adequate)
- ☐ Patient/legal representative consents to home admission

Appendix B — Escalation Contact Tree (template)

Level	Role	Contact SLA
L1	Command-centre monitoring nurse	<= 60 s (Red)
L2	Charge nurse / second nurse	+60 s
L3	Virtual hospitalist / intensivist on duty	<= 5 min (Red)
L4	Medical director on call	<= 15 min
Dispatch	Paramedic rapid-response	Arrive <= 30 min
Transfer	Ambulance → partner ED/ICU	Door-to-door <= 45 min

Appendix C — Initial Condition Order Sets (MVP)

1. Community-acquired pneumonia (IV/oral antibiotics, O_2 titration, CXR, CBC/CRP)
2. Heart-failure exacerbation (IV/oral diuretics, daily weight, electrolytes, fluid balance)

3. COPD exacerbation (nebulisation, steroids, O_2 targets 88–92%, ABG/VBG POC)
4. Cellulitis (IV antibiotics, wound photo tracking)
5. Complicated UTI / pyelonephritis (IV antibiotics, cultures, renal function)

Appendix D — Glossary

Term	Meaning
HaH	Hospital at Home
IPD	In-Patient Department
NEWS2	National Early Warning Score 2
eMAR	Electronic Medication Administration Record
CPOE	Computerised Physician Order Entry
SaMD	Software as a Medical Device
ABDM / ABHA	Ayushman Bharat Digital Mission / Ayushman Bharat Health Account
HIP / HIU	Health Information Provider / User (ABDM)
NHCX	National Health Claims Exchange
FHIR	Fast Healthcare Interoperability Resources (HL7)
SFU	Selective Forwarding Unit (video server)
SBAR / I-PASS	Structured clinical handoff formats
RPO / RTO	Recovery Point / Time Objective

This document is a product and technical specification. Clinical protocols, thresholds and regulatory classifications in it are starting points that must be reviewed and approved by the organisation's clinical governance and regulatory counsel before use with patients.

Annex C — Doctor App: Virtual IPD Screen Specification (D-01…D-32)

MyDoctor24 — Doctor App Master Screen Specification (v2.0)

Virtual Hospitalization (IPD + ICU-level monitoring at home) · Consolidated & corrected · 17 Oct 2026

Item	Detail
Replaces	The 3 ChatGPT PDFs (Doctor-App Executive Summary / Tele-ICU Feature Map / Advanced Virtual ICU Design) and the 3 storyboard canvases (25 screens)
Parent documents	virtual-hospital-ecosystem-and-mobile-app-spec.md (platform), virtual-hospitalization-research.md (evidence)
Scope	The Doctor/Intensivist mobile app only. The Patient, Caregiver, Field-Nurse and Command-Centre apps are covered in the parent spec.
Market	India first (DPDP Act, NMC Telemedicine Guidelines, ABDM, CDSCO, NABH)

PART A — Review of the Attached Material

A1. What's strong (keep)

- The **25-screen flow is complete and logical**: admission → daily care → alerts → escalation → discharge → follow-up. It lines up one-to-one with the care journey in the platform spec.
- The visual language is consistent: MyDoctor24 blue, Red/Amber/Info severity chips, a bottom nav (Home · Patients · Rounds · Alerts · More), and status pills (Stable/Critical/Pending).
- The right clinical building blocks are present: NEWS2/SOFA scoring, SOAP notes, CPOE, MAR, I-PASS-style handover, discharge checklist, and family updates.
- Escalation is handled correctly in the mock data: the patient with SpO_2 82% and BP 78/50 is sent to *Transfer to Hospital*, not managed at home.

A2. Issues that must be fixed before Figma/dev hand-off

#	Severity	Where	Issue	Fix (applied in Part B)
1	Critical	All 3 PDFs	Citations like [45†L7-L10], [34†embed_image] and "graphic courtesy Shutterstock" are **ChatGPT placeholders, not real references**. The "ChatGPT" logo is on every page.	Removed. Real sources are in the research doc. Don't send the PDFs externally as they are.
2	Critical	Emergency/Code-Blue console (PDF 1 #21, PDF 2 #22)	A remote doctor **can't run ACLS/CPR timers at home**. There's no code team at the bedside, so presenting it this way creates medico-legal risk.	Redesigned as the **Emergency Response Console**. It calls 108/112 and the paramedic, guides the caregiver through CPR, notifies the receiving ED, and logs a timestamped record.
3	Critical	PDF 3 #17 "Add remote ventilator control", "start ventilator at home"; PDF 2 "titration guardrails for infusions"	Remote ventilator control and vasopressor titration are **out of scope for safe home care**. They would also count as high-risk medical-device functions.	Removed. Ventilator data is **view-only** (screen D-12). Vasopressors trigger a transfer.
4	Critical	Admission Eligibility mockup (all 3 canvases)	"Exclusion Criteria ⊗" is shown, yet **"Proceed to Consent" is still enabled**.	Any failed hard exclusion blocks the flow. Only an MD override with a written reason is allowed, and it's audited.
5	Critical	Missing screen	No **Goals of Care / Code Status (Full code / DNR / limits)**. This is essential for home-ICU and palliative patients and changes what "emergency" means.	Added **D-07 Goals of Care & Code Status**. It shows as a banner on every patient screen.
6	High	PDF 1 & 2 data mapping	Uses **retired or non-existent FHIR resources**: ReferralRequest, ProcedureRequest, OrderRequest, ClinicalNote, "ImagingStudy requests". It also mixes FHIR v5 with ABDM's FHIR R4.	Remapped to **FHIR R4** (ABDM profiles): ServiceRequest, Task, DocumentReference, Composition and others (Part C).
7	High	PDF 1 #20	"Call **911**".	India: **108 / 112**, or the contracted private ambulance. Make the emergency numbers configurable per region.
8	High	Alert Inbox (PDF 1 #10)	**Swipe-to-acknowledge / swipe-to-escalate** on critical alerts leads to accidental actions. Snooze has no limit.	Use explicit buttons. Red alerts can't be snoozed. Amber snooze is capped at 15 min and needs a reason.
9	High	Family Update	Clinical data goes to family **without a consent check**, which the DPDP Act 2023 requires.	Send only to recipients who have consented (the consent scope is captured in D-04).
10	High	e-Prescription	No **RMP registration number / digital signature / Telemedicine-Guideline drug-list checks**.	Added to D-17.
11	High	Missing screens	No **results viewer** (only ordering), **medication reconciliation & allergy**, **per-patient alarm thresholds**, **device/connectivity status**, **shift handover**, or **ventilator/respiratory telemetry**.	Added D-13, D-16, D-11, D-10, D-28, D-12.
12	Medium	PDF 2 partner table	Lists Surescripts (US only), "MediPro (India)" (unverified), "108 service API" (there's no public API; it's set up through state/EMRI contracts), "BD Heart", "Geneva monitors".	Replaced with a vendor-neutral integration list (Part D). Confirm vendors through procurement.

#	Severity	Where	Issue	Fix (applied in Part B)
13	🟡 Medium	PDF 3	Claims about Mercy SafeWatch "450+ beds / VidyoWorks" and Philips Capsule aren't backed by verifiable citations.	Dropped from the spec. If needed, use them only in a pitch deck with real sources.
14	🟡 Medium	Canvas 3	"Dr. Ramesh Kumar" is both the **logged-in doctor and a 72M patient**. Patient Sita Devi is shown as 68F on one screen and 66F on another.	Use separate, consistent dummy data (Part E).
15	🟡 Medium	Canvas 3 dashboard	Date reads "**Tue**, 24 Sep 2026", but 24 Sep 2026 is a **Thursday**.	Fix in the mock data.
16	🟡 Medium	Three overlapping documents	Screen counts (25 / ~30 / ~35 / "40"), screen names and numbering all disagree.	This document is the **single source of truth**: 32 screens with fixed IDs D-01…D-32.
17	🟢 Low	PDF 2 Gantt	The roadmap chart can't be read at that resolution.	Roadmap is in the parent spec §19.

> **Docx attachment:** the file `Virtual-Hospitalization-RPM-Master-Specification.docx` arrived as raw compressed binary, so I couldn't read it. Upload it into the workspace (or paste its text) and I'll merge anything it has that isn't here.

PART B — Consolidated Screen Specification (32 screens)

B1. Screen Index

ID	Screen	Module	Canvas ref	Status
D-01	Login & Role Selection	Access	C1-01	Kept
D-02	Doctor Dashboard	Home	C1-02, C2-01, C3-01	Kept
D-03	Patient Queue / Census & Filters	Census	C1-03, C2-02, C3-02	Kept
D-04	Admission Eligibility	Admission	C1-04	**Fixed** (hard-stop)
D-05	Consent Review	Admission	C1-05	**Fixed** (consent scopes)
D-06	Baseline Assessment	Admission	C1-06	Kept + NEWS2
D-07	Goals of Care & Code Status	Admission	—	**New**
D-08	Care-Team Assignment	Admission	C1-07	Kept
D-09	Initial Care Plan & Order Set	Admission	C1-08	Kept + order sets
D-10	Device & Connectivity Status	Monitoring	—	**New**
D-11	Alarm Thresholds & Monitoring Profile	Monitoring	—	**New**
D-12	ICU Monitoring Dashboard (waveforms + ventilator view)	Monitoring	C1-11	**Fixed** (view-only)
D-13	Results Viewer (Labs & Imaging)	Clinical	—	**New**
D-14	Unified Patient Chart	Clinical	C1-09, C2-08	Kept
D-15	Real-Time Vitals & Trends	Monitoring	C1-10	Kept
D-16	Medication Reconciliation & Allergies	Clinical	—	**New**
D-17	e-Prescription & MAR	Clinical	C1-16	**Fixed** (RMP/e-sign)
D-18	Order Entry (CPOE)	Clinical	C1-14	Kept + safety checks
D-19	Lab & Imaging Orders	Clinical	C1-15	Kept
D-20	Overnight / Since-Last-Seen Review	Rounds	C2-09	Kept
D-21	Tele-Round Console	Rounds	C1-12	Kept
D-22	SOAP Progress Note	Rounds	C1-13	Kept + AI draft
D-23	Alert Inbox	Alerts	C1-17	**Fixed** (no swipe)
D-24	Alert Detail (Amber / Red)	Alerts	C1-18	Kept
D-25	Emergency Response Console	Alerts	C1-19	**Redesigned**
D-26	Escalation Decision	Alerts	C1-20	Kept
D-27	Hospital Transfer Handover	Alerts	C1-21	Kept
D-28	Shift Handover (I-PASS)	Team	—	**New**
D-29	Care-Team Chat	Team	C2-20	Kept
D-30	Family Update	Team	C1-22	**Fixed** (consent)
D-31	Progress Analytics & History	Clinical	C1-23, C2-22	Merged
D-32	Discharge (Checklist → Summary → Continuity)	Discharge	C1-23/24/25	Merged as 3-step flow

C1/C2/C3 = canvases 1/2/3 in the order they were attached.

B2. Global UI Rules (apply to every screen)

1. **Patient header** on all patient-context screens:

- Photo, name, age/sex, UHID/ABHA, tier badge (T1 IPD / T2 Step-down / T3 Home-ICU), day of stay.
- **Code-status chip** (Full / DNR / Limited) and **allergy chip** (red if any allergy is recorded).

2. **Severity colours** (tokens):
   - `critical` #D92D20
   - `urgent` #F79009
   - `advisory` #EAAA08
   - `technical` #2E90FA
   - `stable` #12B76A
   - Always pair colour with an icon and text so it's colour-blind safe.
3. **Persistent emergency access:** a red **"Emergency"** action in the patient header opens D-25 from anywhere.
4. **Data freshness:** every vital shows a "last updated" time. If data is older than 5 min (T2/T3) or 15 min (T1), it turns grey and shows a "stale data" warning.
5. **States** every screen must design for: Loading (skeletons), Empty, Error (with retry), Offline (read-only cache plus a banner), and Permission denied.
6. **Audit:** every view, order, acknowledgement and message writes an `AuditEvent`. Patient data is only visible to the care team or the on-call roster. Break-glass access needs a reason.
7. **Signing:** orders, prescriptions, notes and discharge are signed with biometric or PIN plus the RMP registration number.
8. **Accessibility:** touch targets >= 48 dp, dynamic text up to 200%, screen-reader labels on charts ("SpO_2 trend, falling from 96 to 89 over 2 hours").

B3. Screen Details

> Format: **Purpose · Key components · Data (FHIR R4) · Rules & states · Acceptance criteria (AC)**

D-01 Login & Role Selection

- **Purpose:** secure entry and role context.
- **Components:**
  - Role cards: Doctor/Intensivist, Consultant, On-call Doctor.
  - Mobile + OTP or SSO, MFA, device binding, biometric unlock for returning users.
- **Data:** `Practitioner`, `PractitionerRole`.
- **Rules:** lock after 5 failed attempts; 15-min idle timeout; shift check-in prompt for the on-call role.
- **AC:** without MFA, no patient data loads. The selected role filters the census.

D-02 Doctor Dashboard

- **Purpose:** what needs my attention right now.
- **Components:**
  - Greeting and shift.
  - KPI tiles: Total / ICU-tier / IPD-tier / Discharge-ready.
  - **Alert tile** (critical/urgent counts, which open D-23).
  - Pending orders to sign; today's rounds (scheduled/done); unread chat.
  - Bottom nav: Home · Patients · Rounds · Alerts · More.
- **Data:** aggregates of `Encounter`, `Flag`, `Task`, `Appointment`.
- **Rules:** critical alerts are pinned at the top with a pulsing badge. Tiles show counts only, with no PHI on the lock-screen widget.
- **AC:** counts match the census within 5 s of a change.

D-03 Patient Queue / Census & Filters

- **Purpose:** the virtual ward list, sorted by acuity.
- **Components:**
  - Tabs: All / ICU-tier / IPD-tier / Alerts / Pending admission / Discharge-ready.
  - Search (name, UHID, ABHA).
  - Patient rows: name, age/sex, tier, day, diagnosis, **NEWS2 score + trend arrow**, status pill, latest HR/BP/SpO_2, data-freshness dot.
  - "+ Admit patient".
- **Default sort:** active Red → Amber → NEWS2 high→low → day of stay.
- **Data:** `Encounter` (status in-progress, class `HH` / `VR`), `Observation` (latest), `RiskAssessment`.
- **AC:** a patient with an active Red alert is always in the first visible rows.

D-04 Admission Eligibility *(fixed)*

- **Purpose:** check the referral is safe before a home admission.
- **Components:**
  - 4-step stepper: Eligibility → Consent → Baseline → Confirm.
  - Three checklist groups:
    - **Clinical inclusion:** approved diagnosis, stable vitals, NEWS2 <= 4.
    - **Hard exclusions:** vasopressors, blood products, new NIV/invasive ventilation, continuous high-risk infusions, active bleed, unstable arrhythmia.
    - **Home & social:** caregiver available, power and network checked, inside the ambulance radius (<= 30–45 min), safe home.

  - Auto-filled from the referral and HIS. Map pin with the drive time to the partner ED.
- **Rules:**
  - Any **hard exclusion = blocked**. The CTA changes to "Refer to IPD/ICU".
  - A soft criterion can be overridden only by the MD, with a written reason that is audited.
- **Data:** `ServiceRequest` (referral), `QuestionnaireResponse` (checklist), `Observation`.
- **AC:** with any exclusion ticked, "Proceed" can't be tapped (automated UI test).

D-05 Consent Review *(fixed)*

- **Purpose:** record informed consent with granular scopes.
- **Components:**
  - Tabs: Patient consent / Family (LAR) consent.
  - Documents:
    - Virtual hospitalization consent
    - Tele-consultation consent (NMC)
    - Data sharing & privacy (DPDP notice in the user's language)
    - Emergency & transfer consent
    - Video recording (optional)
    - **Family data-sharing scope** (who sees what)
  - Each document has View plus e-sign (Aadhaar e-Sign / OTP / drawn signature) and a timestamp.
- **Data:** `Consent` (one per scope), `DocumentReference` (signed PDF).
- **AC:** monitoring can't start until the mandatory scopes are signed. The family update screen (D-30) only lists recipients covered by a consent scope.

D-06 Baseline Assessment

- **Components:**
  - Vitals: BP, HR, SpO_2 (+ O_2 flow), Temp, RR, weight, glucose, plus **auto-calculated NEWS2**.
  - General condition, consciousness (ACVPU/GCS), pain 0–10, frailty (CFS), fall risk, pressure-injury risk (Braden) for T3.
  - Problem list, comorbidities.
- **Data:** `Observation` (LOINC), `Condition`, `RiskAssessment`.
- **Rules:** values outside physiological limits show inline errors; the baseline becomes the reference for trend alerts.
- **AC:** NEWS2 matches the reference calculator for 20 test vectors.

D-07 Goals of Care & Code Status *(new)*

- **Purpose:** record what the patient wants in an emergency, before it happens.
- **Components:**
  - Code status: Full / DNR / DNI / Limited, plus escalation ceiling (Transfer to ICU? Yes/No).
  - Surrogate decision-maker, advance directive upload, palliative flag, discussion note and date.
- **Data:** `Consent` (category = advance directive), `Goal`, `Flag`.
- **Rules:** shown as a chip on every patient header. **D-25 changes its actions** based on code status (for example, DNR means comfort-care pathway instead of CPR guidance). Re-confirm on tier change.
- **AC:** the code-status chip shows on all patient-context screens.

D-08 Care-Team Assignment

- **Components:**
  - Roles: Primary doctor, Nurse (command centre), Field nurse, Physiotherapist, Dietitian, Caregiver, Specialist on call.
  - Each role has Assign / Change / Optional, with staff search showing availability and load.
- **Data:** `CareTeam`, `PractitionerRole`.
- **Rules:** primary doctor and command-centre nurse are mandatory. Assigning someone notifies them.
- **AC:** alerts route to the assigned nurse within 1 s of creation.

D-09 Initial Care Plan & Order Set

- **Components:**
  - Duration estimate, goals (e.g. infection control, SpO_2 >= 94%, mobilisation, nutrition).
  - **Condition order set picker:** Pneumonia, CHF, COPD, Cellulitis, Complicated UTI, Post-op, T3 Home-ICU.
  - Interventions (IV antibiotics, nebulisation, physio, diet), visit frequency, monitoring frequency.
  - "Activate care plan".
- **Data:** `CarePlan`, `Goal`, `PlanDefinition` (order set) → `MedicationRequest` / `ServiceRequest`.
- **AC:** activating creates the scheduled tasks for the field and nurse apps.

D-10 Device & Connectivity Status *(new)*

- **Components:**
  - Home Hub (online/offline, 4G/Wi-Fi, battery or mains power).
  - Each device: battery, last reading time, signal quality.
  - Data-gap timeline. "Request technician" and "Call patient" actions.

- **Data:** `Device`, `DeviceMetric`, hub heartbeat.
- **AC:** a hub that's offline longer than the tier threshold creates a Blue technical alert.

D-11 Alarm Thresholds & Monitoring Profile *(new)*

- **Components:**
  - Per vital: low/high limits, sustained-duration delay, enabled/disabled.
  - Defaults by tier and condition (e.g. COPD SpO_2 target 88–92%).
  - "Why changed" reason, expiry/review time.
- **Rules:**
  - Limits outside the safe bounds need a second sign-off.
  - Every change is audited and shown on the chart.
  - The Critical tier can't be disabled.
- **AC:** a threshold change reaches the alarm engine within 10 s.

D-12 ICU Monitoring Dashboard *(fixed — view-only)*

- **Components:**
  - Live waveforms: ECG lead II, SpO_2 pleth, RR, optional $EtCO_2$.
  - Numerics: HR, SpO_2, RR, NIBP, Temp, plus urine/fluid balance if recorded.
  - **Ventilator/BiPAP telemetry panel** (mode, FiO_2, PEEP/EPAP, Vt, RR, leak, alarms), **read-only**.
  - Freeze/scroll 30 s, full-screen landscape.
- **Data:** `Observation` (sampled-data for waveforms), `DeviceMetric`.
- **Rules: no remote device control.** Changes go out as an order to the on-site nurse or RT.
- **AC:** waveform lag <= 3 s when on 4G.

D-13 Results Viewer *(new)*

- **Components:**
  - Labs table with reference ranges and abnormal flags, plus trend sparklines.
  - Imaging reports (PDF/DICOM viewer link).
  - POC results (blood gas, electrolytes) tagged by source.
  - "Acknowledge result".
- **Data:** `DiagnosticReport`, `Observation`, `ImagingStudy`.
- **Rules:** critical values create an Amber/Red alert and must be acknowledged.
- **AC:** an unacknowledged critical result escalates after 30 min.

D-14 Unified Patient Chart

- **Components:**
  - Tabs: Overview · Vitals · Orders · Notes · Results · Meds.
  - Overview cards: BP, HR, SpO_2, Temp, RR, Pain, NEWS2.
  - Diagnosis, allergies, active meds count, clinical notes count, care team, code status.
- **Data:** `Patient`, `Encounter`, `Condition`, `AllergyIntolerance`, `Observation`.
- **AC:** tapping any vital opens D-15 filtered to that vital.

D-15 Real-Time Vitals & Trends

- **Components:**
  - Range chips: Live / 1h / 6h / 24h / 72h.
  - Per-vital charts with threshold bands.
  - Event markers (meds given, alerts, visits); NEWS2 timeline.
  - Export PDF/CSV.
- **AC:** pinch-zoom works; tapping a point shows value, time and device.

D-16 Medication Reconciliation & Allergies *(new)*

- **Components:**
  - Home meds vs hospital meds vs current orders: continue/stop/modify per line.
  - Allergy list with reaction and severity.
  - Interaction warnings.
- **Data:** `MedicationStatement`, `MedicationRequest`, `AllergyIntolerance`.
- **AC:** the first virtual-ward prescription can't be signed until reconciliation is done.

D-17 e-Prescription & MAR *(fixed)*

- **Components:**
  - Tabs: Active / All / History.
  - Each line: drug, dose, route, frequency (1-0-1 style), day x/y, status (Given / Due / Missed / PRN / Held), administered by and time.
  - "Send e-Prescription".

- **Compliance:** e-Rx carries the RMP name, qualification, **registration number**, digital signature, patient ID and date, and follows the **NMC Telemedicine Practice Guidelines** drug lists. Sent to the partner pharmacy; status is tracked.
- **Data:** `MedicationRequest`, `MedicationAdministration`, `MedicationDispense`.
- **AC:** a missed dose older than 60 min creates an Advisory alert. High-alert drugs need double confirmation.

D-18 Order Entry (CPOE)

- **Components:**
  - Tabs: Medications · Labs · Imaging · Nursing/Visits · Consults.
  - Search with formulary auto-complete, dose/route/frequency/duration pickers, order cart, "Review & send".
- **Safety checks:** allergy, duplicate therapy, dose range by weight and renal function, drug–drug interactions. **Home-ineligible drugs** (vasopressors, heparin/insulin infusions, blood products) are blocked with "Requires hospital admission".
- **Data:** `MedicationRequest`, `ServiceRequest`, `Task`.
- **AC:** a blocked drug can't be ordered; the attempt is logged.

D-19 Lab & Imaging Orders

- **Components:**
  - Checklist panels (CBC, CRP, LFT, KFT, electrolytes, blood culture, ABG/VBG, procalcitonin, ECG 12-lead, portable chest X-ray, POCUS).
  - Priority (Routine/Urgent/STAT), specimen instructions, clinical indication.
  - Collection mode: field phlebotomy / POC / transfer to centre.
- **Data:** `ServiceRequest`, `Specimen`.
- **AC:** an urgent order creates a field task with an SLA (<= 60 min).

D-20 Overnight / Since-Last-Seen Review

- **Components:**
  - Date range, combined trend (HR/SpO_2/BP).
  - **Clinical events list** (e.g. 02:15 SpO_2 ↓ 88%, 02:17 nebulisation given, 04:30 SpO_2 ↑ 95%, 06:00 stable).
  - Nurse notes. "Mark as reviewed".
- **AC:** "Mark as reviewed" clears the "unseen events" badge for this doctor only.

D-21 Tele-Round Console

- **Components:**
  - Full-screen video plus picture-in-picture self view.
  - Multi-party (patient, field nurse, family, interpreter).
  - **Live vitals overlay toggle**, screen-share of results, chat, snapshot (with consent), timer.
  - Mute/Video/Share/Chat/End.
  - Fallback: audio-only, then a PSTN call.
- **Data:** `Encounter` (virtual), `Communication`, `Media` (snapshots).
- **AC:** ending the call opens D-22 with the encounter linked.

D-22 SOAP Progress Note

- **Components:**
  - S / O / A / P tabs. **O auto-fills** with the latest vitals, NEWS2 and new results.
  - Voice dictation; *optional AI draft* (clinician must edit/sign).
  - "Convert plan item to order" (opens D-18). Save / Sign.
- **Data:** `Composition` (progress note) / `DocumentReference`.
- **AC:** an unsigned note older than 12 h shows on the dashboard as pending.

D-23 Alert Inbox *(fixed)*

- **Components:**
  - Filter chips: All / Red / Amber / Advisory / Technical.
  - Row: patient, alert text (e.g. "SpO_2 84% (Critical)"), source device, age.
  - **Explicit "Acknowledge" and "Open" buttons, no swipe actions.**
  - Ack SLA countdown on Red.
- **Rules:**
  - Red: can't be snoozed; auto-escalates after 60 s if not acknowledged.
  - Amber: snooze <= 15 min, reason required.
- **Data:** `Flag`, `DetectedIssue`, `Communication`.
- **AC:** the ack time (median and p95) is recorded per alert.

D-24 Alert Detail (Amber / Red)

- **Components:**
  - Severity banner with ack countdown; triggering value vs threshold.
  - 2-h mini-trend; other vitals (HR, RR, BP); trigger source (device); current status note from the nurse.
  - Code-status chip.

  - Actions: **Start video call** · **Dispatch field team** · **Escalate** · Resolve (with reason).
- **AC:** every action is time-stamped on the patient timeline.

D-25 Emergency Response Console *(redesigned)*

- **Purpose:** coordinate the response to an acute emergency at home. The clinical work at the bedside is done by the people physically there.
- **Components:**
  - **LIVE EMERGENCY** banner with elapsed timer; latest vitals.
  - Response tracker:
    - Nurse at home: En route / On site
    - Doctor: On call
    - **Ambulance (108/112/private): Requested → ETA**
    - Receiving ED: Pre-alerted
  - **Caregiver guidance mode:** pushes step-by-step CPR / recovery-position / suction instructions to the caregiver app and Home Hub (only if code status = Full).
  - One-tap: **Call ambulance**, **Call caregiver**, **Video to home**, **Notify receiving ED**.
  - Event log: auto plus manual, timestamped.
- **Rules:**
  - Code status changes what's offered: DNR shows the comfort-care pathway, the palliative contact, and no CPR prompt.
  - Closing the console requires an outcome (transferred / stabilised at home / death) and generates D-27 or an incident report.
- **Data:** `Encounter` (emergency), `Task` (dispatch), `Communication`, `AuditEvent`.
- **AC:** "Call ambulance" places the call and sends location plus summary in <= 2 taps.

D-26 Escalation Decision

- **Components:**
  - Current status summary.
  - Recommendation (rules-based).
  - Options:
    - Continue home care with new orders
    - Upgrade tier (e.g. T1 → T2) / urgent field visit
    - **Transfer to hospital**
  - Reason (structured: haemodynamic instability, respiratory failure, neuro change, patient/family request, device/home failure) plus free text.
- **Data:** `Task`, `ClinicalImpression`.
- **AC:** choosing Transfer opens D-27 already pre-filled.

D-27 Hospital Transfer Handover

- **Components:**
  - Handover summary: diagnosis, current vitals, meds, recent events, allergies, code status.
  - Receiving hospital (partner list with ED/ICU bed status if integrated).
  - Attachments: clinical summary, latest vitals, lab reports, ECG.
  - "Send handover package".
- **Data:** FHIR `Bundle` (document) with `Composition`, sent via HIS integration / ABDM HIP. A `ServiceRequest` is the transfer request.
- **AC:** the package is delivered or acknowledged by the receiving facility, or it falls back to a secure PDF link with a call.

D-28 Shift Handover (I-PASS) *(new)*

- **Components:**
  - Per patient: **I**llness severity, **P**atient summary, **A**ction list, **S**ituation awareness/contingency, **S**ynthesis by receiver.
  - Receiver must accept the handover.
- **Data:** `Composition` (handoff), `Task`.
- **AC:** the outgoing doctor can't end their shift until every assigned patient has been handed over.

D-29 Care-Team Chat

- **Components:**
  - Tabs: Team / Family / Field team.
  - Patient-context threads, @mentions, images (stored in the chart, not the phone gallery), voice notes, read receipts, priority flag.
- **Rules:** chat is **not used for emergencies**; the composer has a banner linking to D-25. Messages are retained per policy.
- **Data:** `Communication`.

D-30 Family Update *(fixed)*

- **Components:**
  - Update text (templates: Stable / Improving / Plan change / Visit time / Discharge plan, all editable).
  - Recipients (**only consented contacts** from D-05).
  - "Include latest reports" toggle (respects consent scope).
  - Language selection; delivered by app push plus WhatsApp/SMS link.

- **Data:** `Communication`, checked against `Consent`.
- **AC:** an unconsented recipient can't be selected.

D-31 Progress Analytics & History

- **Components:**
  - Tabs: Vitals / Treatment / Notes / Events.
  - SpO_2 and other trends over the stay; average SpO_2; episodes count; improvement %.
  - Key indicators (fever resolved, O_2 requirement ↓, mobility, labs trending normal).
  - Daily NEWS2/SOFA.
  - Timeline of admissions, alerts, escalations, notes.
- **Data:** aggregates of `Observation`, `RiskAssessment`, `AuditEvent`.

D-32 Discharge — 3-step flow

1. **Checklist:**
   - Clinically stable >= 24–48 h
   - Meds reconciled
   - Caregiver trained
   - Follow-up booked
   - Devices pickup scheduled
   - Home-care instructions given
   - Discharge summary prepared

"Mark for discharge" is enabled only when every item is ticked.

1. **Summary:**
   - Auto-generated hospital course, discharge meds, follow-up, red-flag symptoms in the patient's language.
   - Doctor edits and signs. Download PDF / share to family / **push to ABHA (ABDM DischargeSummary)**.
2. **Post-discharge continuity:**
   - Next follow-up (video), remote monitoring toggle (e.g. 14 days, T0), home-care plan, reschedule.
3. **Data:** `Composition` (Discharge Summary), `Appointment`, `CarePlan` (T0), `Encounter` status → finished.
4. **AC:** discharge closes the virtual bed, creates the device-pickup task, and starts the 30-day follow-up schedule.

PART C — FHIR R4 Mapping Corrections

Used in the PDFs	Problem	Use instead (R4 / ABDM)
`ReferralRequest`	Removed after STU3	`ServiceRequest` (intent = order, category = referral/transfer)
`ProcedureRequest`	Removed after STU3	`ServiceRequest` / `Task`
`OrderRequest`	Doesn't exist	`MedicationRequest` / `ServiceRequest`
"ClinicalNote"	Not a resource	`Composition` (structured) or `DocumentReference`
"ImagingStudy requests"	`ImagingStudy` holds results, not orders	`ServiceRequest` (order) → `ImagingStudy` + `DiagnosticReport` (result)
`AdverseEvent` for Code Blue log	Wrong semantics	`Encounter` + `Task` + `AuditEvent`; `AdverseEvent` only for harm incidents
"Alert resource"	Doesn't exist	`Flag` / `DetectedIssue` + `Communication`
`Media` for waveforms	Not the right fit	`Observation` with `valueSampledData`
FHIR v5 references	ABDM is built on R4	Pin **FHIR R4 (4.0.1)** with ABDM profiles

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PART D — Integration Partners (vendor-neutral, India-first)

Category	Options to evaluate	Integration
Wearables / RPM	Clinical-grade patches/armbands with SDK or cloud API	BLE → Home Hub, or cloud-to-cloud → `Observation`
Bedside monitors	Philips, GE, Mindray, Nihon Kohden, Dräger (whichever your DME partner supplies)	HL7 v2 ORU / vendor gateway → `Observation`
Home ventilators/BiPAP	ResMed, Philips, Löwenstein and others (check telemetry access)	Vendor cloud API, **read-only**
Video	LiveKit (self-host), Agora, 100ms, Zoom Video SDK	WebRTC SDK in app; recordings only with consent
HIS/EHR/LIS	Partner hospital HIS	HL7 v2 / FHIR APIs
National stack	**ABDM** (ABHA, HIP/HIU, Consent Manager), **NHCX** claims	ABDM gateway (R4 profiles)
e-Pharmacy	Licensed partner pharmacy chains	e-Rx API / signed PDF
Ambulance	108/112 (via state service agreements), private networks	Phone plus dispatch API where contracted
Cloud	AWS / Azure / GCP India regions	Data stays in India (DPDP)
Auth	Keycloak / Cognito / Auth0 (OIDC, MFA)	RBAC + ABAC, audit

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PART E — Mock Data Fix-list (for Figma)

Field	Current	Use
Logged-in doctor	Dr. Ramesh Kumar (also used as a patient on canvas 3)	Dr. Ramesh Kumar, Intensivist, Reg. No. `MCI-XXXXX` (placeholder)
Patient 1	Sita Devi 68F / 66F (inconsistent)	**Sita Devi, 68F**, T1 IPD Day 3, Pneumonia, Stable
Patient 2	Mohan Yadav 72M ICU Sepsis	**Mohan Yadav, 72M**, T2 Day 1, Sepsis → Critical → **Transfer** (this is the escalation storyline)
Patient 3	Ram Prasad 60M	**Ram Prasad, 60M**, T1 IPD Day 1, Post-surgery, Stable
Patient 4	Sunita Kumari 72F ICU Day 4	**Sunita Kumari, 72F**, **T3 Home-ICU** Day 4, chronic respiratory failure on home ventilator, code status: Limited (no intubation)
Dashboard date	"Tue, 24 Sep 2026"	"Thu, 24 Sep 2026"
Emergency number	911	108 / 112
Code-Blue screen title	"CODE BLUE"	"LIVE EMERGENCY"

PART F — Figma Hand-off Checklist

- ☐ Rebuild the screens using IDs **D-01…D-32**, one Figma page per module.
- ☐ Components with variants:
  - Patient header (tier × code status × allergy)
  - Vital tile (normal/urgent/critical/stale)
  - Alert row (4 severities)
  - Status pill
  - Waveform card
  - Order line
  - MAR line (Given/Due/Missed/PRN/Held)
- ☐ Design tokens for colour, type, spacing and radius, exported for Flutter (Doctor app) and React (Command Centre).
- ☐ Prototype links for the three critical journeys:
  1. Admission D-03→04→05→06→07→08→09
  2. Red alert D-23→24→26→27, with the D-25 branch
  3. Discharge D-32 steps 1→2→3
- ☐ Design every state (Loading/Empty/Error/Offline/Stale) for D-03, D-12, D-15, D-23, D-25.
- ☐ Usability test with >= 5 doctors and >= 5 nurses (IEC 62366 formative). The emergency flows must be completable in under 30 s.
- ☐ Clinical-safety review sign-off on D-04, D-07, D-11, D-18, D-25 before development.

Clinical rules, thresholds and drug restrictions here are design starting points. MyDoctor24's clinical governance committee and regulatory counsel must approve them.

Annex D — Chronic Care Management & Longitudinal RPM

MYDR24 Master Specification — Domain 2: Chronic Care Management (CCM) & Longitudinal Remote Patient Monitoring (RPM)

Item	Detail
Document	Stage 1 consolidated domain specification (**for approval; no screens to be generated yet**)
Version	2.1-draft (adds Domain 2 to master v2.0)
Date	17 October 2026
Parent / sibling documents	virtual-hospital-ecosystem-and-mobile-app-spec.md (Domain 1: Virtual Hospitalization platform), mydoctor24-doctor-app-master-screen-spec.md (Doctor app D-01…D-32), virtual-hospitalization-research.md (evidence)
Figma sources	*Caregiver Flow Final (MyDr.24)* and *MVP-1 Instant Service Flow (MyDr.24)*. **Couldn't open: the links need a Figma login (HTTP 403).** Component reuse below is based on the shared canvases and is marked **"verify"**.
Status of clinical content	**DRAFT. Every disease programme, threshold, interval and alert rule needs sign-off by the MYDR24 Clinical Governance Committee (CGC) before build.**

Delivery-order tracker (from the brief §17)

Step	Description	Status
1	Research chronic-care models, guidelines, regulations	✅ Done (Section 1, Evidence Register in Section 20)
2	Add Chronic Care domain to master specification	✅ This document + change log (Section 19)
3	Update screen inventory and role-permission matrix	✅ Sections 15 and 16
4	Present consolidated document for approval	Pending **Awaiting your approval** (decisions in Section 21)
5–8	Generate Phase 1–4 screens (Doctor → Nurse/Caregiver → Patient/Family → Allied)	Locked Blocked until step 4 is approved
9	Validate the full cross-role CCM ↔ Virtual IPD journey	Locked Blocked

1. Research Summary (what the evidence supports and what it doesn't)

Topic	Key finding	Design implication
Chronic Care Model (Wagner)	Interventions that consistently improved care fell into four categories: changes to the way care is delivered, education and support for patients, educating or reminding providers, and changes to information systems [EV-01]. In diabetes, *delivery-system design* and *self-management support* showed the largest effects [EV-02].	Technology alone isn't the intervention. Every programme combines a care team, scheduled reviews, self-management support and decision support.
Standardised protocols (India)	WHO HEARTS and the India Hypertension Control Initiative use **drug- and dose-specific standard treatment protocols** with team-based care and information systems [EV-03, EV-04]. ICMR publishes Standard Treatment Workflows (STWs), e.g. T2DM and CKD [EV-05].	Programme templates reference CGC-approved protocols (ICMR STW / national programme) and can be configured per partner.
Hypertension	The 2025 ACC/AHA guideline keeps <130/80 mmHg as the general target and gives home BP monitoring a bigger role [EV-06]. Telemonitoring reduced SBP by about 3.7 mmHg vs usual care in a network meta-analysis [EV-07]; self-measured home BP alone gives a smaller reduction [EV-08].	Home BP **plus clinician co-intervention** (titration reviews), not readings alone. The target is set per patient by the clinician.
Diabetes	The ADA Standards of Care 2026 widen CGM eligibility and assess glycaemic status with A1C, BGM and CGM metrics (TIR, TBR, GMI) [EV-09, EV-10].	Support CGM and BGM sources, plus CGM summary metrics. Hypoglycaemia gets its own escalation pathway.
Heart failure	A 2026 meta-analysis found remote monitoring reduced HF hospitalisations (IRR 0.81) and all-cause mortality (RR 0.90) [EV-11]. Another systematic review found **heterogeneous, mostly neutral** results, especially for mortality [EV-12]. Guideline therapy: ESC 2021 + 2023 focused update [EV-13].	HF telemonitoring is reasonable for selected patients, but don't claim it "prevents hospitalisation". Build GDMT-optimisation tasks (clinician-led).
COPD	GOLD 2026 makes *disease activity* a treatment target, lowers the Group E threshold to >=1 moderate/severe exacerbation, and adds a technology section [EV-14].	Exacerbation history drives programme intensity. Symptom diary and action plan are core; SpO_2 only where clinically indicated.
CKD	KDIGO 2024 classifies CKD by Cause, GFR and albuminuria (CGA). Monitoring frequency follows the risk heat map [EV-15].	The CKD programme schedules eGFR and uACR by heat-map category, set by the clinician.
Regulation — India	**DPDP Rules 2025** were notified 13 Nov 2025 with phased commencement: Consent-Manager provisions from **Nov 2026** and core obligations from **May 2027** [EV-16]. Also applicable: NMC Telemedicine Practice Guidelines 2020, ABDM, CDSCO SaMD rules, NABH.	Build for full DPDP compliance now: consent ledger, notice, rights, breach handling. Long-term programmes will outlive the transition period.
Regulation — US (future)	CMS 2026 Physician Fee Schedule added RPM code **99445** (2–15 days of data/30 days) and **99470** (first 10 min management) [EV-17].	Capture monitoring days and clinician time per patient-month. This also supports India payer reporting.

> **Evidence gaps (honest):** I haven't yet reviewed current guidelines for stroke, Parkinson's, arthritis/MSK, obesity, thyroid, frailty/multimorbidity, asthma (GINA) or post-operative pathways. They're marked **EV-TBD**, and the CGC's clinical leads must attach the source before those programmes can be approved.

2. Care Pathways & Status Model

Terminology update (Master v3.0): the master specification now defines four top-level pathways: **A Acute (ACU), B Chronic (CCM), C Virtual IPD (VIPD), D RPM/Elderly Care (RPM)**. The chronic-care *sub-variants* below were renamed to avoid clashing with those letters. "Monitoring-only enrolment" (`RPM-ONLY`) is now **Pathway D**.

2.1 Chronic Care Sub-variants

Code	Sub-variant	Who	Core components	Monitoring mode
CCM-STD	Chronic Care Management	Patients needing planned long-term follow-up	Care plan, scheduled reviews, med review, lifestyle and self-management support, periodic reassessment	Per plan
CCM-IOT	Chronic Care **with** IoT/RPM	CCM-STD **plus** a clinician-approved device monitoring plan	Above + connected devices, data-freshness rules, device support	`IOT` or `HYBRID`
CCM-MAN	Chronic Care **without** IoT	CCM-STD patients managed without connected devices	Consultations, patient-reported outcomes (PROs), manual readings entered by the patient/nurse, home visits, labs	`MANUAL`
CCM-ESC	Deterioration → Acute/Virtual IPD reassessment	Any enrolled patient with acute or worsening illness	Clinical reassessment, then a decision: adjust plan / Acute Care episode / urgent visit / Virtual IPD eligibility (D-04) / ED	Inherits

CCM-IOT and CCM-MAN are one programme with different monitoring modes, not separate products. A patient can switch modes (for example, a device is returned) without being re-enrolled. Every mode change is audited.

2.2 Three Distinct Enrolment Statuses (never merged)

Record	FHIR	Purpose	Owner
Monitoring-only enrolment (Pathway D)	`EpisodeOfCare` type= `RPM`	Time-limited device monitoring under a clinician order with a defined review owner, **without** a comprehensive care plan (e.g. 14-day post-discharge T0, BP diagnosis week)	Ordering clinician
Chronic-care enrolment	`EpisodeOfCare` type= `CCM` + `CarePlan` (one per programme; multimorbidity = a *coordinated* plan across programmes)	Long-term managed care	Primary clinician (responsible)
Virtual IPD admission	`Encounter` class= `HH` (inpatient-level at home), linked to the CCM episode	Acute inpatient-level care at home	Admitting hospitalist

2.3 Status Lifecycle

From	Event / condition	To
(start)	Referral received	Referred
Referred	Intake started	Screening
Screening	Criteria not met (reason recorded)	NotEligible
Screening	Clinician authorises + consent signed	Enrolled
Enrolled	Baseline done + care plan signed	Active
Active	Deterioration trigger (rule or human)	ReassessmentRequired
ReassessmentRequired	Clinician: manage in CCM (plan updated)	Active
ReassessmentRequired	Clinician opens an Acute Care episode (CCM stays Active, linked)	Active + ACU episode
ReassessmentRequired	Clinician admits to Virtual IPD (D-04 passed)	SuspendedForIPD
ReassessmentRequired	ED / hospital admission	SuspendedForHospital
SuspendedForIPD	IPD discharge	PostAcuteReview
SuspendedForHospital	Hospital discharge notified	PostAcuteReview
PostAcuteReview	Med reconciliation + plan review signed	Active
Active ↔ OnHold	Patient request / travel / device gap (reason)	OnHold / Active
Active	Goals met, step-down approved	Graduated
Active	Patient withdraws consent	Withdrawn
Active	Transfer of care / death / programme end	Closed

Emergency bypass (all states): patient SOS, an emergency symptom report, or a caregiver emergency call goes **straight to the Emergency Pathway** (Command Centre + 108/112, from the Domain 1 spec). Routine CCM workflow steps are never inserted in front of an emergency.

2.4 Transition Rules

#	From → To	Who can authorise	Pre-conditions	System effects
T1	Screening → Enrolled	Responsible clinician	Inclusion met, no exclusion, consent scopes signed (incl. family access)	Creates `EpisodeOfCare` ; assigns care team

#	From → To	Who can authorise	Pre-conditions	System effects
T2	Enrolled → Active	Responsible clinician	Baseline assessment, med reconciliation, signed care plan	Schedules tasks and measurements
T3	Active → ReassessmentRequired	**Rule engine or any care-team member**	Trigger fired (Section 8)	Creates reassessment task with an SLA from service policy; **doesn't change care level**
T4	ReassessmentRequired → SuspendedForIPD	**Doctor only, after clinical assessment**	D-04 Virtual IPD eligibility passed; IPD consent	CCM plan **suspended, not cancelled**; chronic meds reconciled into IPD orders; CCM alerts muted *only* for parameters the IPD monitors (traced)
T5	Any → SuspendedForHospital	Any clinician; or auto when a hospital admission notice (ADT) arrives	—	Same as T4; CCM team notified
T6	Suspended* → PostAcuteReview → Active	Responsible clinician	Discharge summary received, med reconciliation, plan review	Restores the plan as a **new version**; the earlier version is kept
T7	Active → Graduated / Withdrawn / Closed	Responsible clinician (Withdrawn: the patient)	Reason; handover letter if care transfers	Stops devices/tasks; data kept per retention policy

> **AI can't trigger T4, T5, T6 or T7.** AI may only raise T3 as a *suggestion*, which a human confirms (Section 13).

3. Disease Programme Catalogue (configurable templates)

Each programme is a versioned `PlanDefinition` with these sections:

- Population
- Inclusion and exclusion criteria
- Assessments
- Monitoring (IoT and manual options)
- Review intervals
- Escalation rules
- Responsibilities
- Evidence
- Approval

The table holds **draft defaults for CGC review**. The responsible clinician individualises them per patient; nothing here is a universal prescription.

Prog.	Population & inclusion (draft)	Key exclusions → other pathway	Baseline & periodic assessments	Monitoring options (if clinically relevant)	Review cadence (draft, configurable)	Example escalation triggers (detail in Section 8)	Core team	Evidence
DM-T2 / DM-T1	Confirmed T1/T2 diabetes	DKA/HHS signs, pregnancy (→ obstetric pathway), paediatric T1 (→ paediatric endocrine)	A1C, lipids, renal (eGFR/uACR), eye and foot exam, hypoglycaemia history	BGM log or CGM (TIR/TBR/GMI); weight; BP	A1C per ADA (more often if not at goal); clinician reviews per protocol	Level 2/3 hypoglycaemia; persistent hyperglycaemia above the patient's plan threshold; foot ulcer	Physician/diabetologist, nurse educator, dietitian	EV-05, EV-09, EV-10
HTN	Confirmed hypertension (HBPM/ABPM/clinic)	Suspected secondary/malignant HTN, pregnancy	BP, CV risk, renal, ECG	Validated home BP cuff (protocol-based measurement weeks); manual entry	Titration reviews while not at target; then periodic	Severe BP with symptoms (emergency); severe BP without symptoms (urgent review); sustained above target	Physician, nurse	EV-03, EV-04, EV-06–08
CKD	CKD by KDIGO CGA	Rapid eGFR decline, dialysis planning (→ nephrology-led)	eGFR, uACR, K^+, Hb, BP, med review (nephrotoxins, dose adjustment)	BP; labs by heat-map category; weight/fluid if clinician directs	Labs per KDIGO heat-map frequency	Critical K^+; AKI-pattern creatinine rise; fluid overload symptoms	Physician/nephrologist, dietitian, pharmacist	EV-05, EV-15
HF	Diagnosed HF (any EF)	NYHA IV at rest / cardiogenic signs (→ hospital)	NYHA, EF, NT-proBNP (where available), renal, K^+, GDMT status	Daily weight, symptoms, BP/HR; SpO_2 optional	GDMT optimisation visits; then periodic	Weight gain above the patient's clinician-set threshold; worsening dyspnoea/orthopnoea; hypotension on titration	Cardiologist/physician, HF nurse, dietitian, pharmacist	EV-11–13
CAD / post-cardiac event	Stable CAD, post-ACS/PCI/CABG	Unstable angina symptoms (→ emergency)	Lipids, BP, glucose, adherence (antiplatelet), cardiac rehab status	BP/HR; activity; symptoms	Per cardiology plan	Chest-pain symptoms (emergency); missed antiplatelet (adherence alert)	Cardiologist, physio (rehab), dietitian	EV-TBD

Prog.	Population & inclusion (draft)	Key exclusions → other pathway	Baseline & periodic assessments	Monitoring options (if clinically relevant)	Review cadence (draft, configurable)	Example escalation triggers (detail in Section 8)	Core team	Evidence
COPD	Spirometry-confirmed COPD	Acute exacerbation needing hospital/NIV	CAT/mMRC, exacerbation history (GOLD ABE), inhaler technique, eosinophils	Symptom diary; SpO_2 where clinician-indicated; activity	Higher intensity for Group E	Exacerbation symptoms per action plan; SpO_2 below the patient's target band	Pulmonologist/physician, respiratory physio, nurse	EV-14
Asthma	Diagnosed asthma	Life-threatening attack features (→ emergency)	Control score, triggers, inhaler technique, action plan	Symptom diary; peak flow optional	Per control level	Action-plan red zone	Physician, nurse	EV-TBD (GINA)
Stroke recovery / neuro	Post-stroke/TIA, other neuro follow-up	New focal deficit (→ emergency stroke pathway)	mRS/Barthel, BP, AF status, swallowing, mood	BP; function PROs; falls	Per neuro/rehab plan	New neuro symptoms (emergency); falls; BP above target	Neurologist, physio, OT, SLP	EV-TBD
Parkinson's & long-term neuro	Diagnosed PD/other	Acute confusion/aspiration (→ urgent)	Motor/non-motor scales, falls, swallowing, cognition	Medication-timing adherence; falls log	Per neurology plan	Missed time-critical doses; falls; swallowing difficulty	Neurologist, physio, SLP, pharmacist	EV-TBD
Arthritis / MSK	OA/RA/chronic MSK	Septic joint signs (→ urgent)	Pain/function scores; DMARD lab monitoring for RA	PROs; activity	Per rheumatology/DMARD schedule	Abnormal DMARD labs; infection signs on immunosuppression	Physician/rheumatologist, physio	EV-TBD
Obesity / metabolic	BMI/waist criteria per CGC	Eating disorder (→ specialist)	Weight, waist, BP, glucose, lipids, comorbidities	Weight scale; activity	Per plan	Rapid unintended change; medication side effects	Physician, dietitian, physio	EV-TBD
Thyroid	Hypo/hyperthyroidism	Thyroid storm / myxoedema signs (→ emergency)	TSH ± FT4	Symptom PROs	Labs after dose change per protocol	Abnormal TSH outside plan; symptoms	Physician/endocrinologist	EV-TBD
Frailty & multimorbidity (elderly)	>=2 chronic conditions and/or frailty (e.g. CFS >= CGC threshold)	—	Comprehensive geriatric assessment, polypharmacy review, falls, cognition, carer status	Mix by conditions; fall sensors optional	Coordinated single plan; regular MDT review	Functional decline; falls; carer breakdown	Geriatrician/physician, nurse, physio, pharmacist, social worker	EV-TBD
Post-op / post-discharge continuity	Recent surgery or hospital/VIPD discharge	Surgical complications (→ surgeon/ED)	Wound, pain, mobility, VTE signs, discharge meds	Vitals per discharge plan; wound photos	Time-limited (e.g. 30 days), then graduate or move to a disease programme	Wound infection signs; uncontrolled pain; VTE symptoms	Surgeon/physician, nurse, physio	EV-TBD

Programme approval checklist (each programme):

- ☐ Evidence source attached
- ☐ Thresholds and intervals signed off
- ☐ Escalation rules signed off
- ☐ Patient education content approved
- ☐ Responsibilities agreed with staffing
- ☐ Version number and review date

4. Longitudinal Patient Profile

Profile section	FHIR R4 resource(s)	History rule
Demographics & verified identity (ABHA/Aadhaar-verified flag, method, date)	`Patient` , `Patient.identifier` , `Provenance`	Version history; identity verification events kept
Active chronic conditions	`Condition` (clinicalStatus, verificationStatus, onset, abatement)	Status changes create new versions; **no deletes** (use `entered-in-error`)
Medical & surgical history	`Condition` , `Procedure`	Append-only
Allergies & ADRs	`AllergyIntolerance` (criticality, reaction)	Append-only, refuted entries kept
Current medications & reconciliation history	`MedicationStatement` , `MedicationRequest` , `List` (reconciliation snapshots)	Every reconciliation saved as a dated `List`
Consultations & admissions (incl. VIPD)	`Encounter` , `EpisodeOfCare`	Immutable once finished
Labs & diagnostics	`DiagnosticReport` , `Observation`	Corrections via `amended` status
Observations & trends	`Observation` (with device, method, data-quality extensions)	Raw readings never edited; annotations separate
Deteriorations & escalations	`Flag` , `DetectedIssue` , `Task` , `Encounter`	Append-only

Profile section	FHIR R4 resource(s)	History rule
Specialist assessments	ServiceRequest → Task → DiagnosticReport / DocumentReference	Recommendations stored separately from orders
Comorbidities & risk factors	Condition, RiskAssessment, Observation (social history)	Versioned
Functional status & mobility	Observation (Barthel, CFS, etc.), QuestionnaireResponse	Time series
Lifestyle & social	Observation (smoking, alcohol, activity), SDOH questionnaire	Time series
Patient-defined goals / clinician goals	Goal (expressedBy patient vs practitioner)	Achievement status history
Care-plan versions & review dates	CarePlan (replaces, period, extension nextReviewDate)	**Each change is a new version; the old one is kept**
Care team & primary clinician	CareTeam	Membership periods
Emergency contacts & family access	RelatedPerson + Consent (scope)	Consent history kept
Consent & communication preferences	Consent, Patient.communication, preference extension	Ledger
Upcoming follow-ups & pending actions	Appointment, Task, ServiceRequest	—

Rule: clinically significant data is **never silently overwritten**. Every change records who, when, why and the source (Provenance). A UI "edit" makes a new version with a reason.

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5. Individualised Chronic Care Plan

5.1 Structure (all 16 required elements)

#	Section	Content rules
1	Diagnoses & comorbidities	Confirmed Conditions only; unconfirmed shown as "provisional"
2	Baseline & risk	Linked baseline observations and risk tools used (name and version)
3	Patient goals	In the patient's own words + clinician-agreed measurable goal
4	Observations & frequency	Per parameter: source (device/manual/lab), frequency, target range, **who reviews**
5	Investigations & schedule	Labs/imaging with due dates
6	Medication plan	Reconciled list; **reconciliation due dates**
7	Lifestyle & self-management	From the approved education library only
8	Dietitian / physio referrals	ServiceRequest with clinical question
9	Doctor & specialist appointments	Appointment schedule
10	Nursing / home care	Visit frequency and task list with **required qualification**
11	Patient education	Assigned modules and completion
12	Warning symptoms & escalation instructions	Patient-language red/amber/green action plan
13	Emergency contact pathway	108/112, command centre number, nearest partner ED
14	Follow-up schedule	Auto-generated, editable
15	Review & reassessment dates	Mandatory next-review date
16	Responsible clinician & team	Named owner per section

5.2 Goal Schema (every goal)

Field	Example (illustrative only)
Goal statement	"Home BP within my target"
Measure	Mean of protocol home BP readings over the measurement week
Target	Set by clinician (e.g. per guideline/individualised)
Owner	Dr X (clinical), Nurse Y (measurement support)
Review interval	e.g. every 4 weeks until achieved
Outcome	Achieved / Improving / No change / Worsening / Not assessable (insufficient data)

5.3 Authorisation

- Care plans move through **Draft → Proposed → Signed (Active) → Superseded**.
- AI or a template may fill a *Draft*. Only the responsible clinician can sign.
- AI **can't** prescribe, change doses, stop treatment or set targets the clinician hasn't verified (Section 13).

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6. Doctor Chronic Care Dashboard — Worklists & Prioritisation

6.1 Worklists

Worklist	Inclusion logic (configurable)
Active patients	Status = Active
By programme	Group by `PlanDefinition`
Follow-up due (next 7 days) / **Overdue reviews**	`CarePlan.nextReviewDate`, `Appointment`
Abnormal / deteriorating trends	Rule-based trend alerts (Section 8) not yet acknowledged
Data gaps (missing scheduled measurements)	Expected vs received per plan
Device connectivity & freshness	Hub/device heartbeat
Pending lab results / unacknowledged results	`ServiceRequest` without result past due; results without ack
Medication reconciliation / review due	Reconciliation due date
Adherence concerns	Patient-reported missed doses per rule
Specialist referrals outstanding / recommendations awaiting authorisation	`Task` status
Nursing visit needed	Plan task due/overdue
Dietitian / physio review needed	Referral status
Repeated emergency escalations	>= N emergency/urgent events in a period (N set by the CGC)
Possible Virtual IPD reassessment	Status = ReassessmentRequired

Filters: programme, clinician, care-plan status, risk category, monitoring mode (IoT/Manual/Hybrid), next review date, location.

6.2 Prioritisation Rules

- The risk category comes **only from a documented, CGC-approved rule set**, which is versioned and shown in an info panel: "Why is this patient here?"
- **Clinical risk and data completeness are separate dimensions.** A missing reading or disconnected device puts the patient in the *Data gaps* worklist with a "Data gap" badge. It **never** marks them high clinical risk on its own.
- Patient-reported concerning symptoms always outrank calculated scores.

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7. Longitudinal RPM & Trend Analysis

7.1 Observation Provenance (shown on every reading)

Value + unit, **measurement time**, receipt time, source (device model/serial, manual entry by patient/caregiver/nurse, lab), method (e.g. protocol BP), signal or data quality flag, and whether it was entered by proxy.

7.2 Freshness States (never treat old data as live)

State	Definition (configurable per parameter and programme)	UI
Live	Streaming device, last sample within the streaming window	Green dot "Live"
Current	Within the expected measurement interval	Timestamp
Due	Expected reading not yet received, inside the grace period	Grey "Due"
Missing	Grace period passed	Hollow marker "Missing"; **no interpolation**
Stale	Last value older than the clinical validity window	Value greyed with "Last: 3 days ago"

Rules:

- Charts **don't interpolate** across missing periods; they show gaps.
- Averages state their n and period ("Mean of 9 of 14 expected readings").
- A missing value is never shown as normal.

7.3 Patient Monitoring Dashboard Components

- Current values (each with its freshness state)
- Historical graphs with clinician target bands
- Baseline line
- **Treatment-change markers** (medication start/stop/dose change)
- Alert history
- Clinician annotations
- Relevant lab trends (e.g. eGFR, A1C, K^+)
- Device info panel
- Data completeness % per parameter

Parameters appear **only if they're in the patient's monitoring plan**.

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8. Chronic Care Alerts & Escalation

8.1 Severity Classes (Domain 2)

Class	Meaning	Routing principle
E – Emergency	Possible life-threatening symptoms (patient/caregiver report or device rule)	**Bypass CCM.** Emergency pathway: command centre + 108/112 guidance
U – Urgent clinical	Needs clinician assessment within the urgent window	Responsible clinician / on-call doctor; may lead to T3 Reassessment
R – Routine clinical	Needs review before the next scheduled review	Care-team queue
A – Adherence / data	Missed measurements, missed doses (patient-reported), overdue tasks	Nurse/care coordinator
T – Technical	Device or connectivity issues	Device support; only clinically escalated if the plan requires active data

Ack deadlines and escalation timers come from the Service Policy configuration (`SP-<class>-<programme>`), which the CGC and operations approve. The spec doesn't fix universal times.

8.2 Alert Rule Definition Schema (every rule must fill all fields)

Field	Description
Rule ID / version	e.g. `HF-WT-01 v1`
Programme	HF
Trigger logic	Exact computable definition, including the data-sufficiency condition
Severity class	E/U/R/A/T
Clinical rationale	Why it matters, with evidence ref
Responsible recipient	Role (not a named person); fallback role
Required response	e.g. "Contact patient; assess symptoms; decide on diuretic plan per protocol"
Ack deadline	Reference to `SP-...`
Escalation route	Next role(s) if not acknowledged
Closure criteria	What must be documented to close
Suppression rules	Duplicate/cool-down only. **Never suppressed by a low risk score.**
Approval	CGC sign-off ID and date

8.3 Draft Rule Catalogue (illustrative — needs CGC approval)

ID	Programme	Trigger (draft)	Class
GEN-SYM-E	All	Patient/caregiver reports an emergency symptom from the approved list (chest pain, stroke signs, severe breathlessness, unresponsive, etc.)	**E**
GEN-SYM-U	All	Patient reports worsening symptoms on the programme symptom check	U
GEN-TGT-R	All	>= X of the last Y valid readings outside the patient's target (data-sufficiency met)	R
GEN-BASE-U	All	Change from established baseline greater than the plan threshold	U
GEN-MISS-A	All	Missed measurements where the plan marks the parameter "clinically consequential"	A (→ U if the plan says so)
GEN-LAB-U	All	Critical lab result (lab's critical list)	U (or E per lab policy)
GEN-FU-A	All	Missed critical follow-up or investigation	A
GEN-MED-U	All	Patient reports a possible serious side effect / interaction flag	U
GEN-FALL-U	Frailty/neuro	Fall reported/detected, or repeated falls	U
GEN-FUNC-R	Frailty/neuro	Functional score decline beyond the threshold	R
GEN-DEV-T	IoT plans	Device disconnected beyond the plan's tolerance	T (→ A if data is required)
GEN-VIPD-U	All	Composite deterioration criteria in the programme → **ReassessmentRequired (T3)**	U
HTN-SEV	HTN	Severe BP reading: with symptoms → E; without → U	E/U
DM-HYPO	DM	Level 2 hypoglycaemia → U; Level 3 (needs assistance) → E	U/E
HF-WT	HF	Weight gain above the patient's clinician-set threshold over the set days	U
COPD-AP	COPD/Asthma	Action plan amber → U; red → E	U/E
CKD-K	CKD	Potassium above the critical threshold	U/E

Hard rule: a calculated low risk score **never** suppresses a concerning symptom report or an emergency request.

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9. Medication & Adherence Management

9.1 Workflows

- Medication reconciliation (at enrolment, after any admission, at each scheduled review)
- Clinician-authorised medication list
- Renewal requests (patient-initiated → clinician queue)
- Reminders
- Patient-reported adherence and missed doses
- Side-effect reporting
- Medication review appointments
- Refill coordination with partner pharmacy (where supported)
- Allergy and interaction checks (only from validated drug-knowledge sources)
- Change history with audit
- Clinician adherence summaries

9.2 Dose-Event States (strictly separate)

State	Meaning	Source
`REMINDER_SCHEDULED`	Reminder planned	System
`REMINDER_SENT` / `DELIVERED`	Push/SMS sent / delivery receipt	System
`NOTIFICATION_OPENED`	User opened the reminder	System, **doesn't mean the dose was taken**
`PATIENT_CONFIRMED_TAKEN`	Patient or caregiver tapped "Taken"	Patient-reported
`PATIENT_REPORTED_MISSED` / `SKIPPED (reason)`	Patient-reported	Patient-reported
`NO_RESPONSE`	No confirmation in the window	System, shown as **Unknown**, not missed and not taken
`ADMINISTERED_BY_PROFESSIONAL`	Nurse administered (eMAR, with ID scan)	Professional

Rules:

- No automatic dose changes.
- **No compensatory or "double-up" advice from the system.** After a missed dose the app shows only the *clinician-authored* missed-dose instruction attached to that prescription, if there is one. Otherwise it shows "Contact your care team".
- Adherence % always states its basis: "patient-reported, 22 of 28 doses confirmed; 4 unknown".

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10. Nursing & Home Visits

10.1 Visit Workflow

Scheduled visit → identity verification (photo + DOB/ABHA, or QR) → observations and symptom check → device setup/troubleshooting → med reconciliation support (within scope) → education → care-plan tasks → functional/wound assessment (if in plan) → escalate to responsible clinician if needed → schedule next visit → documentation and handover.

10.2 Task Scope Matrix (draft — confirm with nursing council norms and CGC)

Task	RN (GNM/B.Sc)	ANM / LPN	Trained care assistant	Family caregiver	Patient
Clinical assessment, triage decision	✅	◐ (under protocol)	❌	❌	❌
Vital signs measurement	✅	✅	✅	✅ (trained)	✅
Device setup / troubleshooting	✅	✅	✅	◐	◐
Medication reconciliation (data gathering)	✅	◐	❌	◐ (provide info)	◐
Medication administration (injectables/IV)	✅	◐ per scope	❌	❌	❌
Oral medication assistance	✅	✅	◐	✅	✅
Wound assessment & dressing	✅	◐	❌	❌ (simple dressing if trained)	❌
Functional assessment (scored tools)	✅	◐	❌	❌	PRO only
Patient/family education	✅	✅	◐ (scripted)	—	—
Escalation to clinician	✅	✅	✅	✅	✅

✅ allowed · ◐ limited/under protocol · ❌ not allowed. Every task in the system carries a `requiredQualification`. The app blocks completion by a user without that qualification.

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11. Specialist & Multidisciplinary Care

- **Referral package (mandatory fields):** clinical question, urgency, longitudinal summary (auto-compiled), current meds, allergies, relevant observations and trends, investigations, previous specialist input.
- **Statuses:** Draft → Sent → Accepted / Declined (reason) → Scheduled → Completed (report) → **Recommendations pending authorisation** → Incorporated / Not adopted (reason).
- **One coordinated care plan** with clear section ownership. Specialist recommendations are stored as *recommendations*. Changes to treatment need the responsible clinician (or the specialist, if delegated) to sign before they become orders.
- **MDT review:** scheduled case conferences for frailty/multimorbidity and complex patients, with documented outcomes.

12. Patient & Family Experience (screen list → Section 15.4)

Principles:

- Plain language in the patient's language.
- Every number paired with context ("within your target set by Dr X").
- No automated risk score shown as a diagnosis.
- **No promises** ("will keep you out of hospital").
- An emergency help button on every screen.
- Family access only within the consent scope (DPDP).

13. AI-Assisted Chronic Care — Capability Register

ID	Capability	Inputs	Intended use	Limitations	Validation before release	Bias assessment	Human oversight	Failure handling
AI-01	Longitudinal clinical summary	Structured record (Section 4), recent notes	Draft summary for clinician review	Can omit or misstate facts	Clinician-rated accuracy on a sample set; zero-tolerance check for meds/allergies	Check performance across languages and note styles	Clinician edits/signs; labelled "AI draft"	Show "Summary unavailable"; full record always accessible
AI-02	Trend & significant-change detection	Observations + plan targets	Highlight changes for review	Depends on data completeness	Retrospective comparison with clinician-labelled events	Device accuracy differences (e.g. SpO_2 by skin tone)	Suggestion only; rules engine remains authoritative	Falls back to rule-based alerts
AI-03	Missing-data identification	Plan schedule vs received	Data-gap worklist	— (deterministic preferred)	Unit tests	n/a	Nurse acts	Deterministic job
AI-04	Follow-up prioritisation	Worklists, risk rules	Order the queue	May deprioritise wrongly	Shadow mode vs the clinician ranking	Equity check (rural, elderly, female)	**Can't remove items; U/E alerts are pinned above AI ordering**	Default to the rules-based sort
AI-05	Patient-friendly education	**Approved content library only**	Personalise and translate approved text	Translation errors	Clinical and linguistic review of templates	Language coverage	Content team approves templates	Show the original approved text
AI-06	Adherence summary	Dose-event states (Section 9.2)	Clinician overview	Patient-reported bias	Matches the deterministic calculation	—	Clinician	Show the raw table
AI-07	Review-note preparation	Since-last-review data	Draft review note	As AI-01	As AI-01	As AI-01	Sign-off required	Blank template
AI-08	Overdue care-plan tasks	Tasks	Flag overdue	— (deterministic)	Unit tests	n/a	Coordinator	—
AI-09	Suggestions for clinician review	Record + guideline rules	"Consider" prompts (e.g. GDMT not optimised)	Guideline mismatch for the individual	Rule review by the CGC	Subgroup review	Accept/dismiss with reason	Hidden if unavailable
AI-10	Inconsistent-record detection	Meds vs conditions vs labs	Flag possible errors (e.g. duplicate drugs)	False positives	Precision/recall on audit sample	—	Pharmacist/clinician resolves	—

Prohibited for AI (system-enforced):

- Diagnosing
- Prescribing or changing/stopping treatment
- Suppressing or downgrading U/E alerts
- Autonomous enrolment, admission, discharge or graduation
- Patient-facing risk predictions presented as fact

If any capability starts to influence clinical decisions, it's assessed for **SaMD classification under CDSCO** before release.

14. Outcomes & Quality Metrics Framework

Metric	Numerator	Denominator	Period	Source	Exclusions	Type
Follow-up completion	Scheduled clinical reviews completed within window	Scheduled reviews due	Monthly	`Appointment` / `Encounter`	Patient-cancelled with rebook <= window	Process
Care-plan review completion	Plans reviewed by due date	Plans with review due	Monthly	`CarePlan`	On-hold/suspended	Process

Metric	Numerator	Denominator	Period	Source	Exclusions	Type
Monitoring adherence	Patient-days with >= required readings	Patient-days requiring readings	Monthly	Observation	Suspended days	Process
Data completeness	Received readings	Expected readings	Monthly	Plan vs Observation	Device-fault days (reported separately)	Data quality
Time to acknowledgement (U/E)	Median / p90 minutes, alert → ack	All U/E alerts	Monthly	Alert log	Test alerts	Safety
Alert resolution	Alerts closed meeting closure criteria within policy time	Alerts raised	Monthly	Alert log	Duplicates	Safety
Med reconciliation completion	Reconciliations done by due date (incl. post-discharge)	Reconciliations due	Monthly	List	—	Safety
Pending-investigation follow-up	Results acknowledged within policy time	Results received	Monthly	DiagnosticReport	—	Safety
Condition indicators (examples)	e.g. HTN: patients with latest valid BP at their individual target; DM: latest A1C at individual target	Enrolled >= 90 days with valid measure	Quarterly	Observation	Insufficient data (reported separately, **not counted as controlled**)	Outcome (intermediate)
PROs / QoL / function	Change in validated score from baseline	Patients with baseline + follow-up	6-monthly	QuestionnaireResponse	—	Outcome
Unplanned hospital transfers / ED visits	Events	Per 1,000 enrolled patient-days	Quarterly	Encounter , ADT feeds, patient report	Planned admissions	Outcome
30-day readmission (where measurable)	Readmissions <= 30 d of discharge	Discharges of enrolled patients	Quarterly	ADT/partners	Planned	Outcome
Patient / caregiver experience	Validated survey scores	Respondents	Quarterly	Survey	—	Experience
Equity of access	All of the above stratified by rural/urban, sex, age band, language, monitoring mode, payer	—	Quarterly	All	—	Equity

Evaluation rules:

- Engagement metrics (app opens, messages) are **never** reported as clinical improvement.
- Outcome comparisons need a defined baseline, a comparison method (e.g. matched cohort or interrupted time series), case-mix adjustment, and a missing-data analysis.
- Every metric definition is versioned.

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15. Chronic Care Screen Inventory (specification only — screens not to be designed until approval)

15.1 Standard Applied to Every Screen (avoids repeating it 60 times)

- **States:**
  - Loading: skeleton
  - Empty: guidance + primary action
  - Offline: cached read-only + banner; write actions queued only where safe (no orders offline)
  - Error: message + retry + support code
  - Permission-denied: explanation
- **Alert state banner:** an E/U alert for this patient shows a sticky banner on any patient-context screen.
- **Data freshness:** per Section 7.2 on any observation.
- **Audit:** view and act events are logged.
- **Accessibility:** >= 48 dp targets, dynamic type 200%, screen-reader labels, colour + icon + text for severity.
- **Localisation:** EN/HI + regional; numbers and units are locale-aware.
- **Figma reuse key** (from the shared canvases, **verify against the library**):

Key	Component
F-HDR	Patient header card
F-VT	Vital tile
F-PILL	Status pill
F-ALR	Alert row with severity chip
F-NAV	Bottom nav
F-STEP	Stepper
F-CHK	Checklist row
F-DOC	Consent document row
F-MAR	MAR line
F-CHAT	Chat thread
F-CHART	Trend chart card

Key	Component
F-TAB	Segmented tabs
F-CTA	Primary button

15.2 Doctor (Phase 1) — CC-D-xx (in the Doctor app, alongside D-01…D-32)

ID	Screen	Entry	Data displayed	Actions & permissions	Validation / safety	APIs	AC (key)	Figma reuse
CC-D-01	Chronic Care Dashboard	Home tab switch "Virtual Ward / Chronic Care"	Worklist counts (Section 6.1), programme tiles	Open worklist; filter	Clinical risk and data gaps shown separately	GET /cc/worklists	Counts update <= 60 s; E/U pinned	F-VT, F-PILL, F-NAV
CC-D-02	Worklist / Patient List	CC-D-01 tile	Patient rows: programme(s), mode, risk category + "why", next review, data completeness, badges	Open patient; bulk assign (coordinator)	No "high risk" from data gaps alone	GET /cc/patients?filters	"Why" panel lists the rule ID/version	F-HDR(row), F-PILL
CC-D-03	Referral & Programme Screening	"+ Enrol" / referral inbox	Referral, programme criteria checklist	Accept → Enrol; Not eligible (reason); Redirect (VIPD/ED/specialist)	Hard exclusions block; emergency features → emergency pathway	POST /cc/enrolments, GET /cc/programmes/{id}	Can't enrol with an exclusion ticked	F-STEP, F-CHK
CC-D-04	Enrolment Consent & Family Access	CC-D-03	Consent scopes (CCM, RPM devices, tele-consult, data sharing, family scope, communication prefs)	Capture e-consent (patient/LAR)	Mandatory scopes before Active	POST /consents	Family recipients restricted to scope	F-DOC
CC-D-05	Longitudinal Profile	Patient row	All Section 4 sections, history toggles	View; add/annotate per role	Edits create versions + reason	GET /patients/{id}/longitudinal	History visible for conditions/meds/plans	F-HDR, F-TAB
CC-D-06	Baseline & Risk Assessment	Enrolment flow	Programme-specific baseline form, risk tools	Enter/confirm; calculate tool scores	Range checks; tool name/version stored	POST /observations, /riskassessments	Calculated scores match reference vectors	F-VT, F-CHK
CC-D-07	Medication Reconciliation	Enrolment, post-acute review, scheduled review	Sources side by side (patient-reported, previous Rx, discharge)	Continue/stop/modify/flag; sign	Allergy/interaction checks; signing requires reasons for changes	POST /medrec	Plan can't be signed without reconciliation	F-MAR
CC-D-08	Care Plan Builder (Draft → Sign)	Enrolment / review	16 sections (Section 5.1), goal schema, programme template, optional AI draft	Edit, assign owners, sign	AI content labelled; targets need clinician entry; mandatory next-review date	POST/PUT /careplans (versioned)	Signing creates schedule; old version kept	F-STEP, F-CHK, F-CTA
CC-D-09	Monitoring Plan & Targets	Care plan	Parameters, source, frequency, target band, consequential flag, review owner	Set/modify	Out-of-safe-bounds needs second sign-off; audit	PUT /careplans/{id}/monitoring	Change reaches the rule engine <= 60 s	F-VT
CC-D-10	Longitudinal Trends	Profile / alert	Section 7.3 components	Annotate; change range; export	No interpolation; n/period on averages	GET /observations?...	Missing periods visible as gaps	F-CHART
CC-D-11	Chronic Alert Inbox & Detail	Alerts tab (filter CC)	Alerts by class E/U/R/A/T; rule rationale; data	Ack, contact, task, escalate, close (criteria)	E never actionable here except "Open emergency"; low score never hides	GET/POST /alerts	Closure blocked without criteria	F-ALR
CC-D-12	Reassessment Decision (Pathway D)	Alert / worklist ReassessmentRequired	Summary, triggers, recent trends, patient-reported symptoms	Manage in CCM (update plan) / Urgent visit / **Assess for Virtual IPD → D-04** / ED referral	Requires documented clinical assessment; **no AI-only admission**	POST /cc/reassessments	T4 only via D-04 pass	F-CHK, F-CTA
CC-D-13	Post-Acute Review	Discharge notice	Discharge summary, med changes, plan diff	Reconcile, update plan → Active	Mandatory reconciliation	POST /cc/postacute	Plan v(n+1) created	F-MAR, F-DOC
CC-D-14	Referral & MDT	Profile / plan	Referral package, status, recommendations	Create referral; authorise/decline recommendation	Mandatory package fields	POST /servicerequests, /tasks	Recommendation != order until authorised	F-CHAT, F-PILL
CC-D-15	Scheduled Review (tele/in-person)	Calendar	Since-last-review summary (AI-07 draft), goals progress	Document review, update goals/plan, next review	Sign-off; AI draft labelled	POST /encounters, /compositions	Review closes the due task	F-TAB
CC-D-16	Programme Outcomes (my panel)	More menu	Section 14 metrics for the clinician's panel	View; drill-down	Shows denominators and data-sufficiency	GET /cc/metrics	Metrics match definitions v-x	F-CHART

15.3 Professional Caregiver / Nurse (Phase 2) — `CC-N-xx` (extends the *Caregiver Flow* Figma)

ID	Screen	Entry	Data	Actions & permissions	Validation / safety	APIs	AC	Figma reuse
CC-N-01	My Day (CCM + VIPD visits)	Home	Route, visits with type (CCM/VIPD), tasks with required qualification	Start visit, navigate	Tasks above the user's qualification are locked	`GET /visits?date`	Locked tasks can't be completed	F-NAV, F-PILL (verify)
CC-N-02	Visit Check-in & Identity Verification	Visit	Patient photo, ID checks	Verify (QR/DOB/ABHA)	Can't document before verification	`POST /visits/{id}/checkin`	Mismatch → stop + report	F-HDR
CC-N-03	Observations & Symptom Check	Visit	Programme-specific form; device auto-import	Capture; flag	Range checks; source recorded (nurse)	`POST /observations`	Readings marked "nurse-measured"	F-VT
CC-N-04	Device Setup & Troubleshooting	Visit / technical alert	Device list, pairing, test reading	Pair, replace, return	Asset scan; test reading required	`POST /devices/pair`	Test reading received before close	F-CHK
CC-N-05	Medication Review Support	Visit	Current list, pill-pack photo	Record what the patient takes; flag discrepancies	Data gathering only; no changes	`POST /medrec/observations`	Discrepancy → clinician task	F-MAR
CC-N-06	Education Session	Visit	Assigned modules	Mark delivered; teach-back check	Approved content only	`POST /education/events`	Completion logged per module	F-CHK
CC-N-07	Functional / Fall / Wound Assessment	Visit (if in plan)	Scored tools; wound photo	Score; photo (chart storage)	Qualification check; photo consent	`POST /questionnaireresponses`, `/media`	Score saved with tool version	F-CHK
CC-N-08	Care-Plan Task Completion	Visit	Due tasks	Complete/not done (reason)	—	`PATCH /tasks/{id}`	Not-done needs a reason	F-CHK
CC-N-09	Escalate to Clinician (SBAR)	Any	SBAR auto-filled	Send urgent/routine; call	Emergency → emergency pathway button	`POST /escalations`	Recipient ack tracked	F-ALR, F-CTA
CC-N-10	Visit Note & Handover	End visit	Summary	Sign; next visit	Mandatory fields	`POST /compositions`	Visit can't close unsigned	F-DOC
CC-N-11	Adherence & Data-Gap Outreach	Worklist	Patients with A-class alerts	Call/message, record outcome	No clinical advice beyond scripts	`GET /alerts?class=A`	Outcome recorded	F-ALR
CC-N-12	Offline Sync Status	Header	Pending uploads	Retry	No silent data loss	Local queue	Unsynced count visible	—

15.4 Patient / Family (Phase 3) — `CC-P-xx` (extends the *Instant Service* / patient Figma)

ID	Screen	Purpose	Key data / actions	Safety / validation	AC
CC-P-01	Chronic Care Home	Today at a glance	Today's tasks, next appointment, last readings (with freshness), **Emergency help**	No risk scores shown as diagnosis	Emergency button on every screen
CC-P-02	My Conditions	Understand conditions	Confirmed conditions, plain-language explanations (approved library)	Provisional diagnoses labelled	—
CC-P-03	My Care Plan	See the plan	Goals, schedule, team, review date (read-only)	Shows version/date signed	Matches the active signed version
CC-P-04	Today's Tasks	Do today's tasks	Measurements, meds, exercises, education	—	Completion syncs <= 60 s online
CC-P-05	My Vitals & Trends	Track readings	Charts with *clinician* targets, gaps visible	"Your target set by Dr X"; no interpolation	Missing shown as missing
CC-P-06	Add Reading (manual / device)	Enter data	Guided measurement (e.g. BP technique), device auto-capture	Range checks; measurement time required	Source = patient/caregiver
CC-P-07	Medication Schedule	Take meds	Schedule, Taken / Missed / Skipped (reason), side-effect report	Section 9.2 states; no double-dose advice	"Opened" != "Taken"
CC-P-08	Symptom Diary	Report symptoms	Programme symptom check; free text	Emergency symptoms show the **Call 108/112 + notify team** screen immediately	E-trigger reaches the command centre <= 10 s
CC-P-09	Appointments & Consultations	Manage visits	Upcoming/past, join video, reschedule	—	—
CC-P-10	Lab & Diagnostic Schedule & Results	Tests	Due tests, booking, released results with clinician comment	Results released per clinician policy	—
CC-P-11	Device Status	Keep devices working	Battery, last sync, help	—	—
CC-P-12	Diet & Nutrition Plan	Follow diet	Dietitian plan, regional foods	Dietitian-authored only	—
CC-P-13	Physio & Activity Plan	Exercise	Video exercises, logs, pain check	Stop-and-report instructions	—

ID	Screen	Purpose	Key data / actions	Safety / validation	AC
CC-P-14	Care Team & Contacts	Reach the team	Team, message, call; hours	Banner: "Not for emergencies"	—
CC-P-15	Warning Symptoms & Emergency Help	Act fast	Personal action plan (green/amber/red), 108/112, nearest ED	Always accessible offline	Works offline
CC-P-16	Education Library	Learn	Assigned modules, language	Approved content only	—
CC-P-17	Progress Towards Goals	Motivation	Goal status (Section 5.2 outcome scale)	"Not assessable" when data is insufficient	No promises of outcome
CC-P-18	Consent & Family Access	Control data	Who sees what; add/revoke family; download data	DPDP rights (access, correction, withdrawal)	Revocation effective <= 5 min

Family/caregiver role: same app, role `FAMILY`, sees only consented scopes. Can add readings and confirm doses on the patient's behalf, and these are labelled "entered by [name]".

15.5 Allied Professionals (Phase 4) — `CC-A-xx`

ID	Role	Screen	Key content	Safety / AC
CC-A-01	All allied	Referral Inbox	Referral package, accept/decline	Decline needs a reason
CC-A-02	Dietitian	Nutrition Assessment	Anthropometry, diet history, renal/diabetic constraints	Uses latest labs with dates
CC-A-03	Dietitian	Diet Plan Builder	Plan → patient CC-P-12	Conflicts with the medical plan flagged (e.g. K⁺ in CKD)
CC-A-04	Physiotherapist	Functional Assessment	Validated scales, goals	Tool version stored
CC-A-05	Physiotherapist	Exercise Programme Builder	Exercises, progression → CC-P-13	Contraindication checklist
CC-A-06	Pharmacist	Medication Review	Polypharmacy, interactions, adherence data	**Recommendations only** → doctor authorisation
CC-A-07	All allied	Session Note & Recommendations	Note, recommendations to the plan	Recommendation != plan change until authorised
CC-A-08	All allied	My Caseload & Outcomes	Panel, scores over time	Denominators shown

15.6 Specialist & Programme Admin — `CC-S-xx`, `CC-M-xx`

ID	Role	Screen	Notes
CC-S-01	Specialist	Referral Review (consult)	Longitudinal package, video consult, report
CC-S-02	Specialist	Recommendations & Delegated Orders	Delegation flag set by the responsible clinician
CC-M-01	Programme manager	Programme Configuration	`PlanDefinition` versions, approval workflow (CGC sign-off)
CC-M-02	Programme manager	Alert Rule & Service Policy Config	Section 8.2 schema; SP timers; approval required to publish
CC-M-03	Programme manager	Enrolment & Capacity	Panels per clinician/nurse, waitlists
CC-M-04	Quality lead	Quality & Equity Dashboard	Section 14 metrics, stratified
CC-M-05	Admin	Education Content Library	Upload, clinical review, publish, versions

Total Domain 2 screens: 16 (Doctor) + 12 (Nurse) + 18 (Patient/Family) + 8 (Allied) + 2 (Specialist) + 5 (Admin) = 61.

16. Role-Permission Matrix (Domain 2)

Legend: **C** create · **R** read · **U** update · **S** sign/authorise · **—** none · *(s)* = within consent/assignment scope

Capability	Resp. clinician	Other doctor (on-call)	Specialist	Nurse (RN)	Care assistant	Dietitian / Physio	Pharmacist	Patient	Family	Prog. manager	Admin
Enrol / close enrolment	C S	C S	—	R	—	—	—	R (withdraw)	R(s)	R	—
Care plan	C U **S**	U S (cover)	R, recommend	R, U tasks	R tasks	R, U own section (recommend)	R, recommend	R	R(s)	R	—
Monitoring targets	C U **S**	U S	recommend	R	—	—	—	R	R(s)	—	—
Orders / prescriptions	C S	C S	C S if delegated	—	—	—	recommend	—	—	—	—
Med reconciliation	C U S	C U S	R	C U (gather)	—	—	C U (recommend)	C (report)	C (report) (s)	—	—
Observations	C R	C R	R	C R	C R	R	R	C R	C R(s)	—	—

Capability	Resp. clinician	Other doctor (on-call)	Specialist	Nurse (RN)	Care assistant	Dietitian / Physio	Pharmacist	Patient	Family	Prog. manager	Admin
Alerts – acknowledge/close	C U	C U	R	C U (A/T, U per policy)	R (A only)	R	R	—	—	R (analytics)	—
Reassessment → Virtual IPD	**S**	**S**	recommend	trigger T3	trigger T3	trigger T3	trigger T3	report symptoms	report	—	—
Referrals	C S	C S	accept, report	C (nursing referrals)	—	accept, report	accept, report	R	R(s)	R	—
Consent & family access	R	R	—	assist capture	—	—	—	C U (revoke)	R own	R	—
Programme / rule config	R	R	—	—	—	—	—	—	—	C U (publish after CGC **S**)	—
Content library	R	R	R	R	R	C (draft)	C (draft)	R	R	U	C U
Audit logs	R own actions	R own	R own	R own	R own	R own	R own	access history (DPDP)	—	R	R (security)

17. Platform Updates Required (Domain 1 → Domain 2 integration)

17.1 Alert & Escalation Engine

- Adds **scheduled/batch evaluation** (daily or hourly windows, data-sufficiency checks, missed-measurement jobs) next to the real-time stream path used by Virtual IPD.
- Adds severity classes E/U/R/A/T, the `SP-*` service-policy timers, and a rule registry with versioning and approval state. Only *approved* rules are executable.
- **Emergency bypass**: patient/caregiver E-symptom → command centre queue (Domain 1, Red tier) at the highest priority.

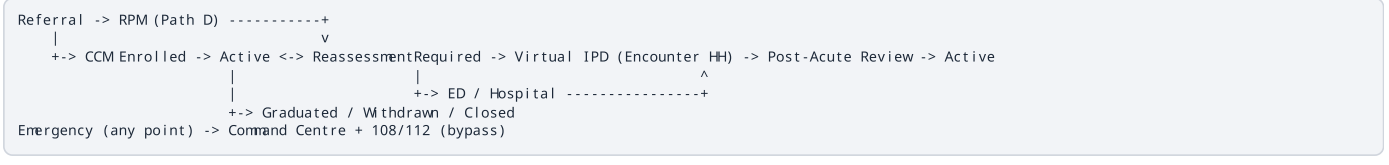
17.2 IoT/RPM Architecture

- Supports low-frequency consumer-grade but validated devices (BP, glucometer, CGM via vendor cloud, scale, peak-flow) as well as the Virtual IPD devices.
- **Device validation registry** (clinical validation status, e.g. validated BP device lists). Only registry devices can feed clinical rules.
- Device lifecycle: assigned → active → returned/lost. The data-gap logic is aware of the device state.

17.3 Data Model & APIs (additions)

Resource / endpoint	Purpose
`EpisodeOfCare` types `CCM`, `RPM` (Pathway D); status per Section 2.3	Enrolment state
`PlanDefinition` (programme), `CarePlan` (versioned), `Goal`, `List` (med-rec)	Programme & plan
`GET /cc/programmes`, `POST /cc/enrolments`, `PATCH /cc/enrolments/{id}/status` (transition with reason, validated against Section 2.4)	Enrolment
`POST/PUT /careplans` (version), `PUT /careplans/{id}/monitoring`	Plan
`GET /cc/worklists`, `GET /cc/patients?filters`	Dashboard
`POST /cc/reassessments`, `POST /cc/postacute`	Pathway D
`GET/POST /alerts` (class, ruleId, ruleVersion, closureCriteria)	Alerts
`POST /dose-events` (state per Section 9.2)	Adherence
`GET /cc/metrics?metricId&version`	Quality
`GET /rules`, `POST /rules/{id}/approve`	Rule governance
Webhooks/ADT: `encounter.admitted`, `encounter.discharged` from partner hospitals	Triggers T5/T6

17.4 Patient Lifecycle (unified across Domains 1 and 2)



18. Clinical Safety Requirements & Test Cases (Domain 2)

ID	Requirement	Test case (summary)	Priority
CC-SR-01	Emergency symptoms bypass CCM	Patient selects "chest pain" in diary → emergency screen shown + command centre E-alert within <= 10 s; no routine task created first	P0
CC-SR-02	Low risk score never suppresses symptoms	Patient with low risk category reports worsening symptoms → U alert raised and visible	P0

ID	Requirement	Test case (summary)	Priority
CC-SR-03	No AI-only admission	AI suggests reassessment → status changes only after a human confirms; T4 impossible without D-04 pass	P0
CC-SR-04	Missing != normal	Withhold readings for 3 days → chart shows gaps; goal shows "Not assessable"; no "controlled" in metrics	P0
CC-SR-05	Stale != live	Disconnect device → value greys out with age label after the validity window	P0
CC-SR-06	Data gap != high risk	Disconnected device only → patient appears in Data-gap list, risk category unchanged	P0
CC-SR-07	Dose-event integrity	Open reminder without tapping "Taken" → state = NOTIFICATION_OPENED, adherence unchanged	P0
CC-SR-08	No compensatory dosing	Report a missed dose → only the clinician instruction or "contact care team" shown	P0
CC-SR-09	History preserved	Change diagnosis/target → previous version retrievable with who/when/why	P0
CC-SR-10	Plan suspension on IPD	Admit to VIPD → CCM plan Suspended; on discharge, Post-Acute Review task created; reconciliation mandatory	P0
CC-SR-11	Qualification enforcement	Care assistant tries a wound assessment → blocked	P0
CC-SR-12	Recommendation != order	Pharmacist recommendation doesn't change the active med list until authorised	P0
CC-SR-13	Family scope	Family member without lab scope → lab results hidden; revocation applies <= 5 min	P0
CC-SR-14	Only approved rules run	A draft rule doesn't execute in production	P0
CC-SR-15	Alert closure criteria	Can't close a U alert without the documented closure fields	P1
CC-SR-16	Equity reporting	Metrics dashboard shows stratification; suppresses cells with small n	P1

19. Change Log to the Master Document

Master section (Domain 1 spec)	Change
§2 Vision, Scope & Care Tiers	Adds chronic sub-variants CCM-STD/IOT/MAN/ESC. Tier T0 ("Remote Monitoring / Post-discharge") is now formalised as **Pathway D (RPM)** or the post-op/post-discharge programme.
§4 Clinical Operating Model	Adds the unified lifecycle (Section 17.4) and transition rules (Section 2.4)
§5–6 Product Suite / Apps	Adds 61 Domain 2 screens (Section 15) across Doctor, Nurse/Caregiver, Patient/Family, Allied, Specialist, Admin
§7 Device Ecosystem	Adds the validated low-frequency device registry and CGM via vendor cloud (Section 17.2)
§9 Monitoring & Escalation	Adds batch evaluation, severity classes E/U/R/A/T, service-policy timers, rule governance (Sections 8 and 17.1)
§10–11 Data Model & APIs	Section 17.3 additions
§13 Compliance	**Correction:** DPDP Rules 2025 commence in phases (Consent Managers Nov 2026; core obligations May 2027) [EV-16]. Build for full compliance at launch.
§17 AI/ML Roadmap	Replaced for Domain 2 by the AI register (Section 13) and its prohibitions
§18 Testing	Adds CC-SR-01…16
§19 Roadmap	Adds Domain 2 stage gates (below)
§22 KPIs	Adds the Section 14 metrics framework
Doctor app spec (D-xx)	D-02 Dashboard gets a "Chronic Care" switch; D-03 census adds a CCM filter; D-04 Eligibility is reachable from CC-D-12

Domain 2 roadmap (stage-gated):

Stage	Scope	Gate
S1 (now)	This document	**Your approval + CGC review plan**
S2	CGC sign-off of 3 launch programmes (recommended: **HTN, T2DM, HF**) plus the post-discharge programme	Signed programme packs
Phase 1	Doctor screens CC-D-01…16 (Figma)	Clinician usability test
Phase 2	Nurse/Caregiver CC-N-01…12	Nursing usability test
Phase 3	Patient/Family CC-P-01…18	Patient usability test (elderly, regional language)
Phase 4	Allied CC-A, Specialist CC-S, Admin CC-M	—
Validation	End-to-end CCM ↔ VIPD journey simulation (CC-SR suite)	Safety case sign-off

20. Evidence Register

ID	Source	Used for
EV-01	Wagner et al., *Organizing Care for Patients With Chronic Illness Revisited* — PMC6739608	CCM components
EV-02	CCM-oriented interventions in diabetes, systematic review — Primary Health Care Research & Development; also Health Affairs evidence review, PMC5091929	Which CCM components matter
EV-03	WHO HEARTS technical package — WHO	Standard protocols, team care
EV-04	India Hypertension Control Initiative, protocol at scale — PMC10959138	India implementation
EV-05	ICMR Standard Treatment Workflows (T2DM, CKD) — ICMR STW T2DM, ICMR STW CKD	India protocols
EV-06	2025 ACC/AHA High Blood Pressure Guideline — ACC summary	HTN targets, HBPM
EV-07	Telemonitoring & HBPM network meta-analysis — PubMed 40156340	Effect size
EV-08	Self-measured home BP meta-analysis — Hypertension Research 2024	Effect size
EV-09	ADA Standards of Care 2026 — ADA press release	Diabetes programme
EV-10	ADA 2026 §6 Glycemic goals — PMC12690178; §7 Technology — PMC12690173	CGM metrics, hypoglycaemia
EV-11	HF telemonitoring modalities meta-analysis (2026) — npj Digital Medicine	HF RPM benefit
EV-12	HF telemonitoring systematic review (heterogeneous/neutral) — PMC13338606	Balanced claims
EV-13	ESC 2023 Focused Update of 2021 HF guidelines — summary, PMC11562572	GDMT
EV-14	GOLD 2026 COPD report — Guideline Central, Healio	COPD programme
EV-15	KDIGO 2024 CKD guideline — KDIGO, NKF heat map	CKD monitoring frequency
EV-16	DPDP Rules 2025 phased commencement — PwC India, Deloitte India	Compliance timing
EV-17	CMS CY2026 PFS RPM changes (99445, 99470) — Nixon Law Group, CMS MM14315	Future US billing data capture
EV-TBD	Stroke, Parkinson's, MSK/RA, obesity, thyroid, frailty/CGA, asthma (GINA), post-op, CAD/cardiac rehab	**Clinical leads to attach before programme approval**

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21. Decisions Needed for Approval (Step 4)

#	Decision	Recommendation
1	Launch programmes for Phase 1	**HTN, T2DM, HF + Post-discharge continuity** (strongest evidence and India burden; reuses Domain 1 devices)
2	Protocol basis	ICMR STWs / national programme protocols as the default, with partner-hospital overrides via CGC
3	Where Doctor CCM screens live	Inside the existing Doctor app (shared patient record), with a Virtual Ward ↔ Chronic Care switch
4	"Caregiver Flow" Figma scope	Confirm whether it covers **professional caregivers (nurse/attendant)**, family caregivers, or both. The spec assumes professional = CC-N and family = CC-P role.
5	Service-policy timers (ack deadlines)	Operations + CGC set them per class/programme before build of the alert engine
6	Figma access	Share view access (or exported frames/library) so the reuse column can be verified and Phase 1 screens built with existing components
7	Clinical Governance Committee	Name the members and the sign-off process for programme packs and rules

This specification is a product and technical design document. All clinical content is draft, and the MYDR24 Clinical Governance Committee and regulatory counsel must approve it before use with patients.